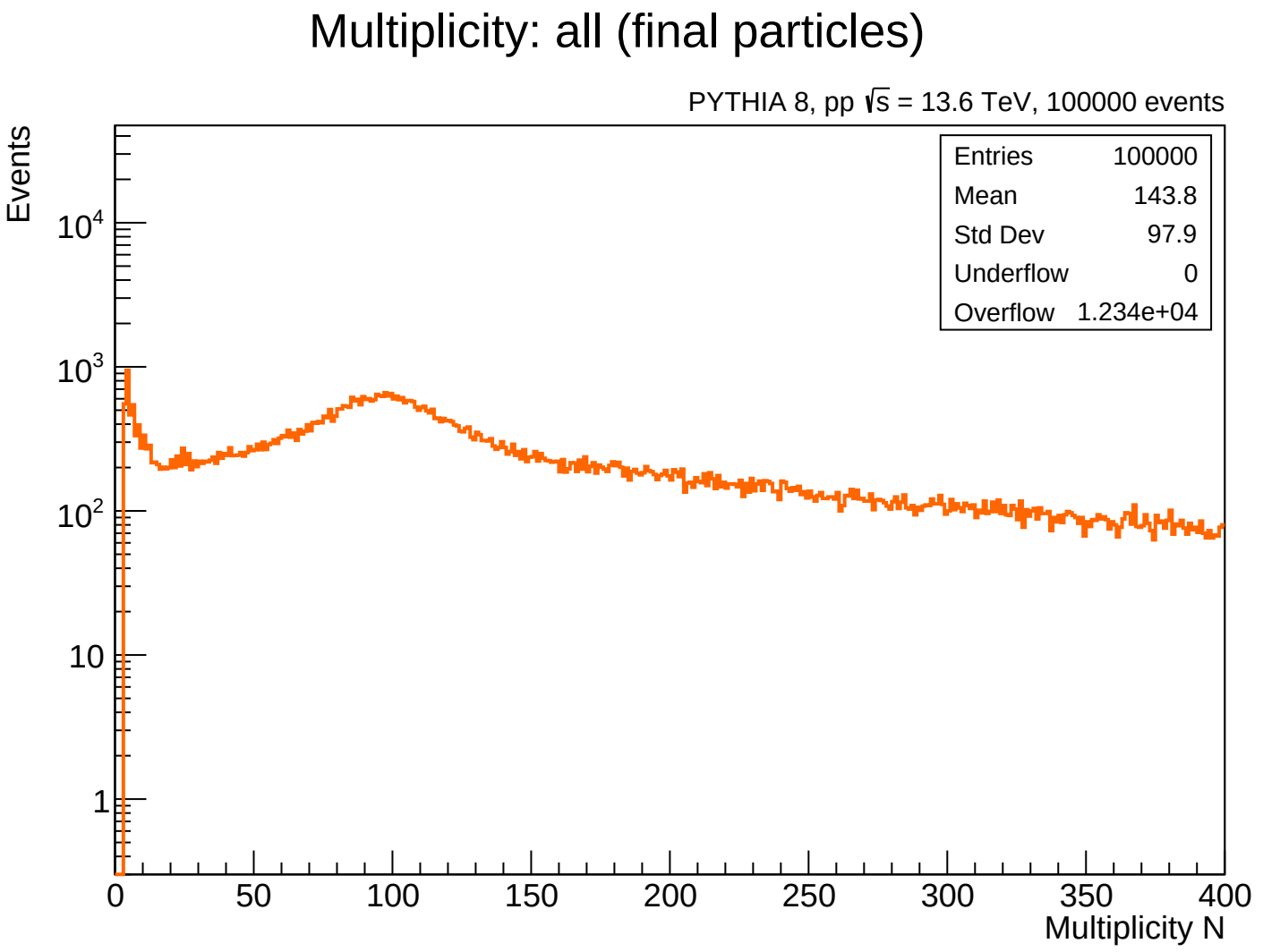
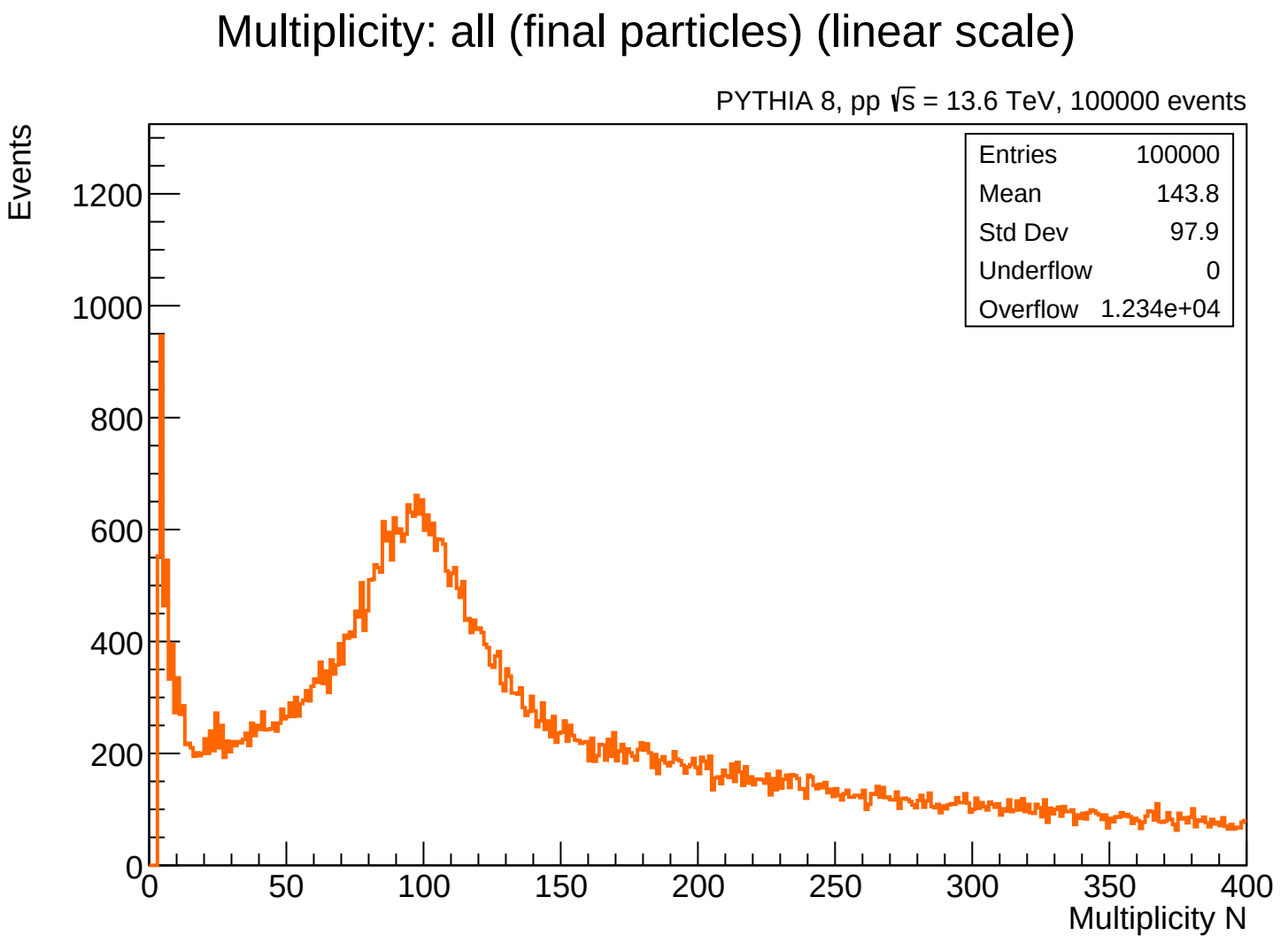
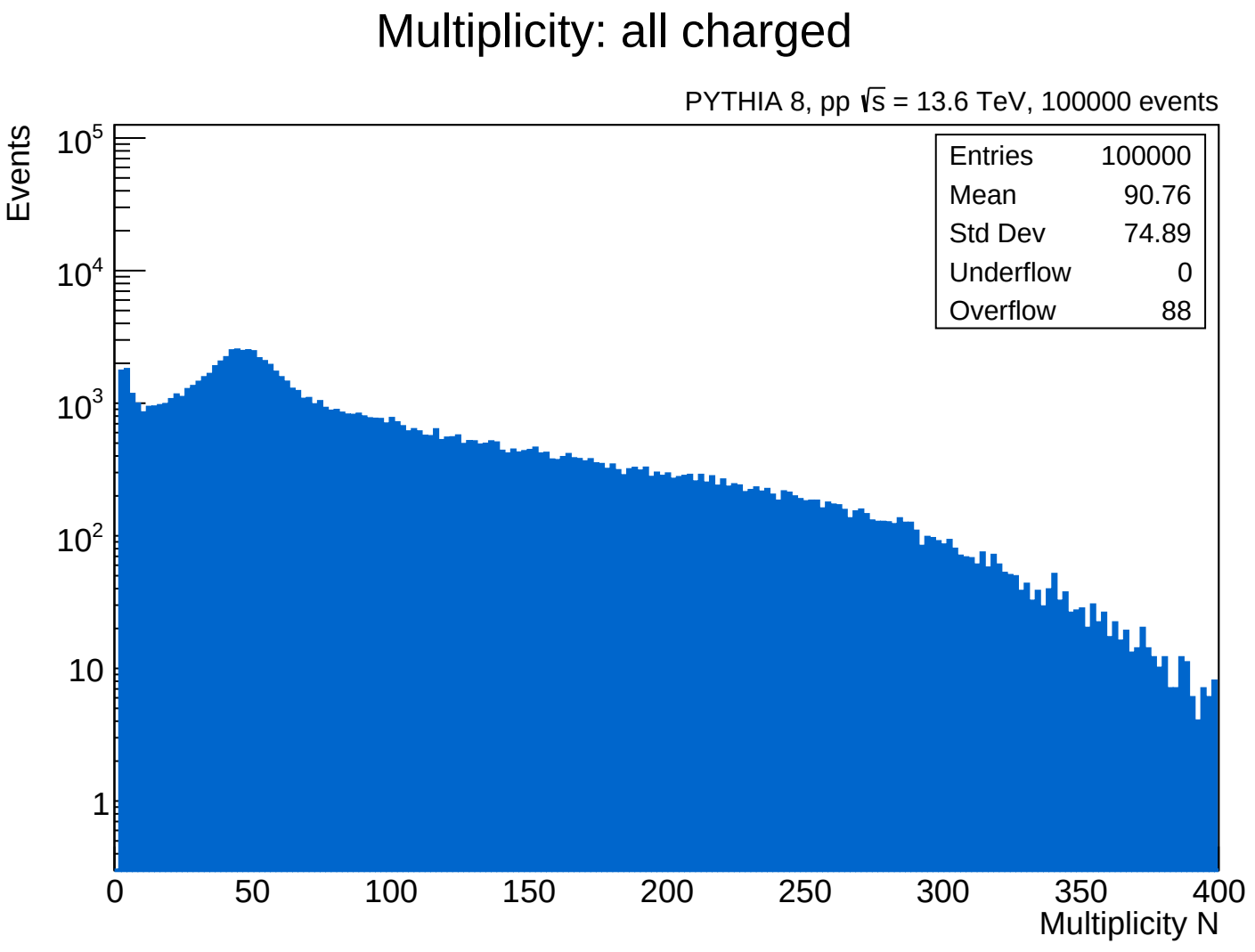
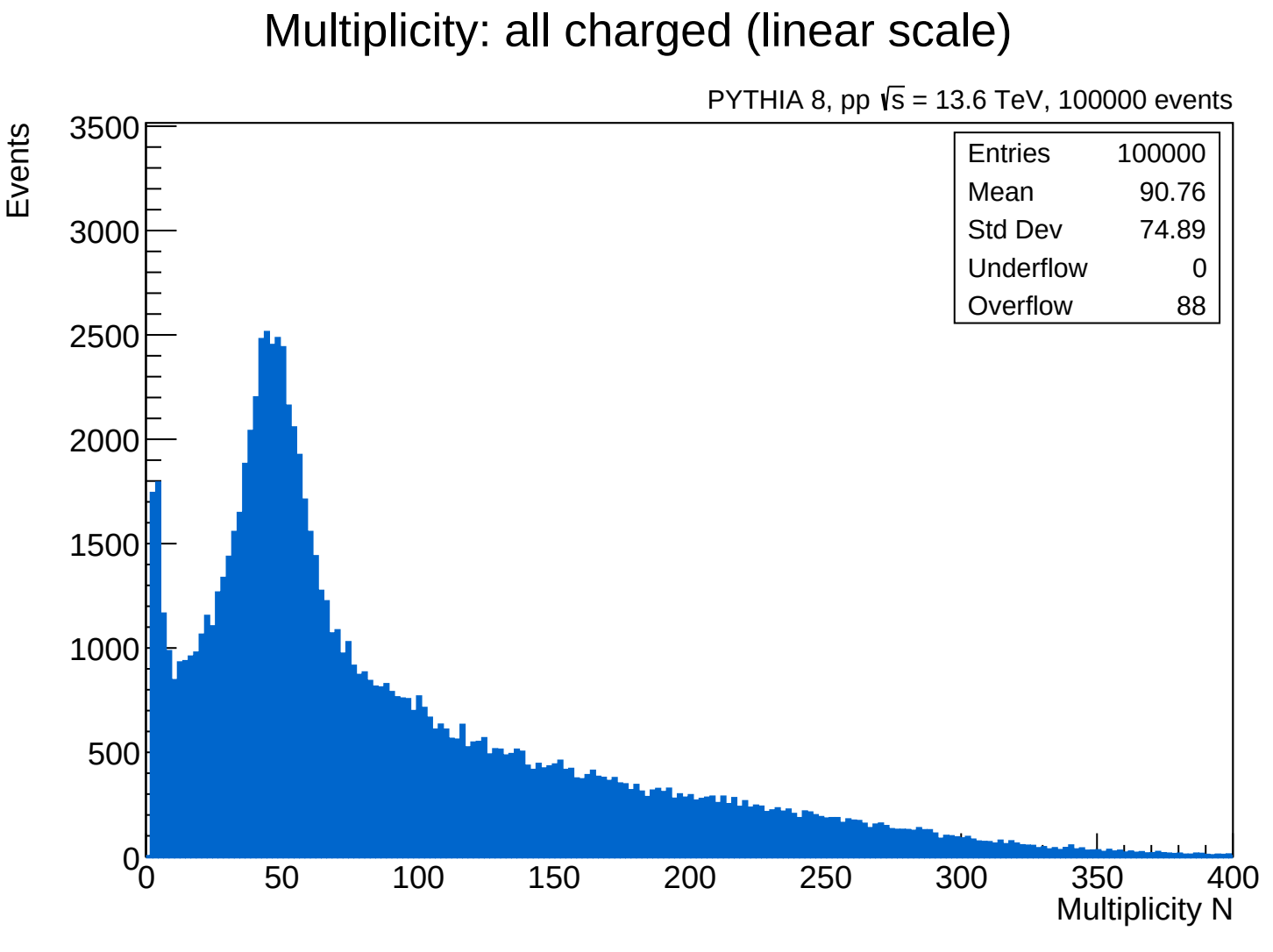
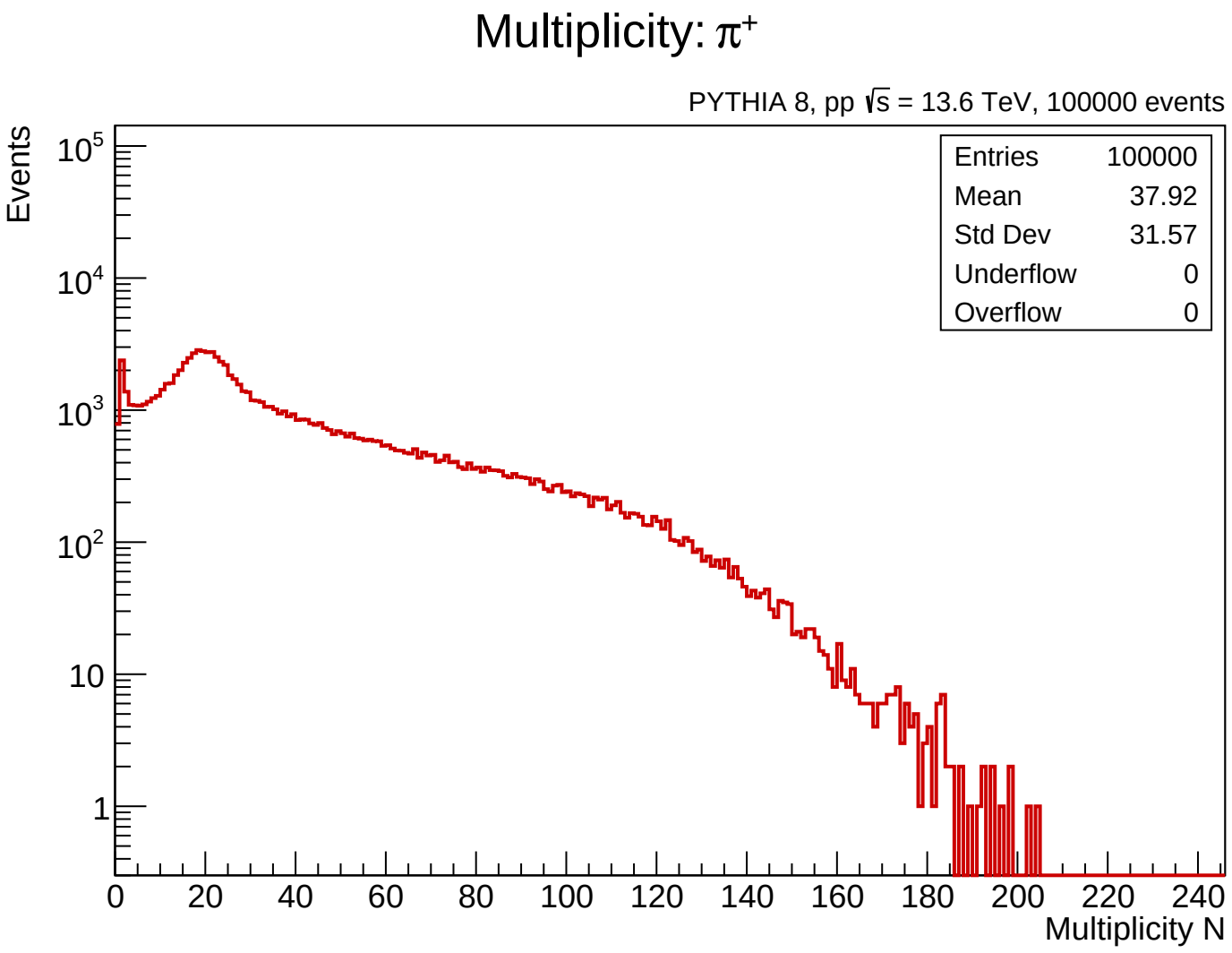
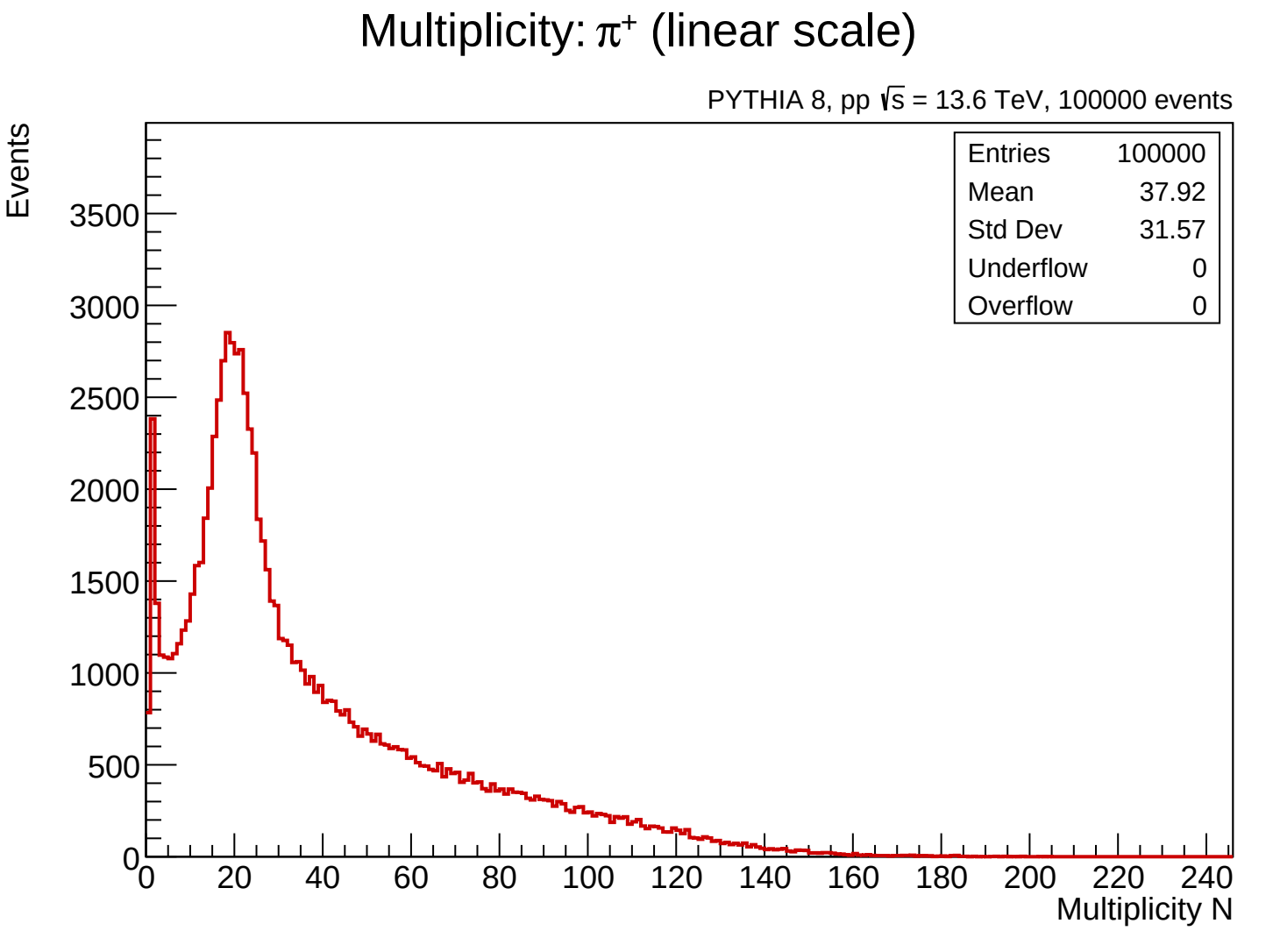

Multiplicity: all (final particles)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 143.8
Std Dev 97.9
Underflow 0
Overflow 1.234e+04
Events
10^4
10^3
10^2
10
1
0
50
100
150
200
250
300
350
400
Multiplicity N

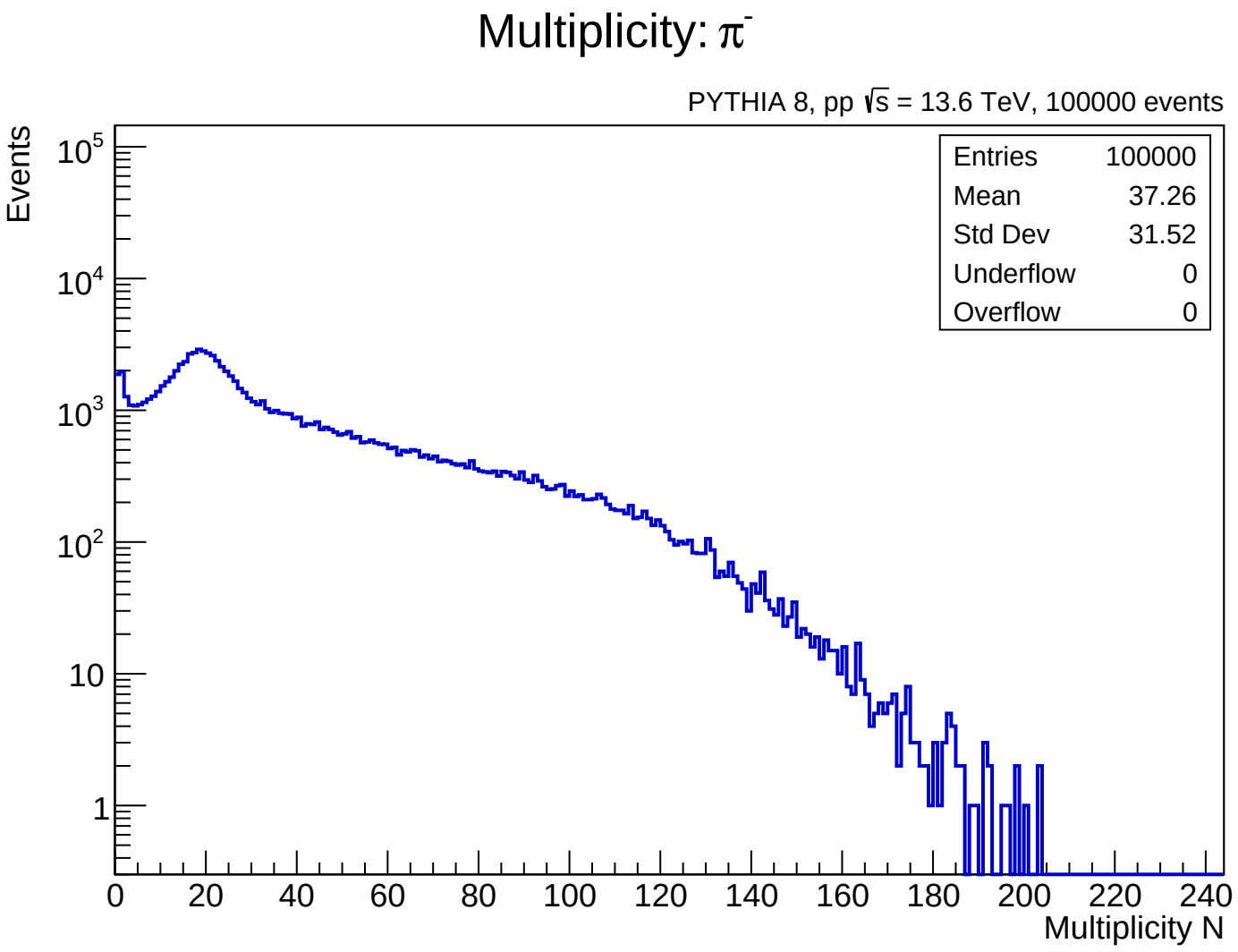

Multiplicity: all (final particles) (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 143.8
Std Dev 97.9
Underflow 0
Overflow 1.234e+04
Events
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
Multiplicity N

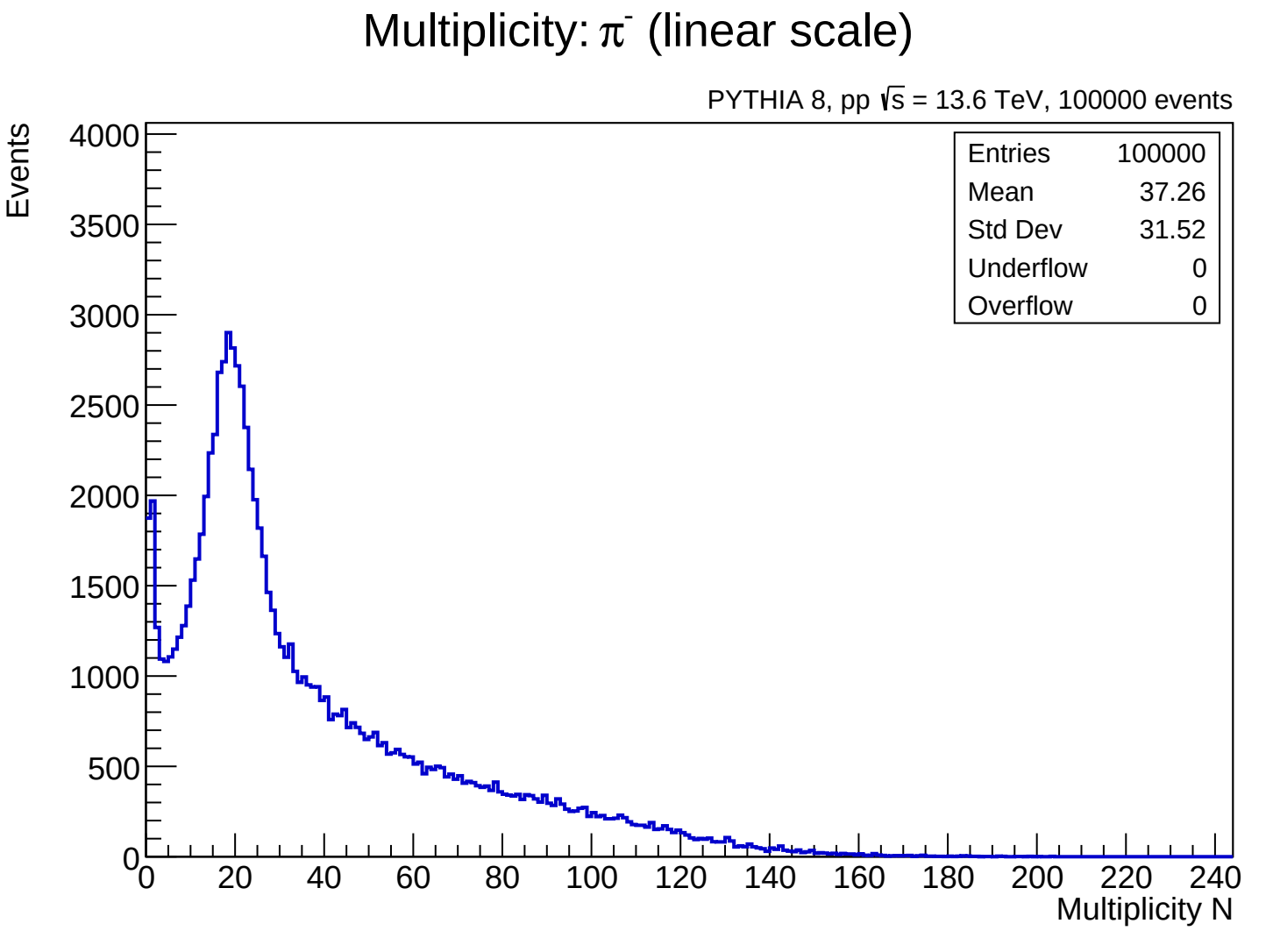

Multiplicity: all charged
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 90.76
Std Dev 74.89
Underflow 0
Overflow 88
Events
10^5
10^4
10^3
10^2
10
1
0
50
100
150
200
250
300
350
400
Multiplicity N

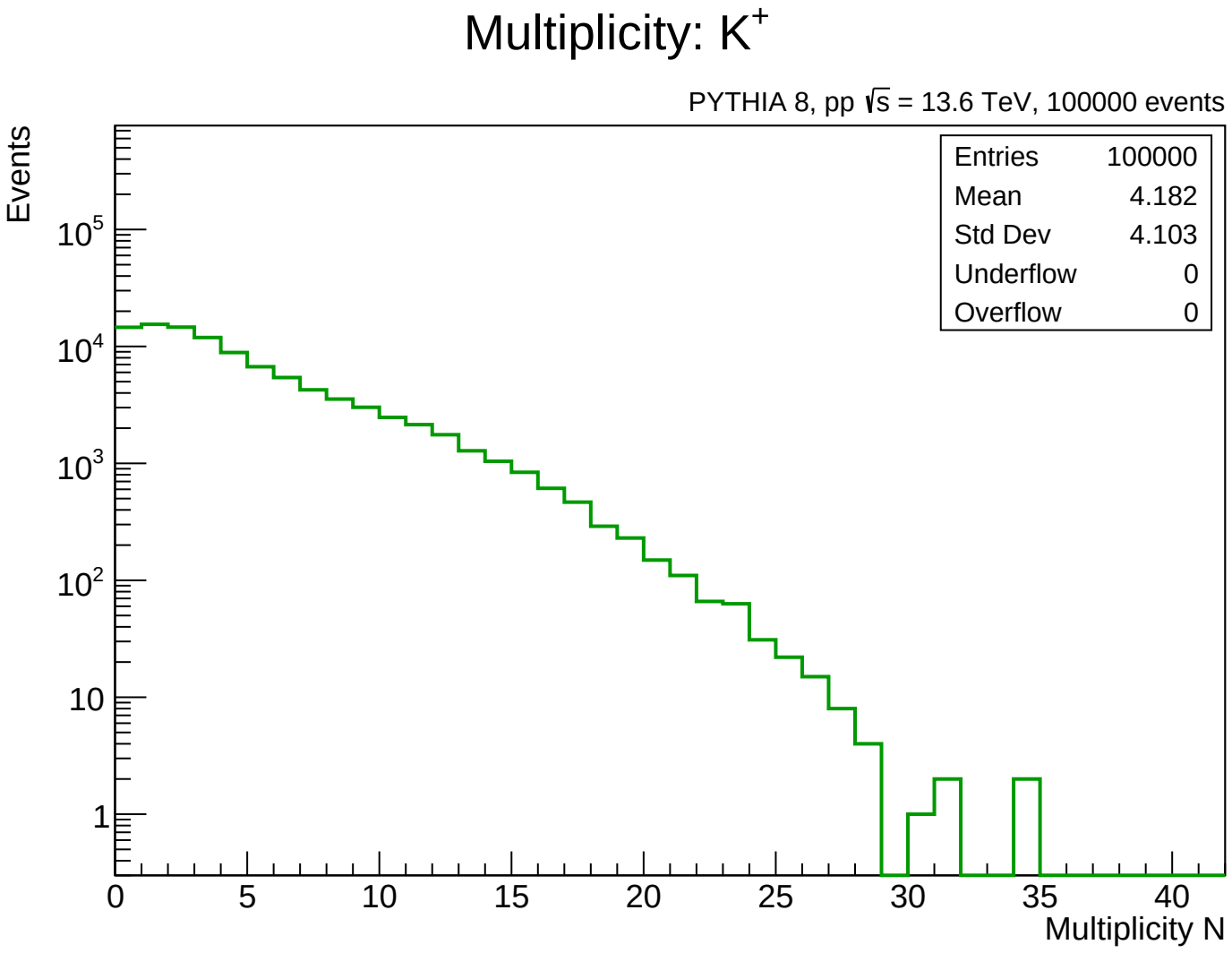

Multiplicity: all charged (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 90.76
Std Dev 74.89
Underflow 0
Overflow 88
Events
Multiplicity N
0
500
1000
1500
2000
2500
3000
3500
0
50
100
150
200
250
300
350
400

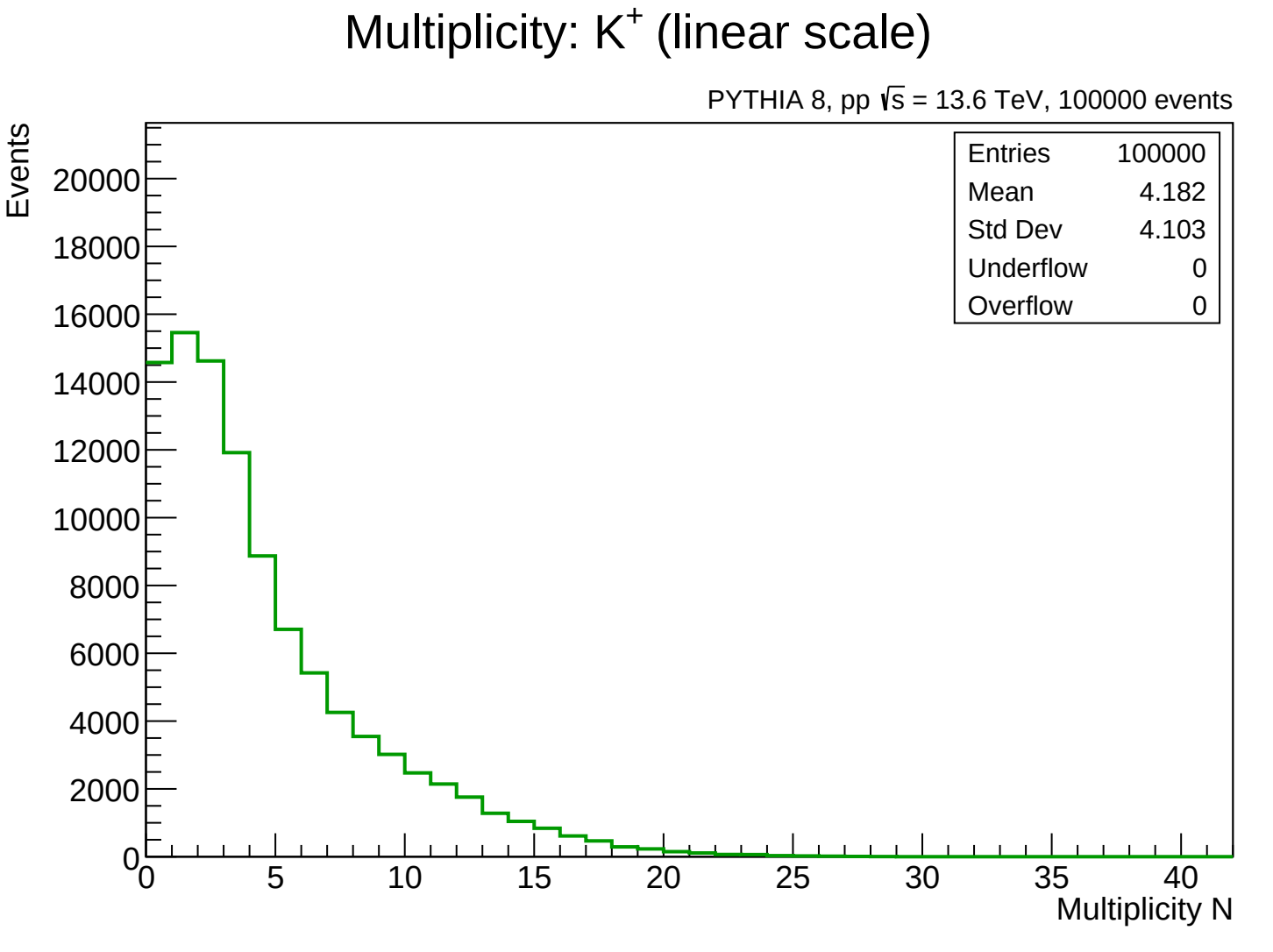

Multiplicity: π+
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 37.92
Std Dev 31.57
Underflow 0
Overflow 0
Events
10^5
10^4
10^3
10^2
10
1
0
20
40
60
80
100
120
140
160
180
200
220
240
Multiplicity N

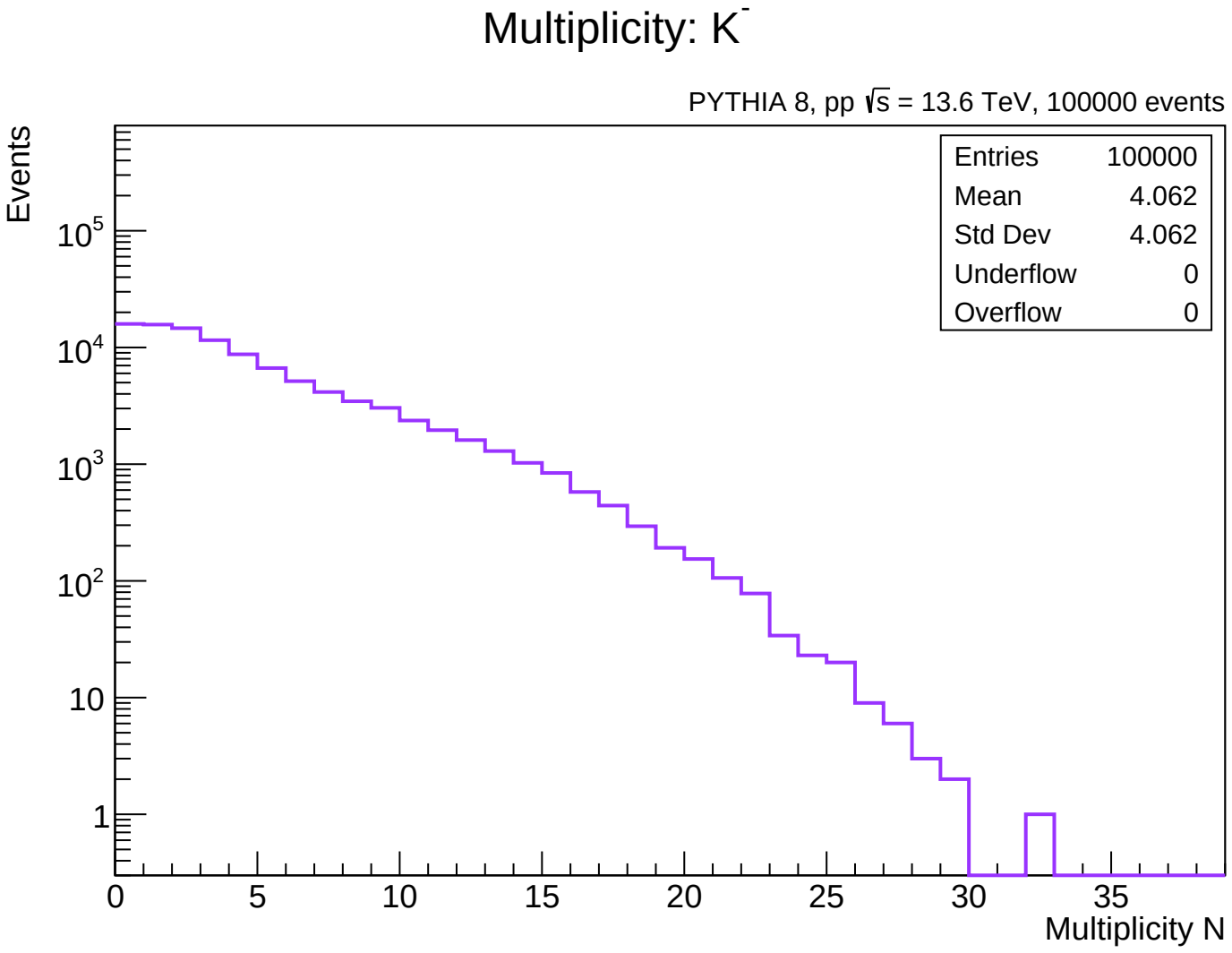

Multiplicity: π⁺ (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 37.92
Std Dev 31.57
Underflow 0
Overflow 0
Events
0
500
1000
1500
2000
2500
3000
3500
0
20
40
60
80
100
120
140
160
180
200
220
240
Multiplicity N

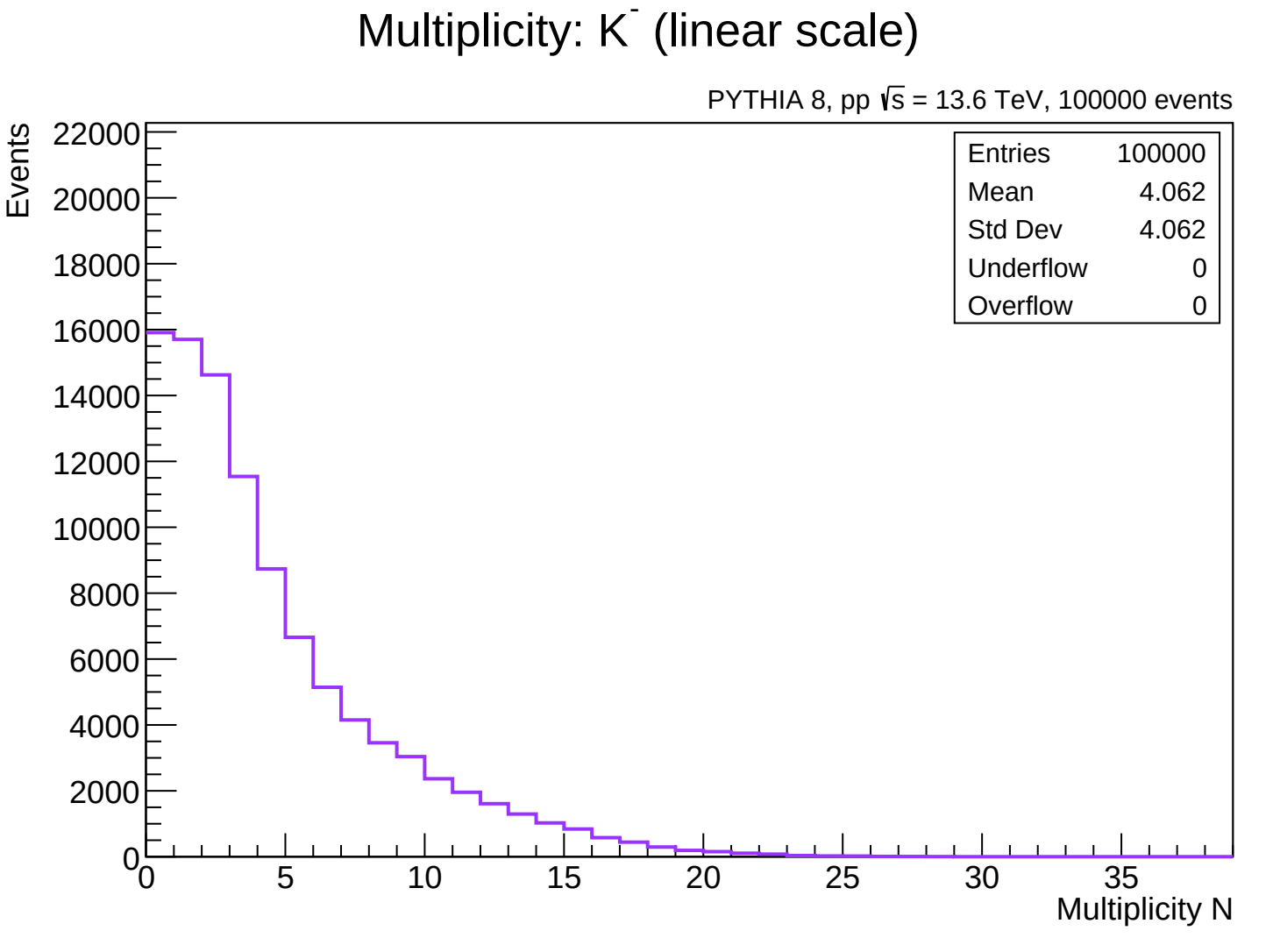

Multiplicity: π⁻
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 37.26
Std Dev 31.52
Underflow 0
Overflow 0
Events
10^5
10^4
10^3
10^2
10
1
0
20
40
60
80
100
120
140
160
180
200
220
240
Multiplicity N

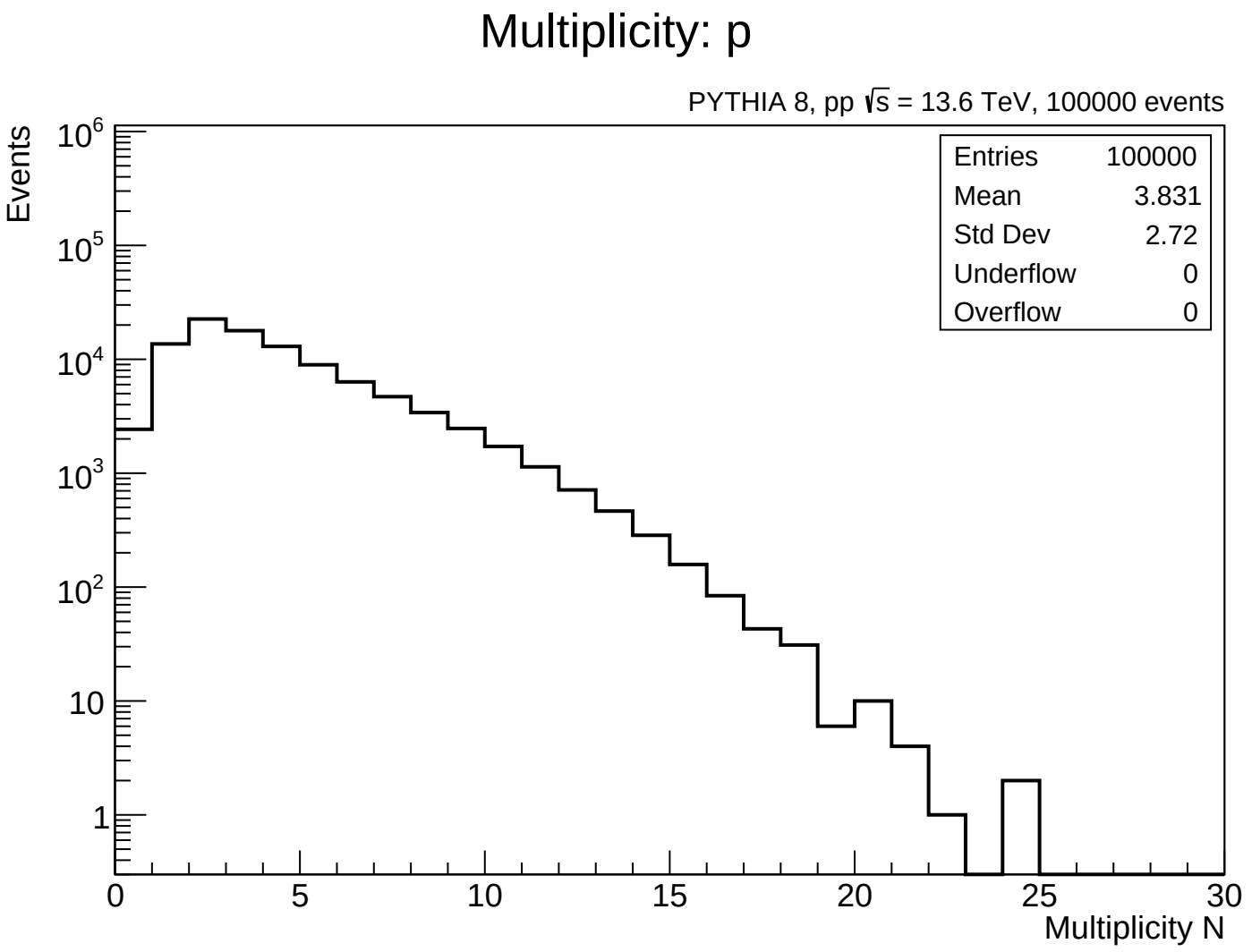

Multiplicity: π⁻ (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 37.26
Std Dev 31.52
Underflow 0
Overflow 0
Events
Multiplicity N
0
500
1000
1500
2000
2500
3000
3500
4000
0
20
40
60
80
100
120
140
160
180
200
220
240

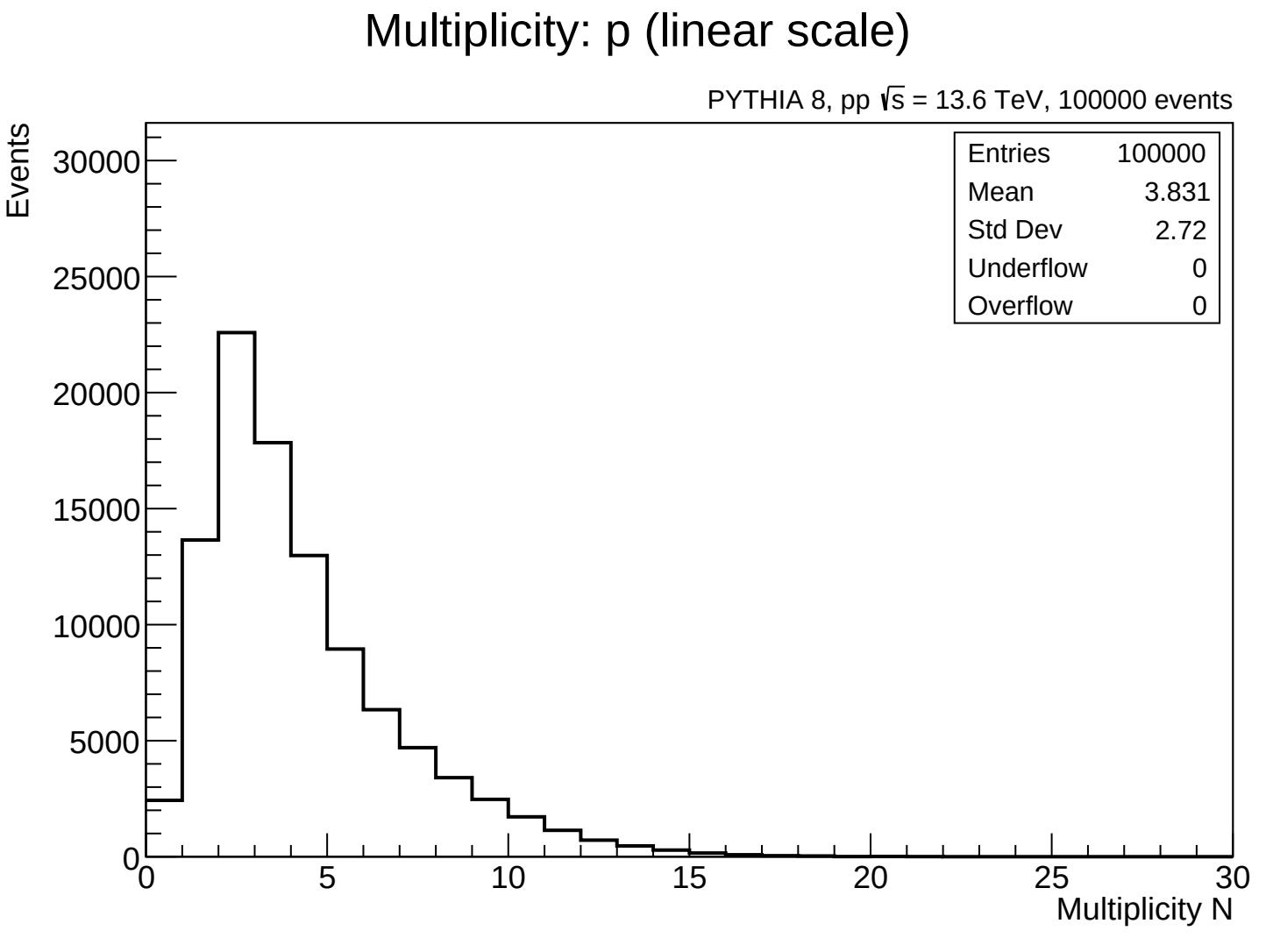

Multiplicity: K+
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 4.182
Std Dev 4.103
Underflow 0
Overflow 0
Events
10^5
10^4
10^3
10^2
10
1
0
5
10
15
20
25
30
35
40
Multiplicity N

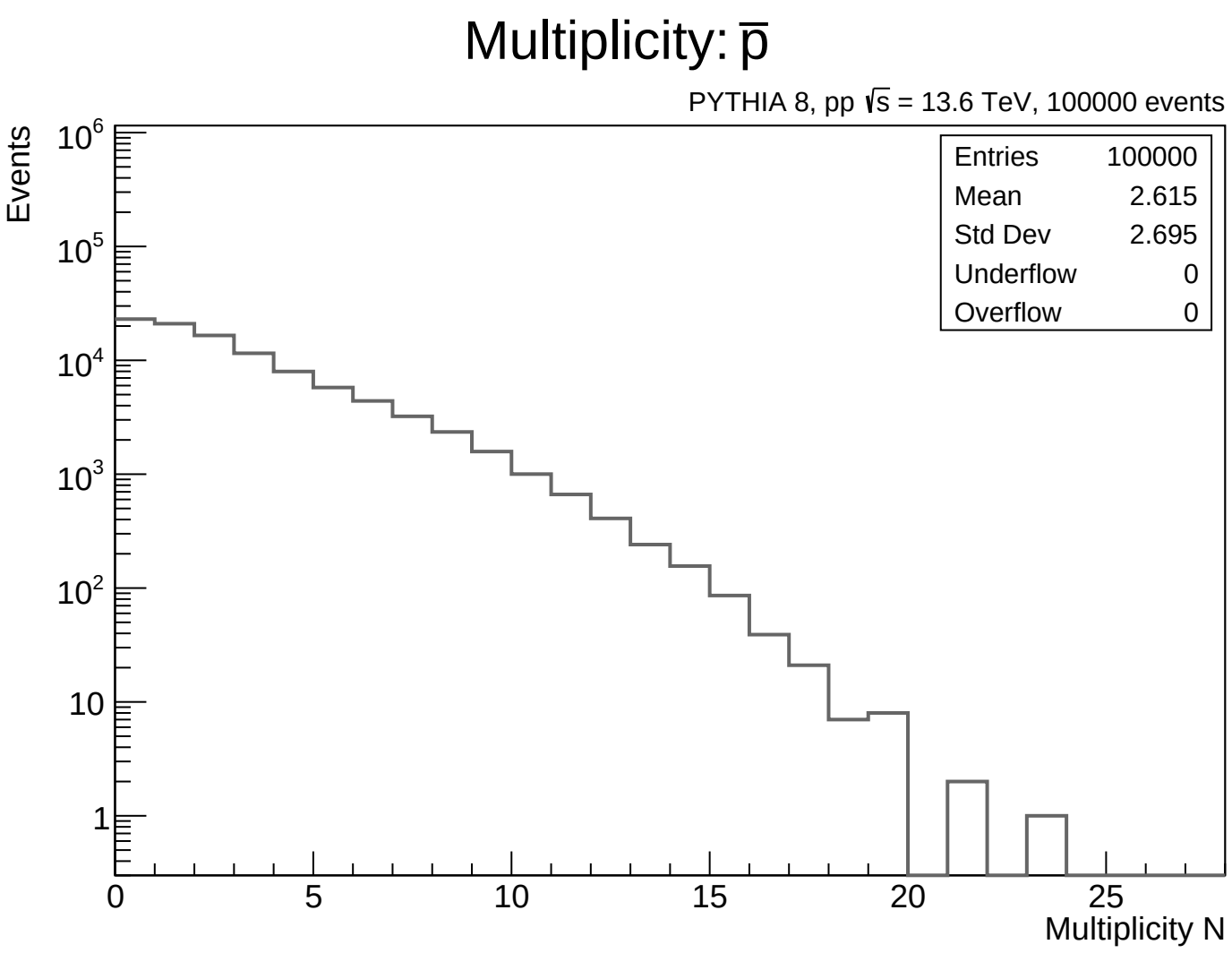

Multiplicity: K+ (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 4.182
Std Dev 4.103
Underflow 0
Overflow 0
Events
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
5
10
15
20
25
30
35
40
Multiplicity N

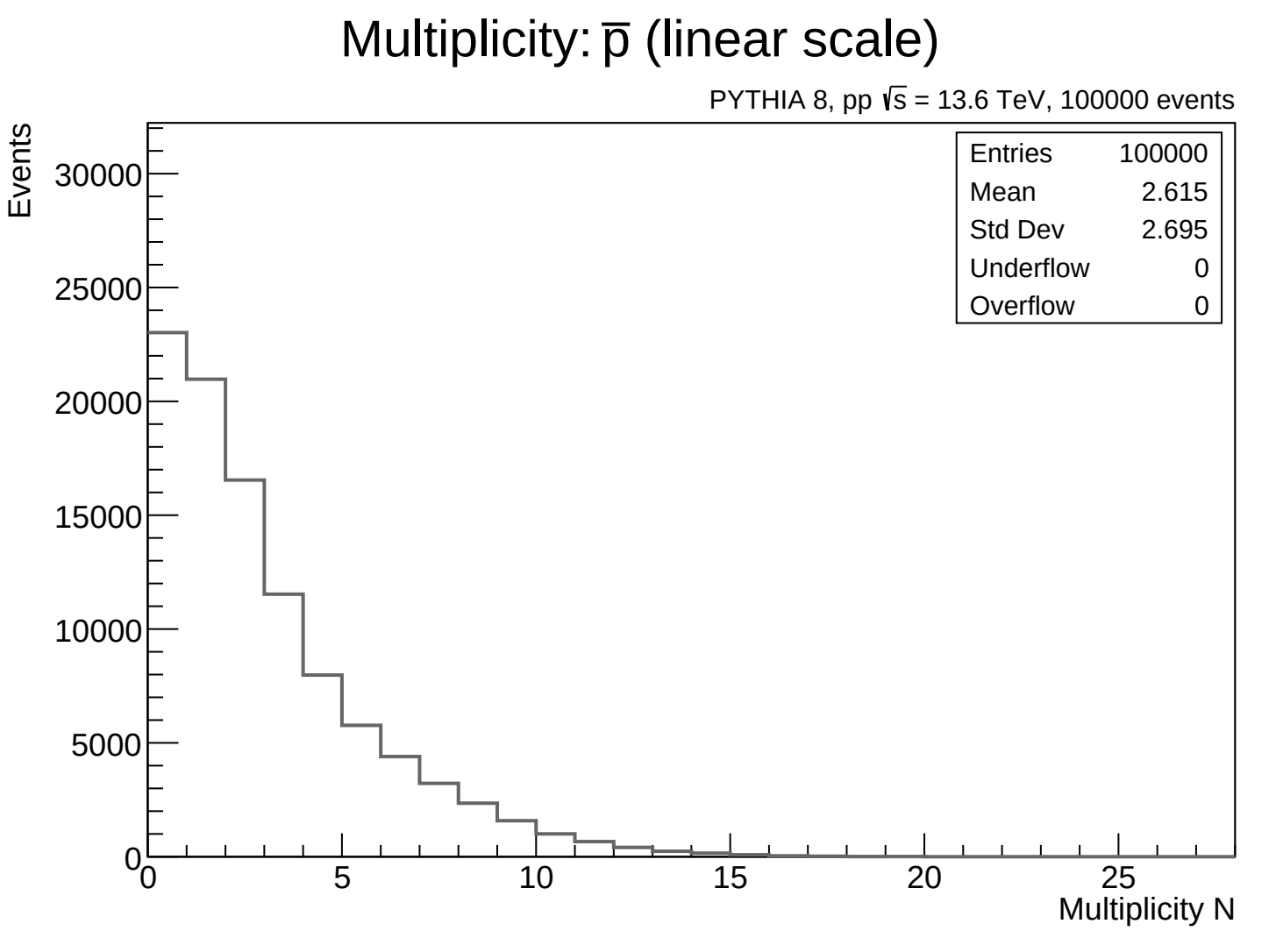

Multiplicity: K-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 4.062
Std Dev 4.062
Underflow 0
Overflow 0
Events
10^5
10^4
10^3
10^2
10
1
0
5
10
15
20
25
30
35
Multiplicity N


Multiplicity: K⁻ (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 4.062
Std Dev 4.062
Underflow 0
Overflow 0
Events
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
5
10
15
20
25
30
35
Multiplicity N

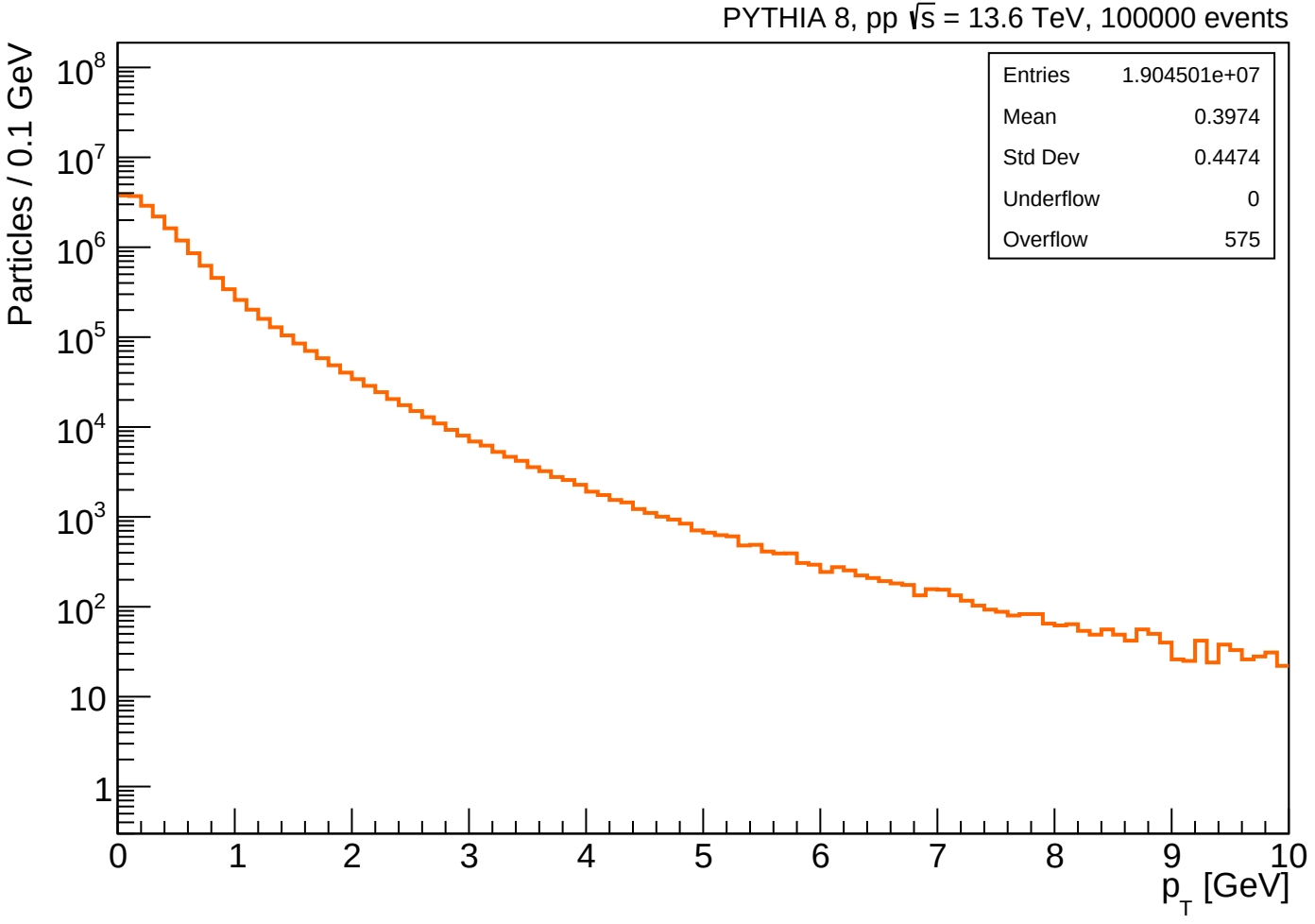

Multiplicity: p
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 3.831
Std Dev 2.72
Underflow 0
Overflow 0
Events
10^6
10^5
10^4
10^3
10^2
10
1
0
5
10
15
20
25
30
Multiplicity N

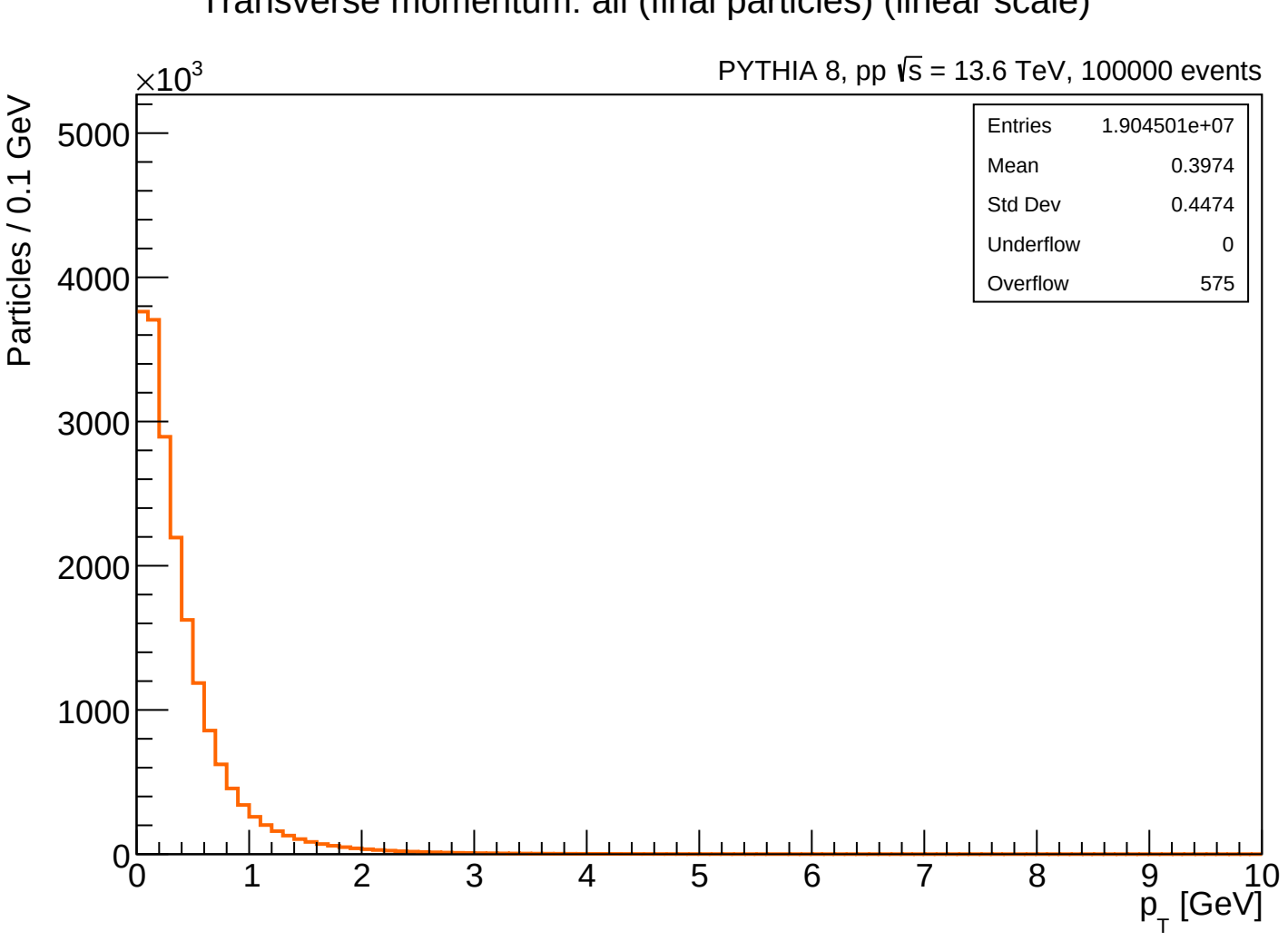

Multiplicity: p (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 3.831
Std Dev 2.72
Underflow 0
Overflow 0
Events
30000
25000
20000
15000
10000
5000
0
0
5
10
15
20
25
30
Multiplicity N

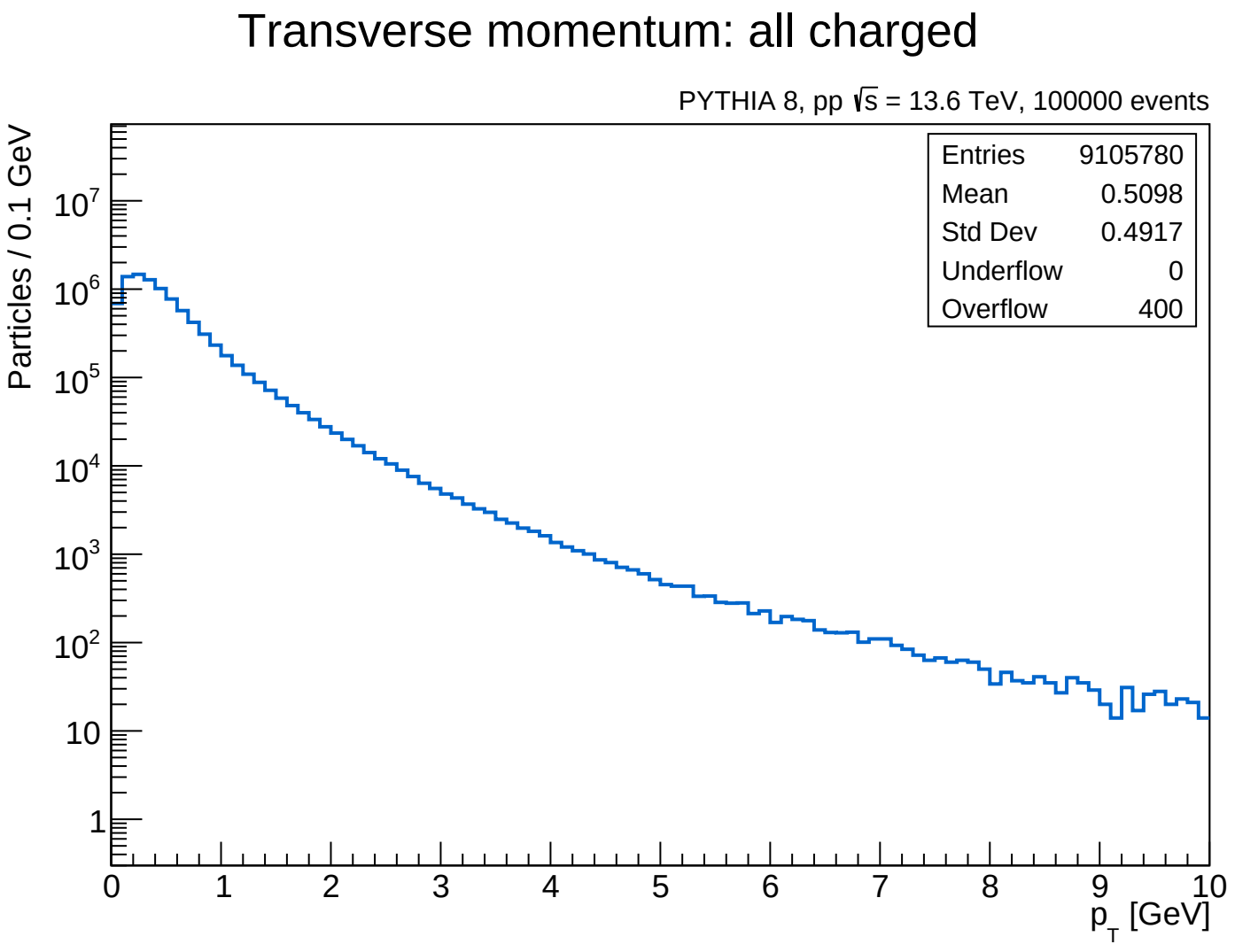

Multiplicity: $\overline{p}$
PYTHIA 8, pp $\sqrt{s}$ = 13.6 TeV, 100000 events
Entries 100000
Mean 2.615
Std Dev 2.695
Underflow 0
Overflow 0
Events
10^6
10^5
10^4
10^3
10^2
10
1
0
5
10
15
20
25
Multiplicity N

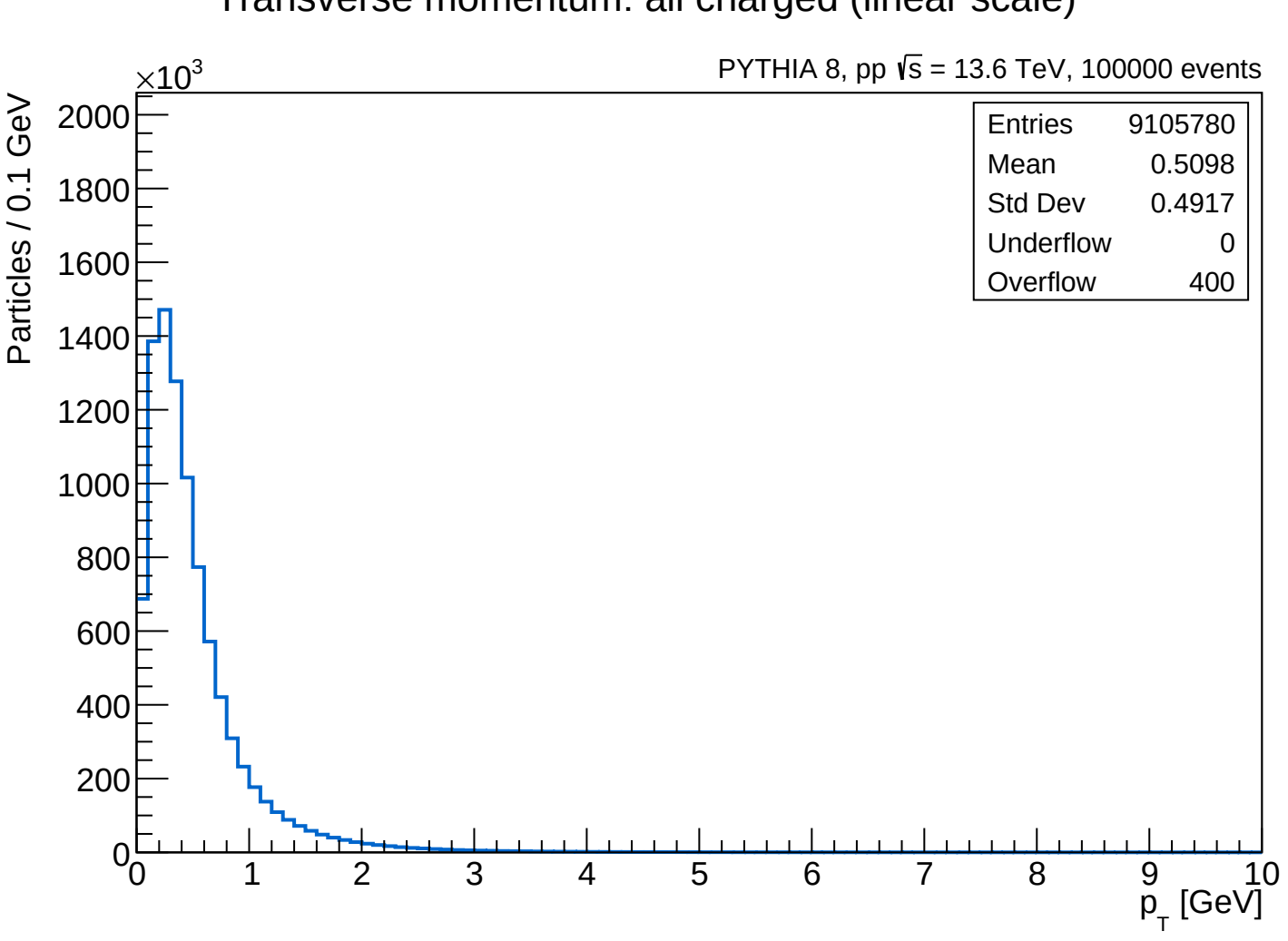

Multiplicity: p̄ (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 100000
Mean 2.615
Std Dev 2.695
Underflow 0
Overflow 0
Events
30000
25000
20000
15000
10000
5000
0
0
5
10
15
20
25
Multiplicity N

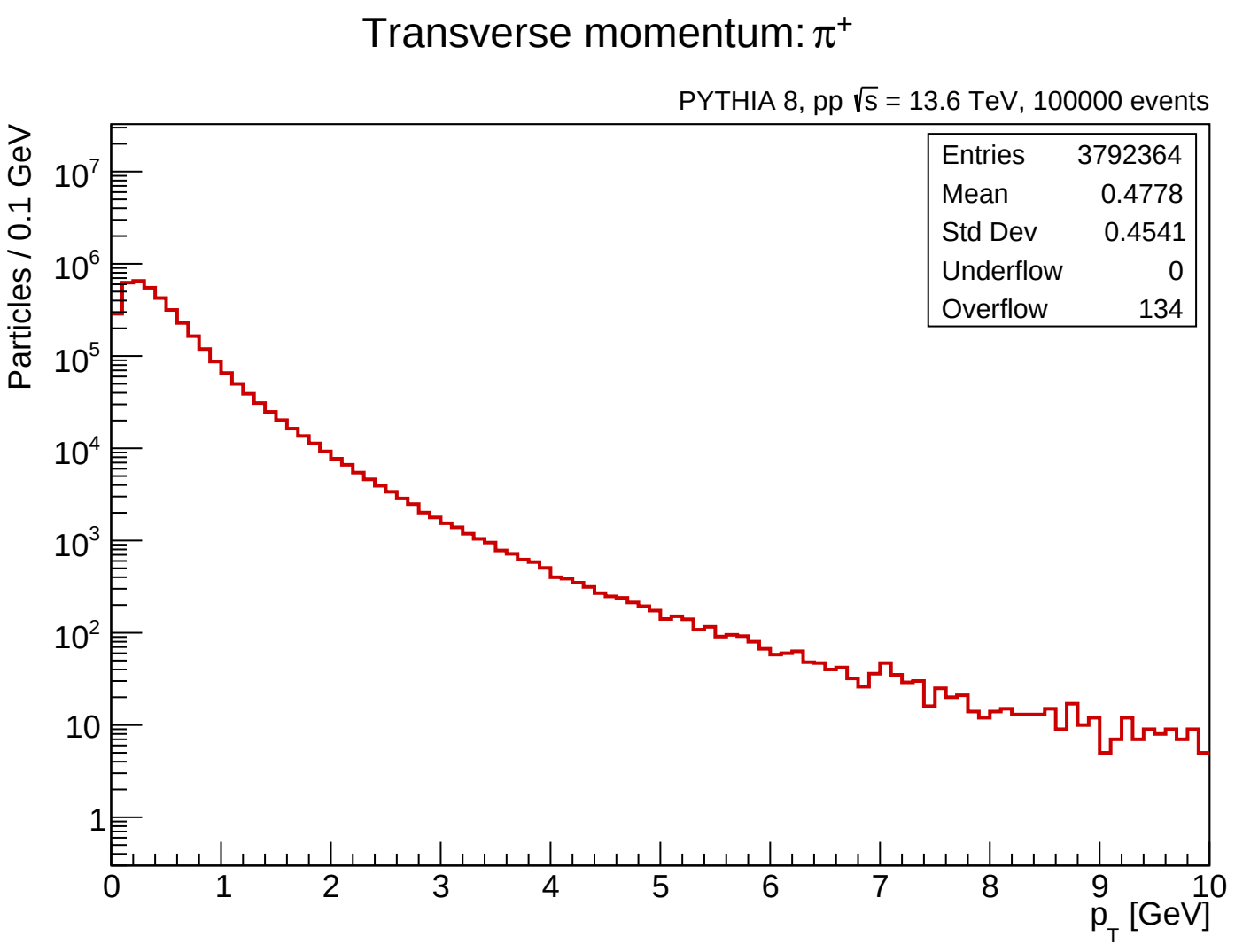
Transverse momentum: all (final particles)

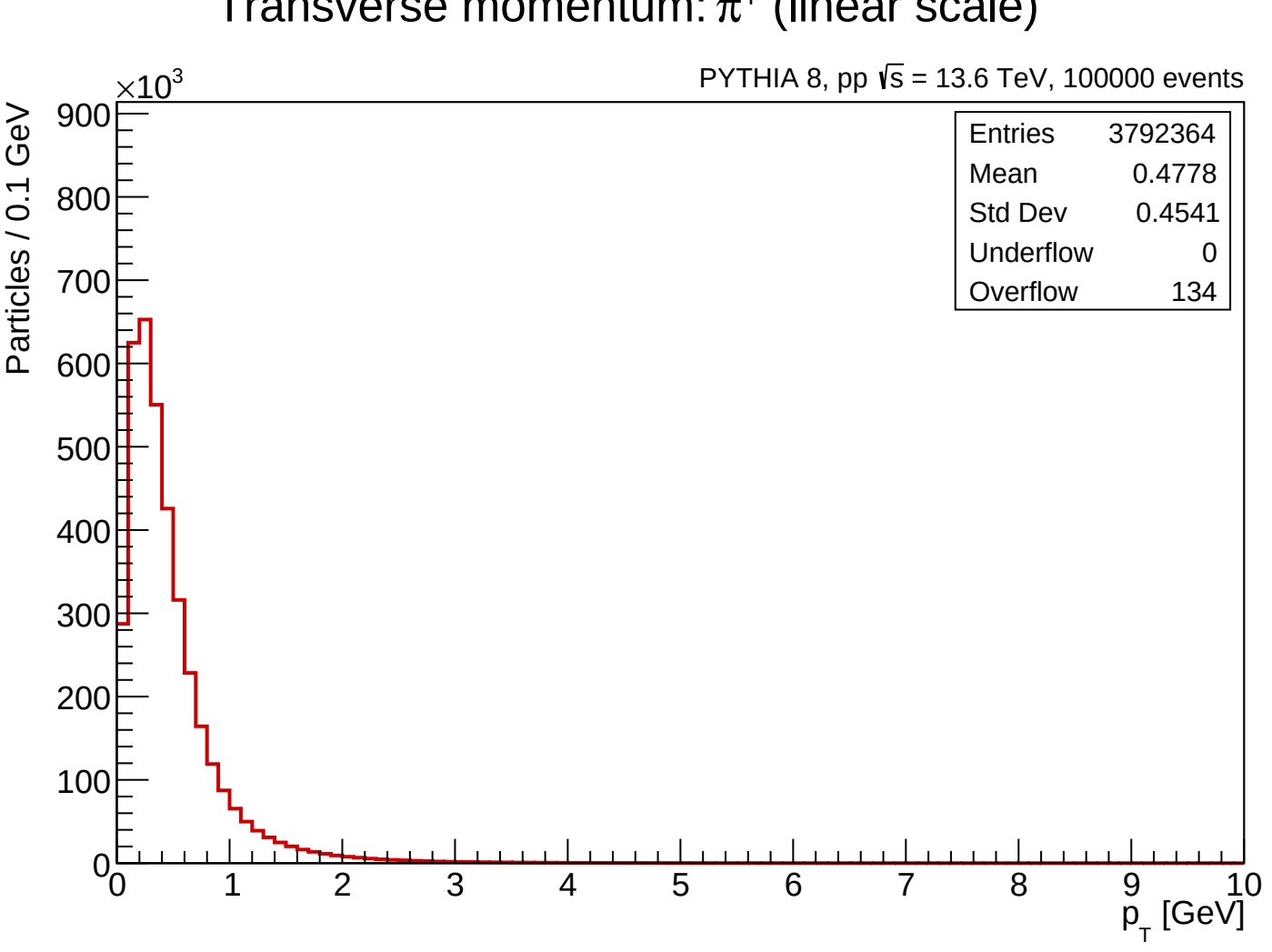


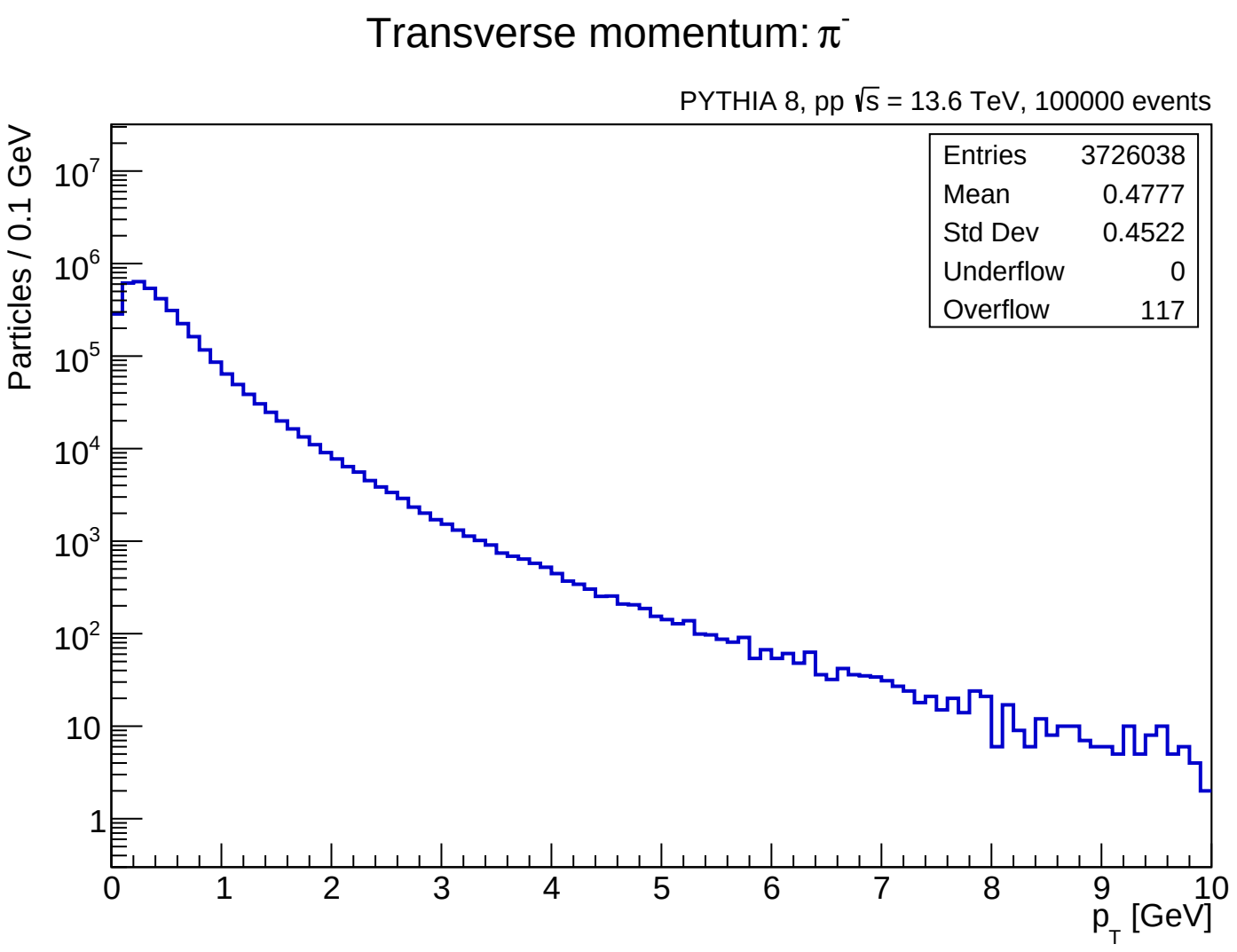

Transverse momentum: all (final particles) (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
×10^3
Particles / 0.1 GeV
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]
Entries 1.904501e+07
Mean 0.3974
Std Dev 0.4474
Underflow 0
Overflow 575

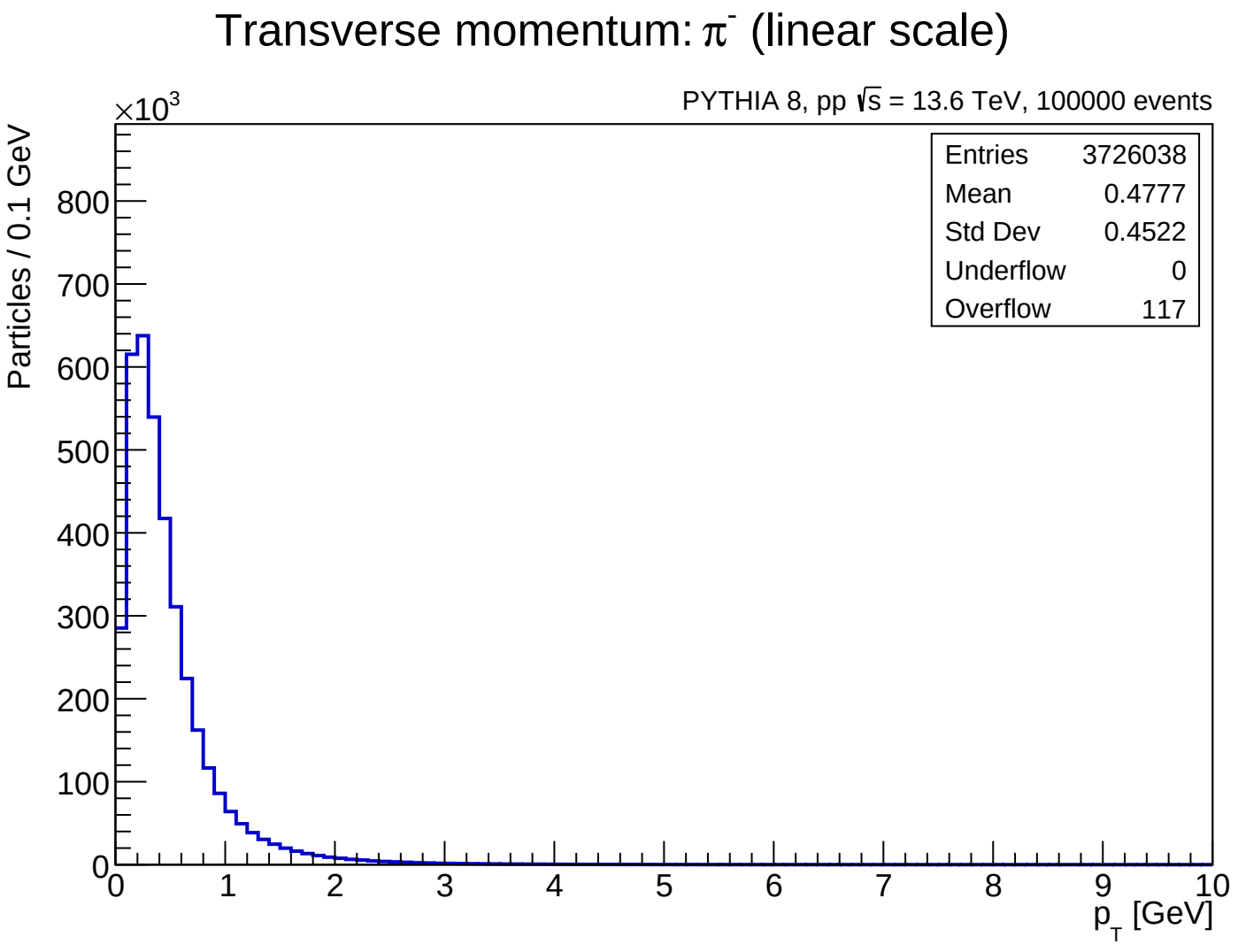

Transverse momentum: all charged
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 9105780
Mean 0.5098
Std Dev 0.4917
Underflow 0
Overflow 400
Particles / 0.1 GeV
10^7
10^6
10^5
10^4
10^3
10^2
10
1
0 1 2 3 4 5 6 7 8 9 10
p_T [GeV]

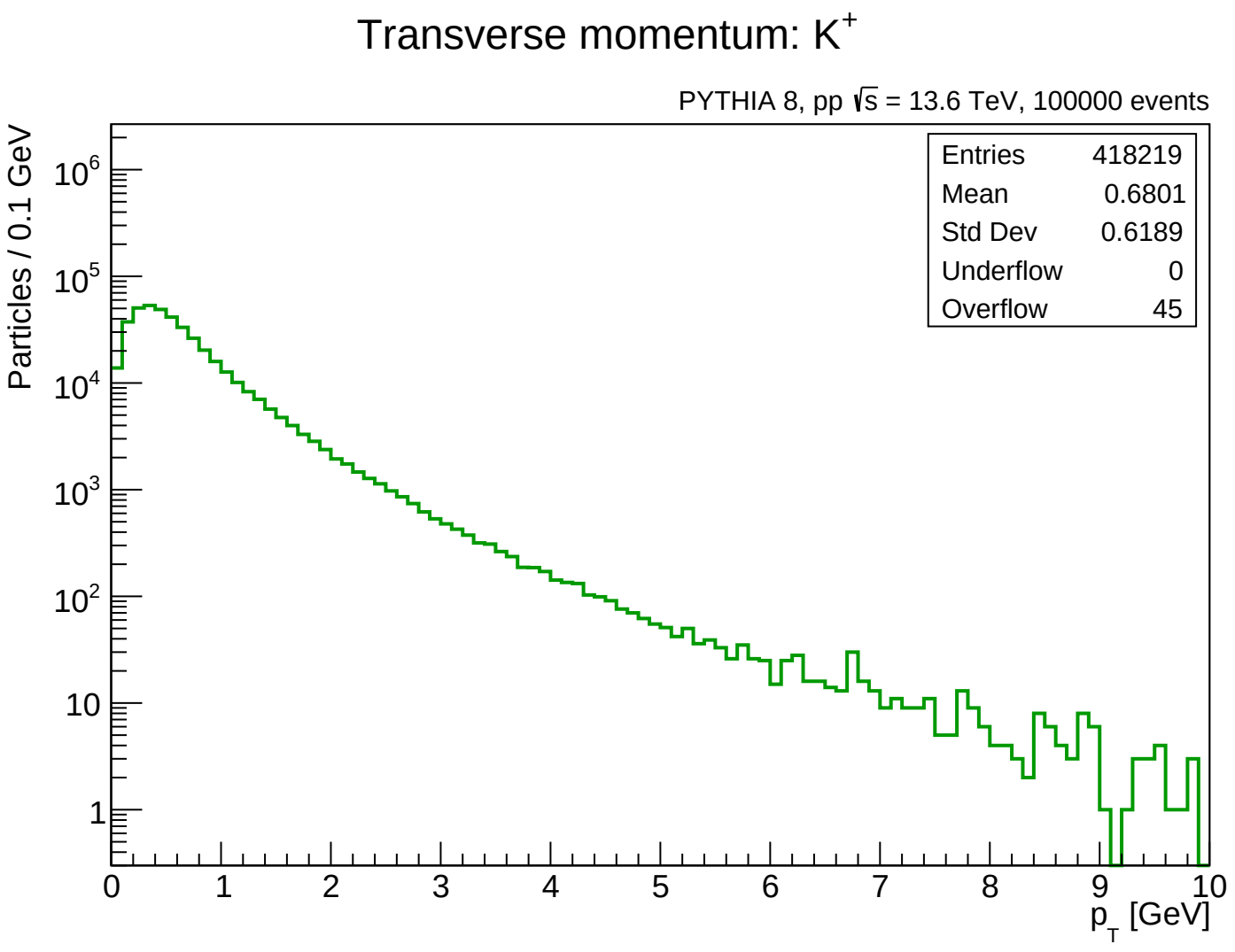

Transverse momentum: all charged (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
×10^3
Particles / 0.1 GeV
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]
Entries 9105780
Mean 0.5098
Std Dev 0.4917
Underflow 0
Overflow 400

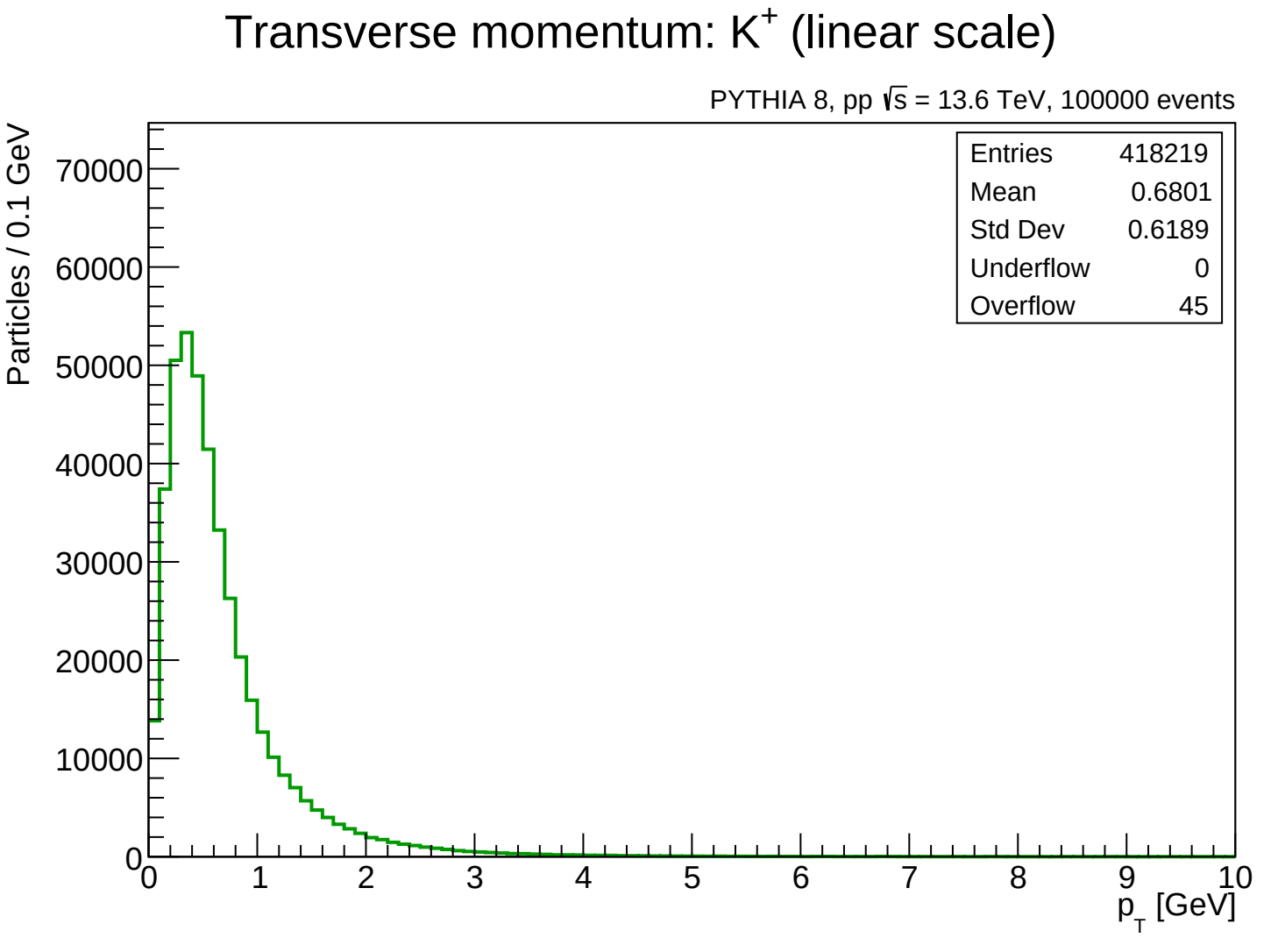

Transverse momentum: π^+
PYTHIA 8, pp $\sqrt{s}$ = 13.6 TeV, 100000 events
Entries 3792364
Mean 0.4778
Std Dev 0.4541
Underflow 0
Overflow 134
Particles / 0.1 GeV
10^7
10^6
10^5
10^4
10^3
10^2
10
1
0 1 2 3 4 5 6 7 8 9 10
p_T [GeV]

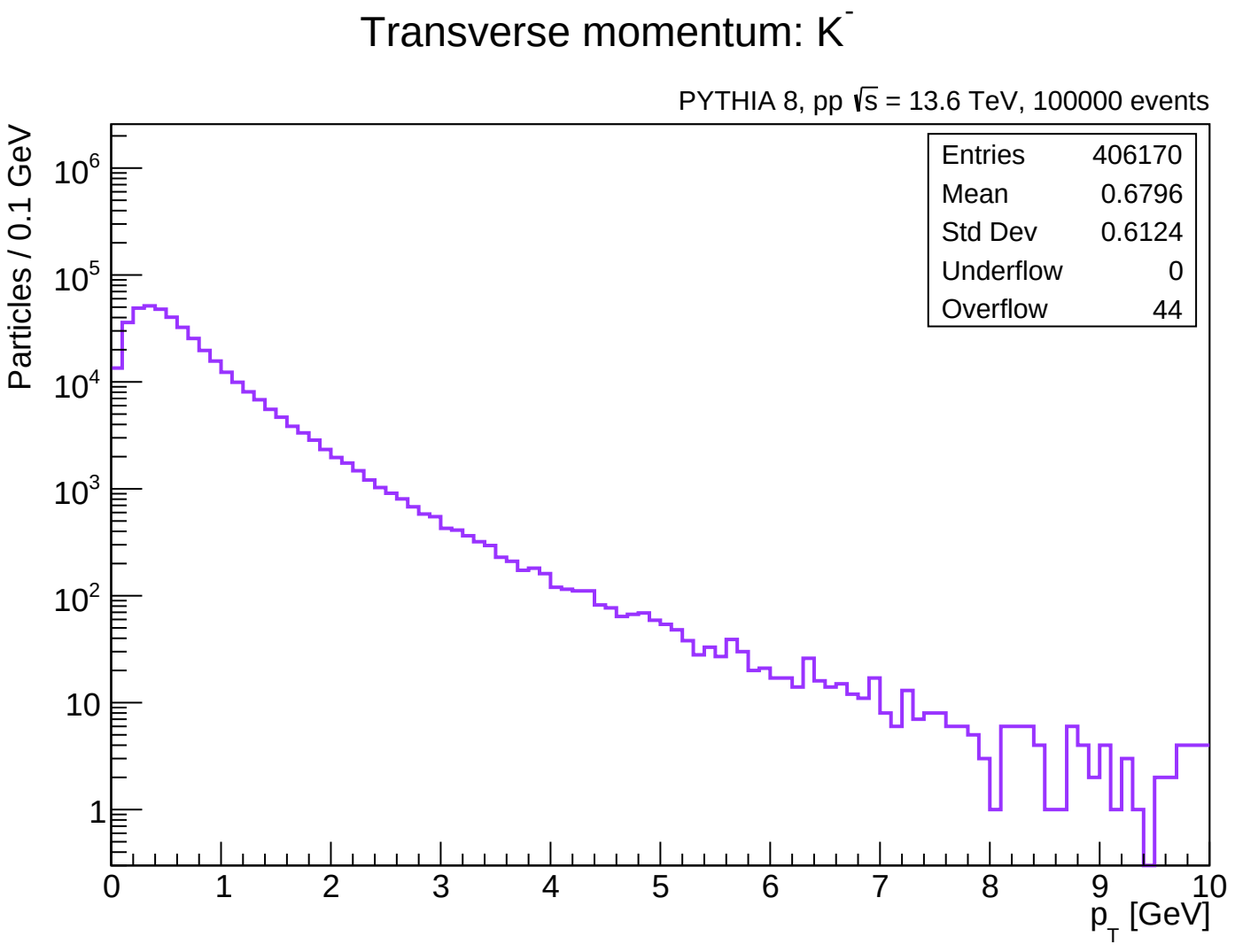

Transverse momentum: π+ (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
×10^3
Particles / 0.1 GeV
Entries 3792364
Mean 0.4778
Std Dev 0.4541
Underflow 0
Overflow 134
p_T [GeV]

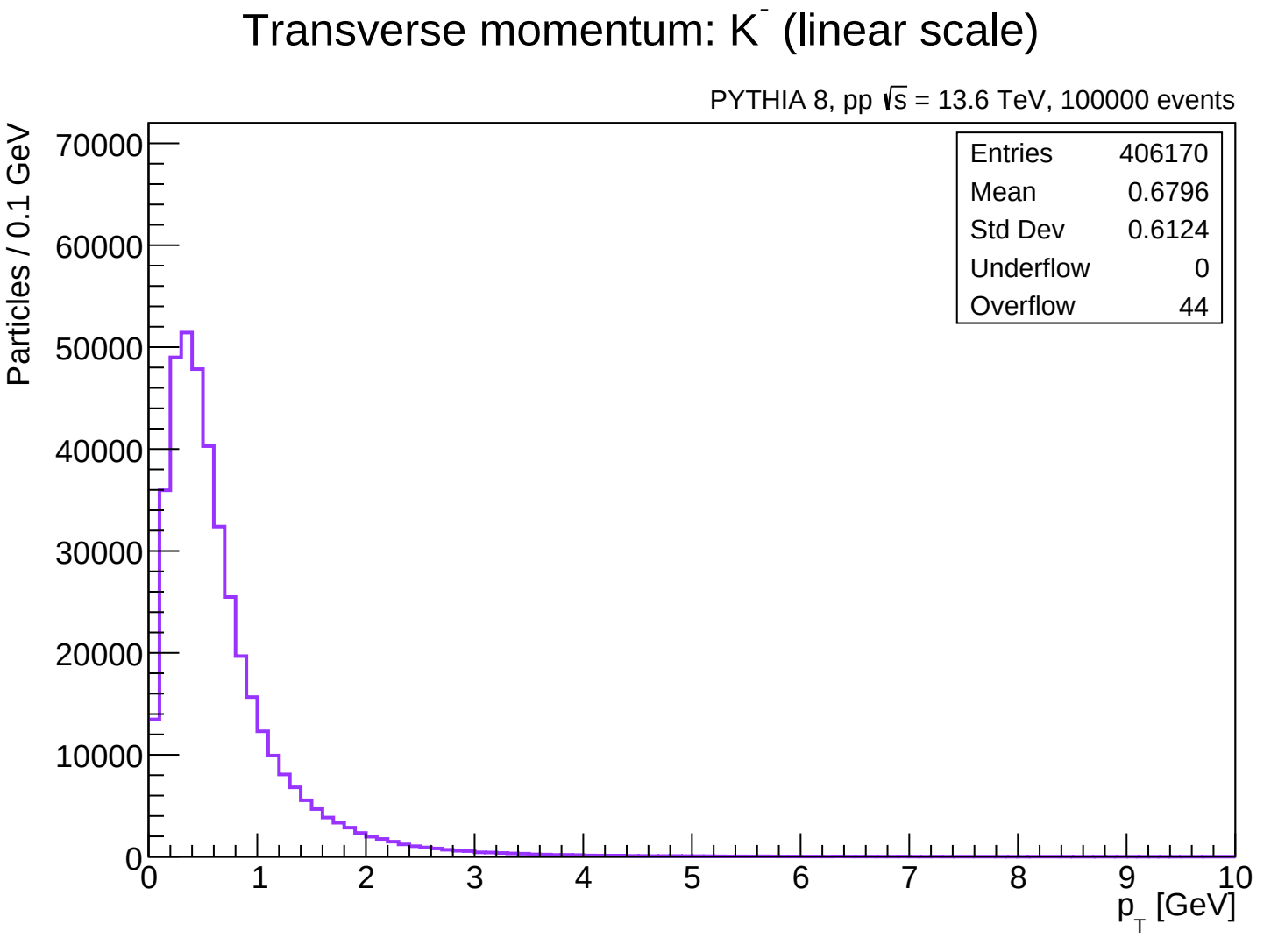

Transverse momentum: π-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 3726038
Mean 0.4777
Std Dev 0.4522
Underflow 0
Overflow 117
Particles / 0.1 GeV
10^7
10^6
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]


Transverse momentum: π⁻ (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
×10³
Particles / 0.1 GeV
Entries 3726038
Mean 0.4777
Std Dev 0.4522
Underflow 0
Overflow 117
0
100
200
300
400
500
600
700
800
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]

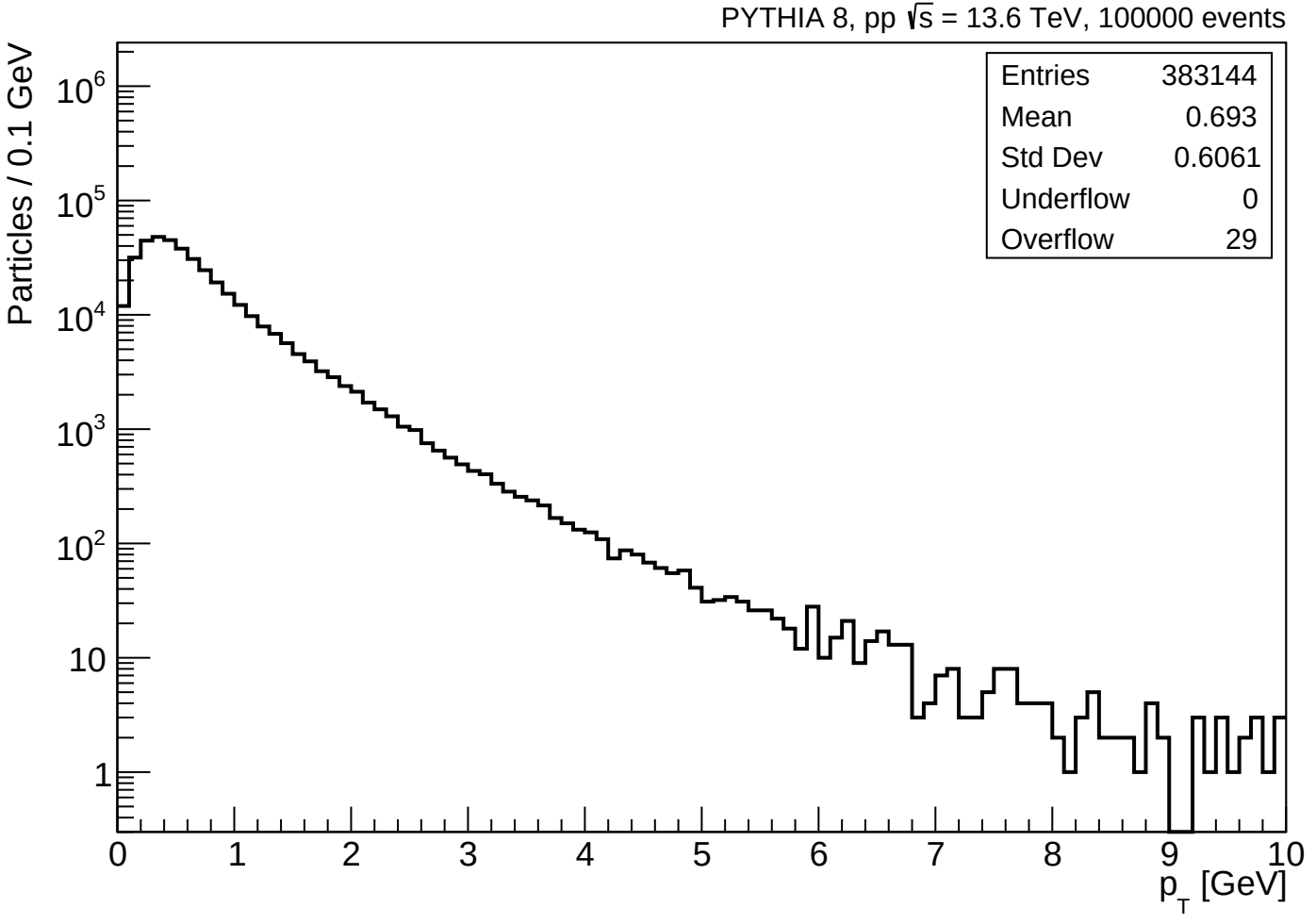

Transverse momentum: K+
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 418219
Mean 0.6801
Std Dev 0.6189
Underflow 0
Overflow 45
Particles / 0.1 GeV
10^6
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]


Transverse momentum: K+ (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 418219
Mean 0.6801
Std Dev 0.6189
Underflow 0
Overflow 45
Particles / 0.1 GeV
70000
60000
50000
40000
30000
20000
10000
0
0
1
2
3
4
5
6
7
8
9
10
pT [GeV]

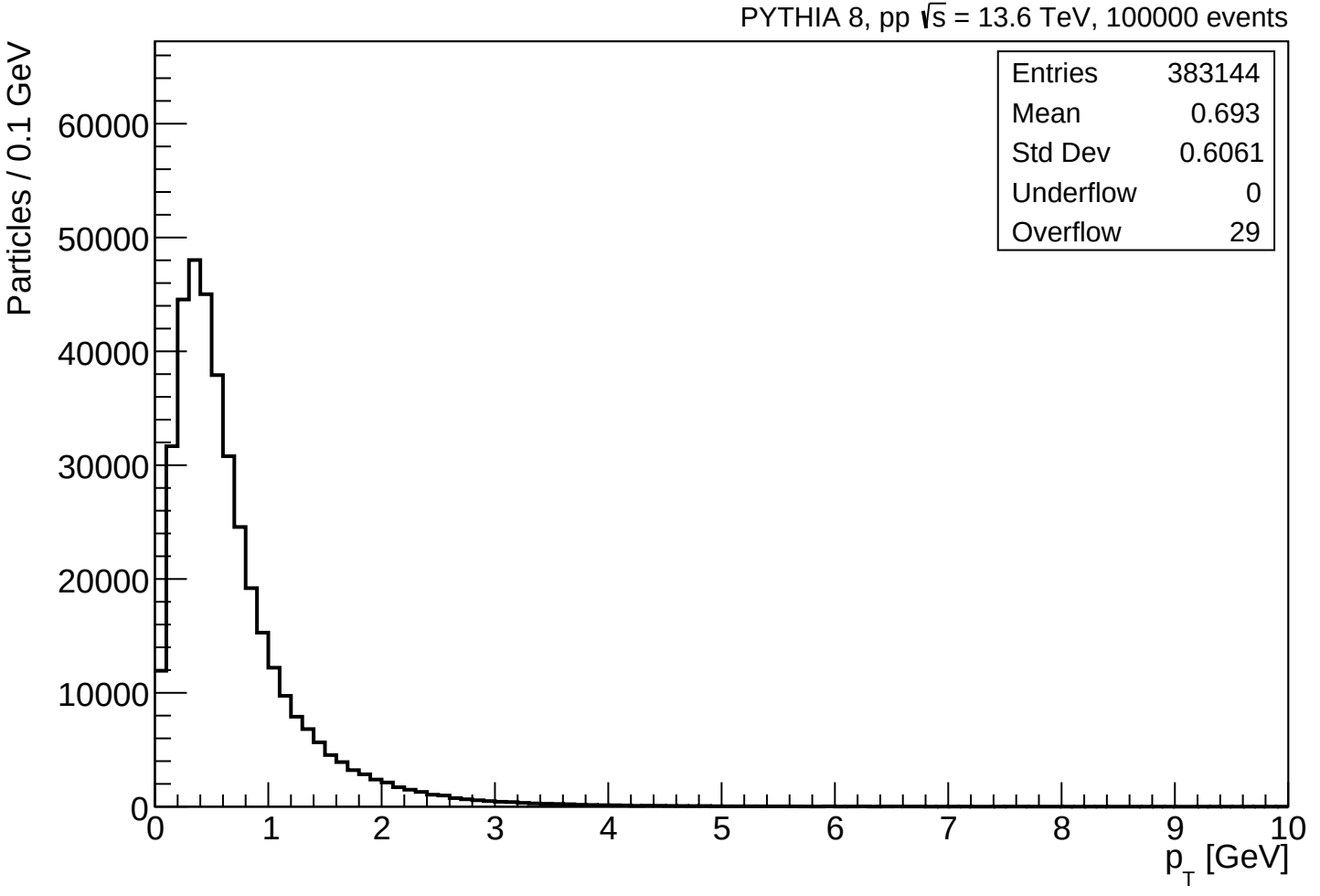

Transverse momentum: K-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 406170
Mean 0.6796
Std Dev 0.6124
Underflow 0
Overflow 44
Particles / 0.1 GeV
10^6
10^5
10^4
10^3
10^2
10
1
0 1 2 3 4 5 6 7 8 9 10
p_T [GeV]

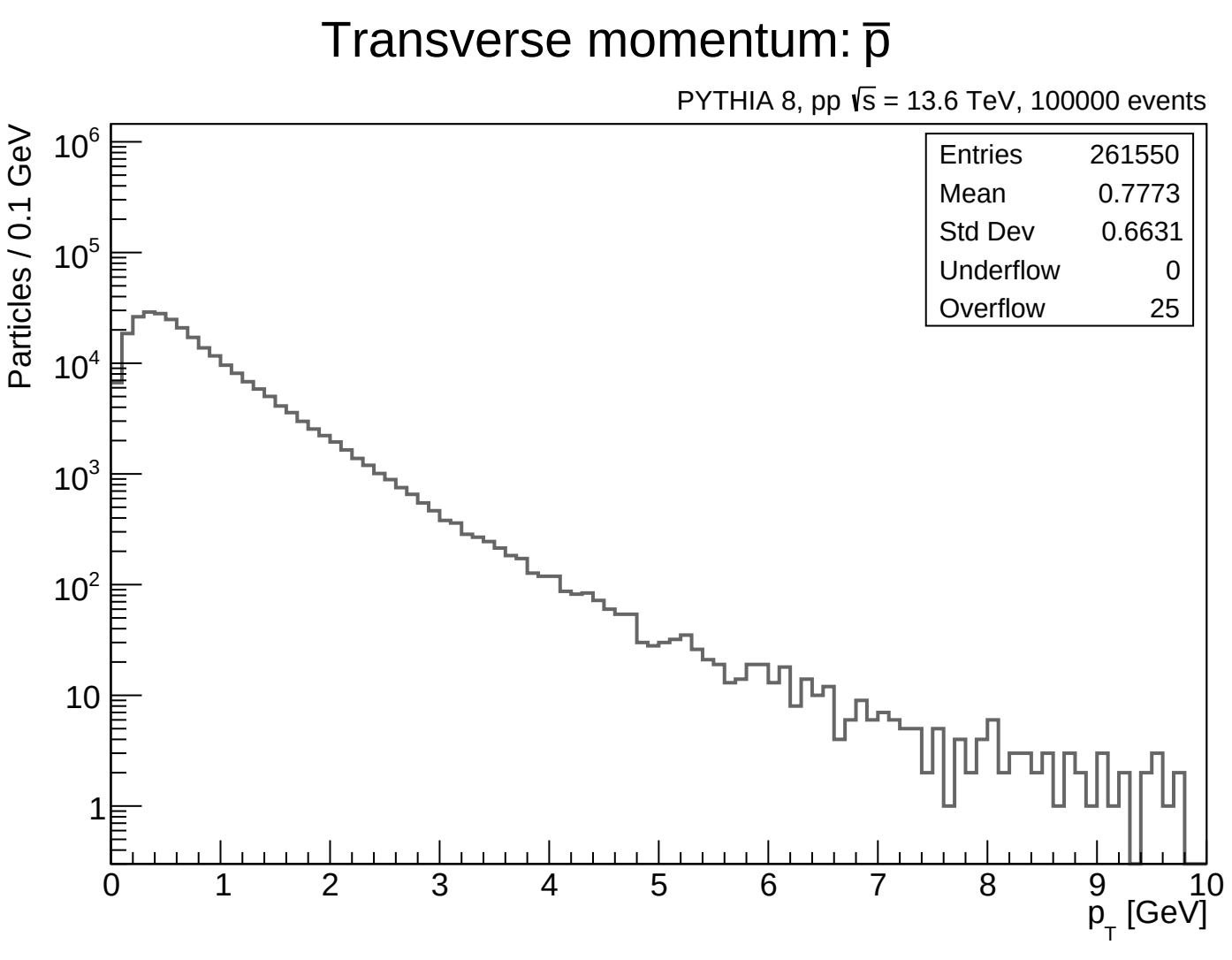

Transverse momentum: K- (linear scale)
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 406170
Mean 0.6796
Std Dev 0.6124
Underflow 0
Overflow 44
Particles / 0.1 GeV
70000
60000
50000
40000
30000
20000
10000
0
0 1 2 3 4 5 6 7 8 9 10
pT [GeV]

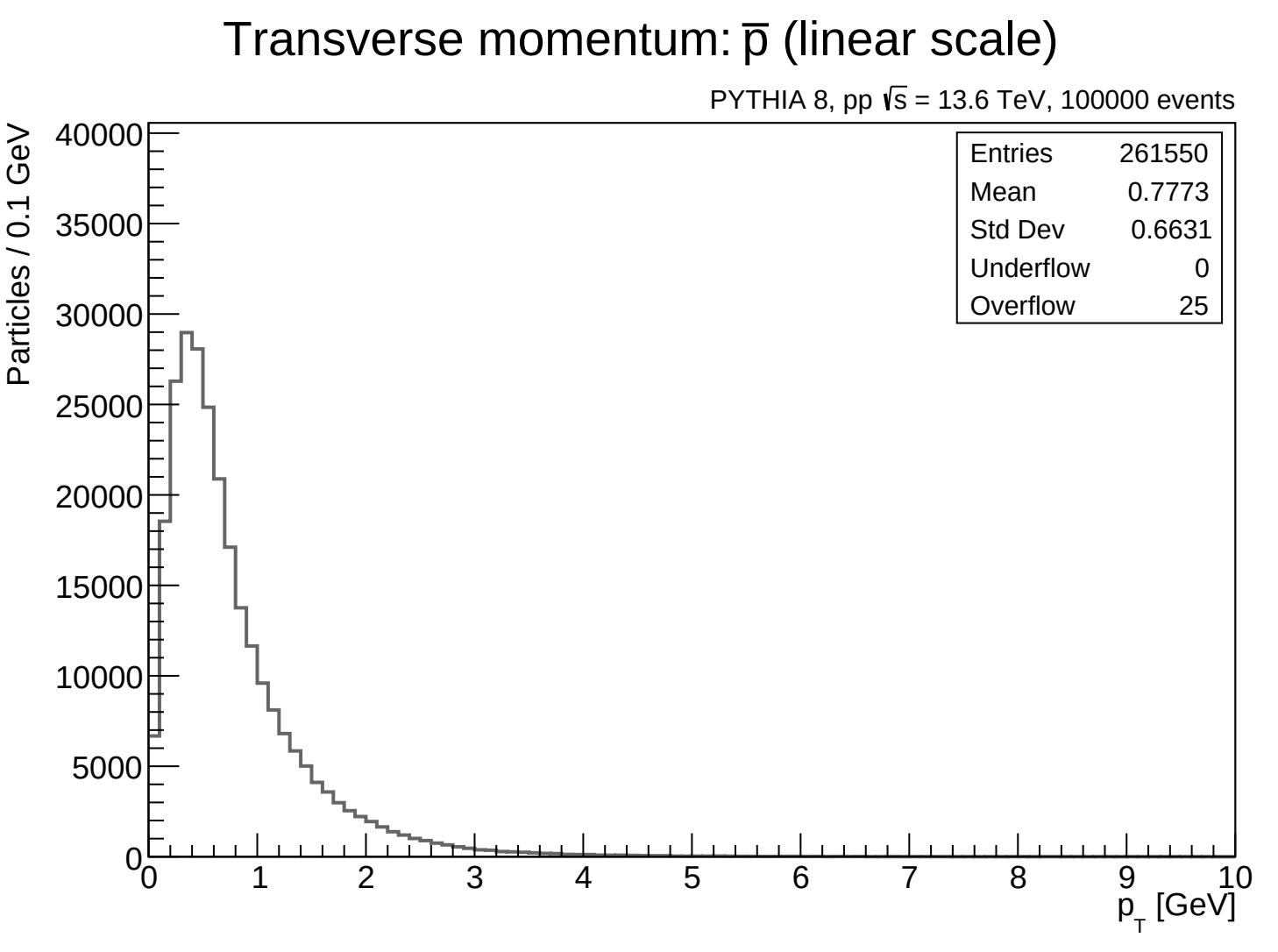
Transverse momentum: p



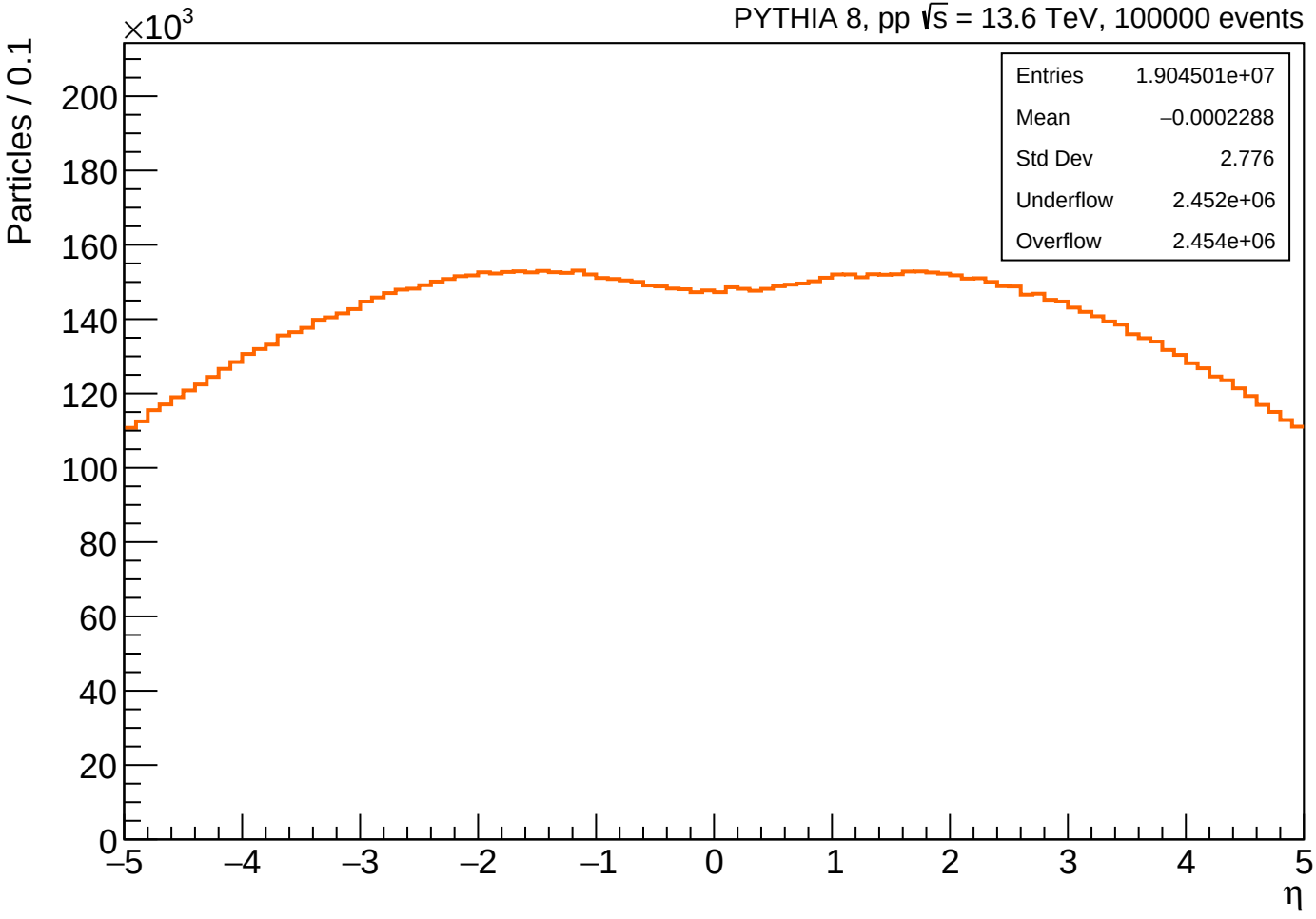
Transverse momentum: p (linear scale)



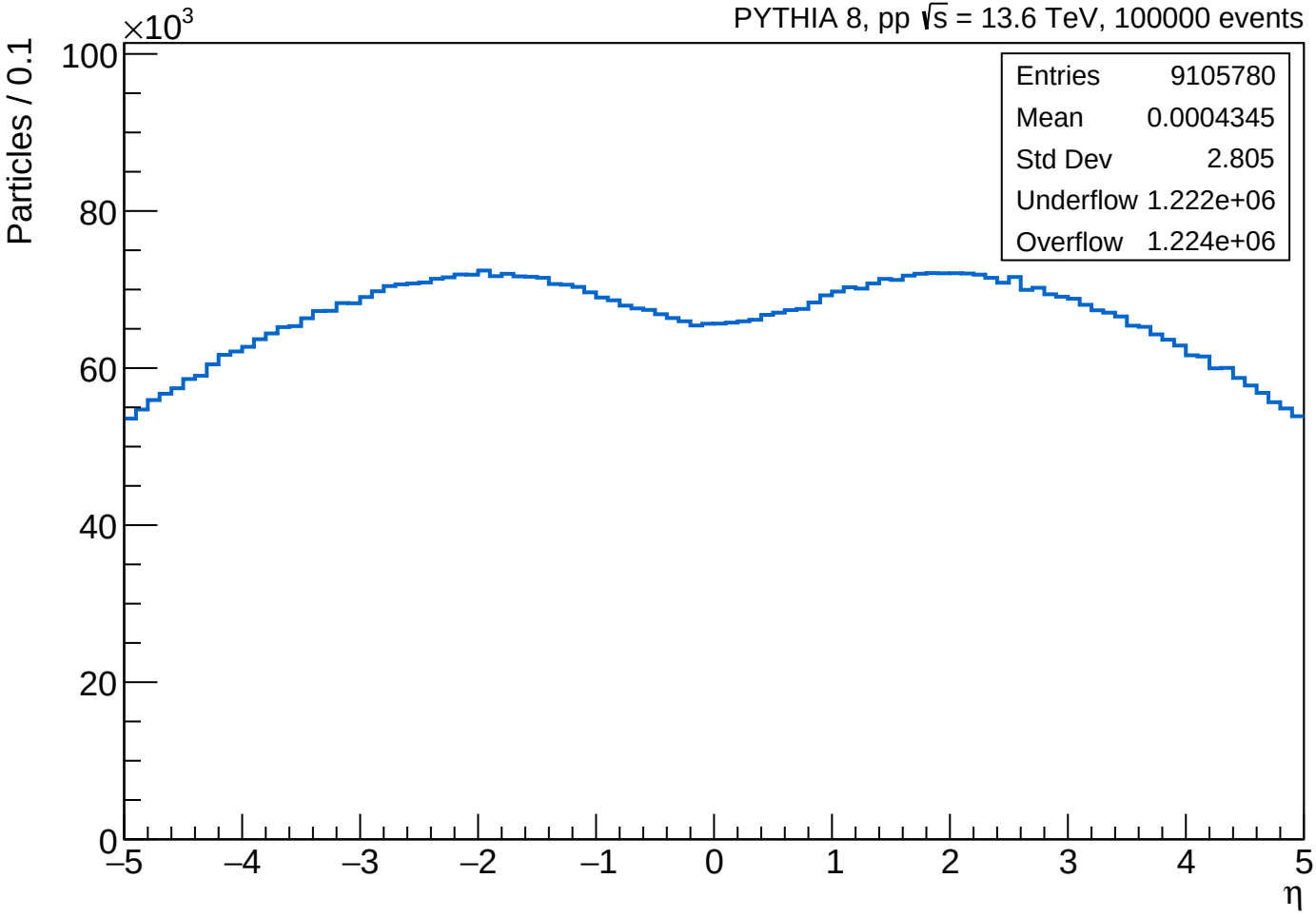

Transverse momentum: $\bar{p}$
PYTHIA 8, pp $\sqrt{s}$ = 13.6 TeV, 100000 events
Entries 261550
Mean 0.7773
Std Dev 0.6631
Underflow 0
Overflow 25
Particles / 0.1 GeV
10^6
10^5
10^4
10^3
10^2
10
1
0 1 2 3 4 5 6 7 8 9 10
p_T [GeV]

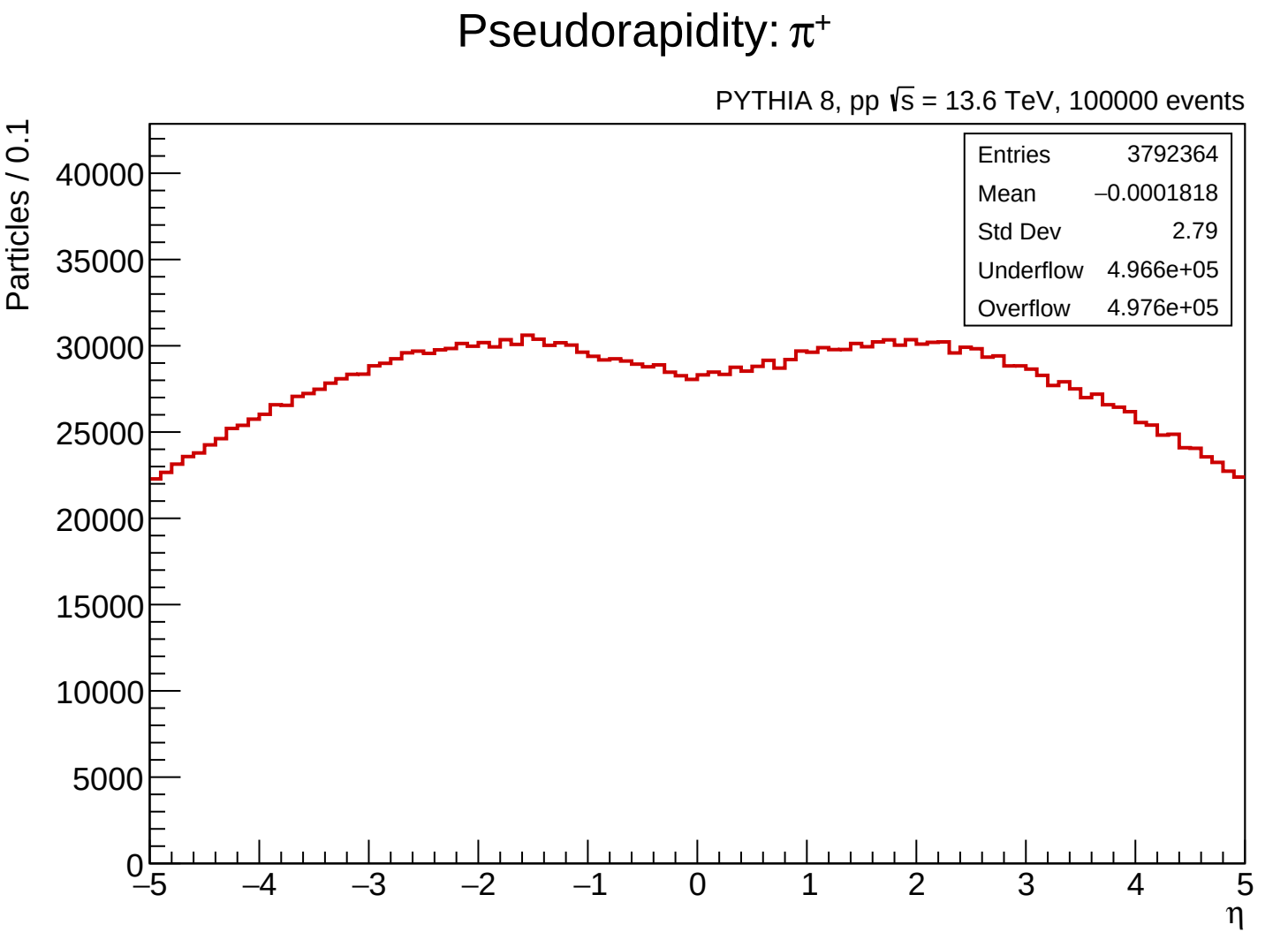

Transverse momentum: $\bar{p}$ (linear scale)
PYTHIA 8, pp $\sqrt{s}$ = 13.6 TeV, 100000 events
Entries 261550
Mean 0.7773
Std Dev 0.6631
Underflow 0
Overflow 25
Particles / 0.1 GeV
40000
35000
30000
25000
20000
15000
10000
5000
0
0 1 2 3 4 5 6 7 8 9 10
p_T [GeV]

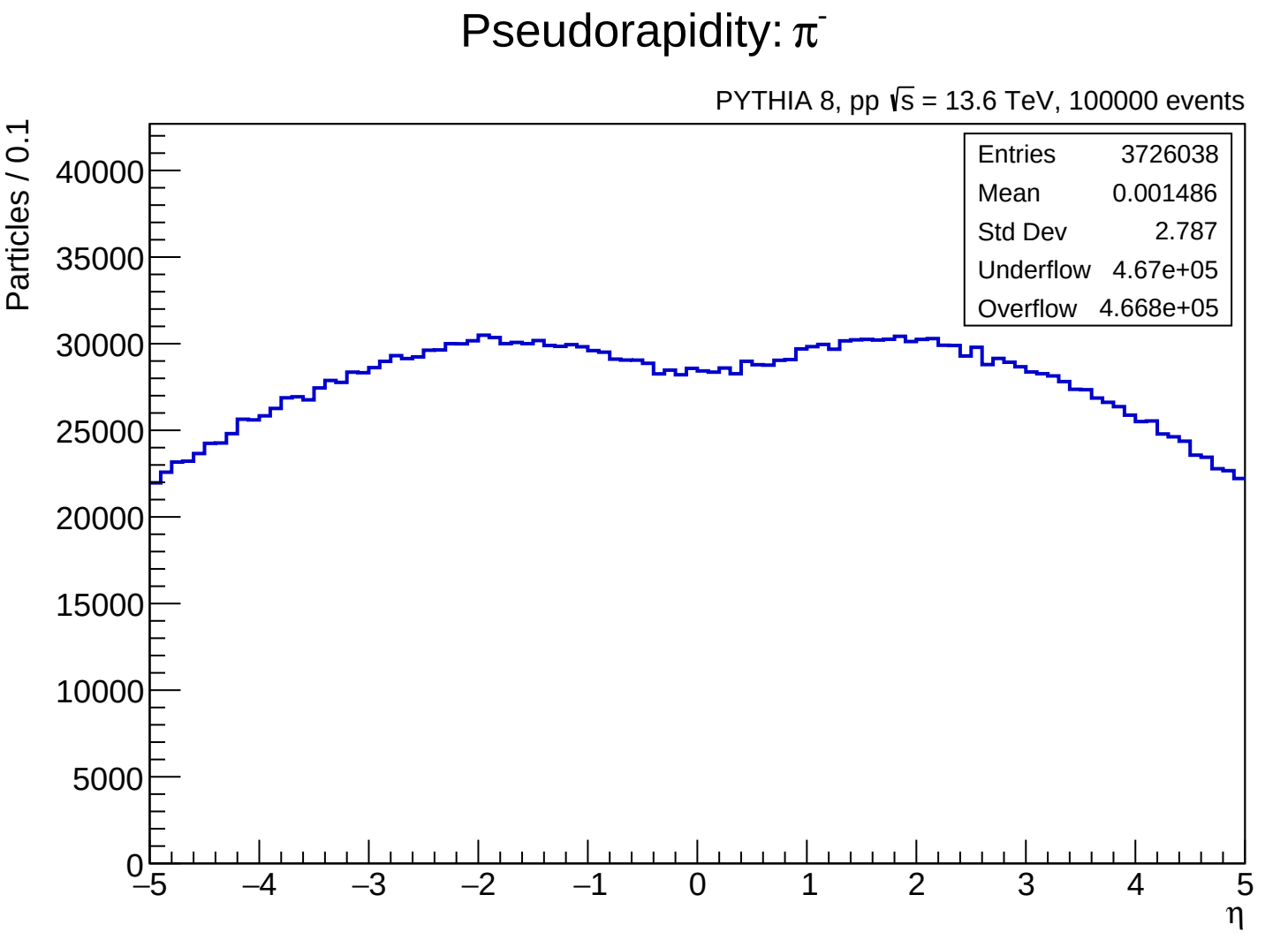
Pseudorapidity: all (final particles)

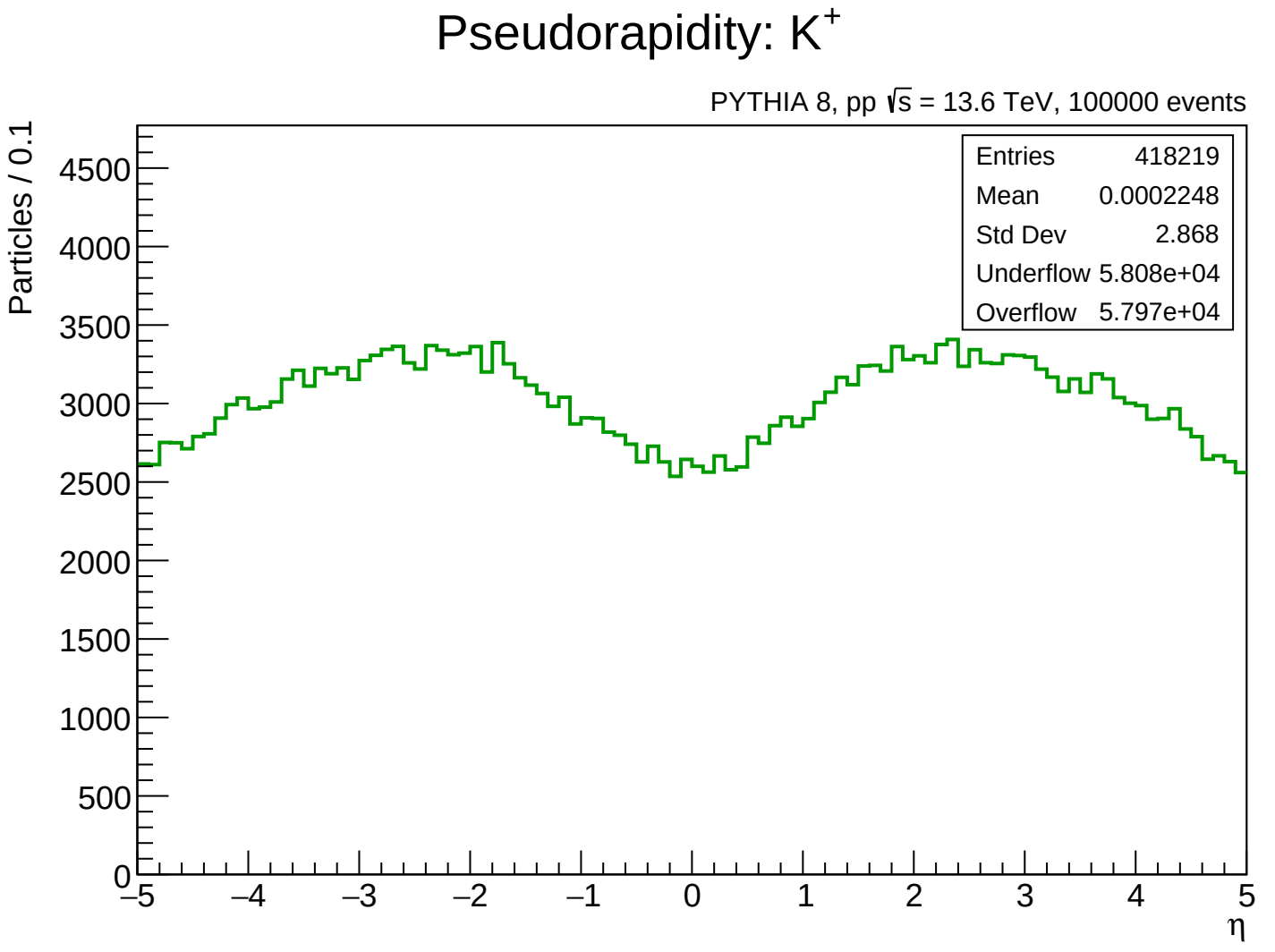


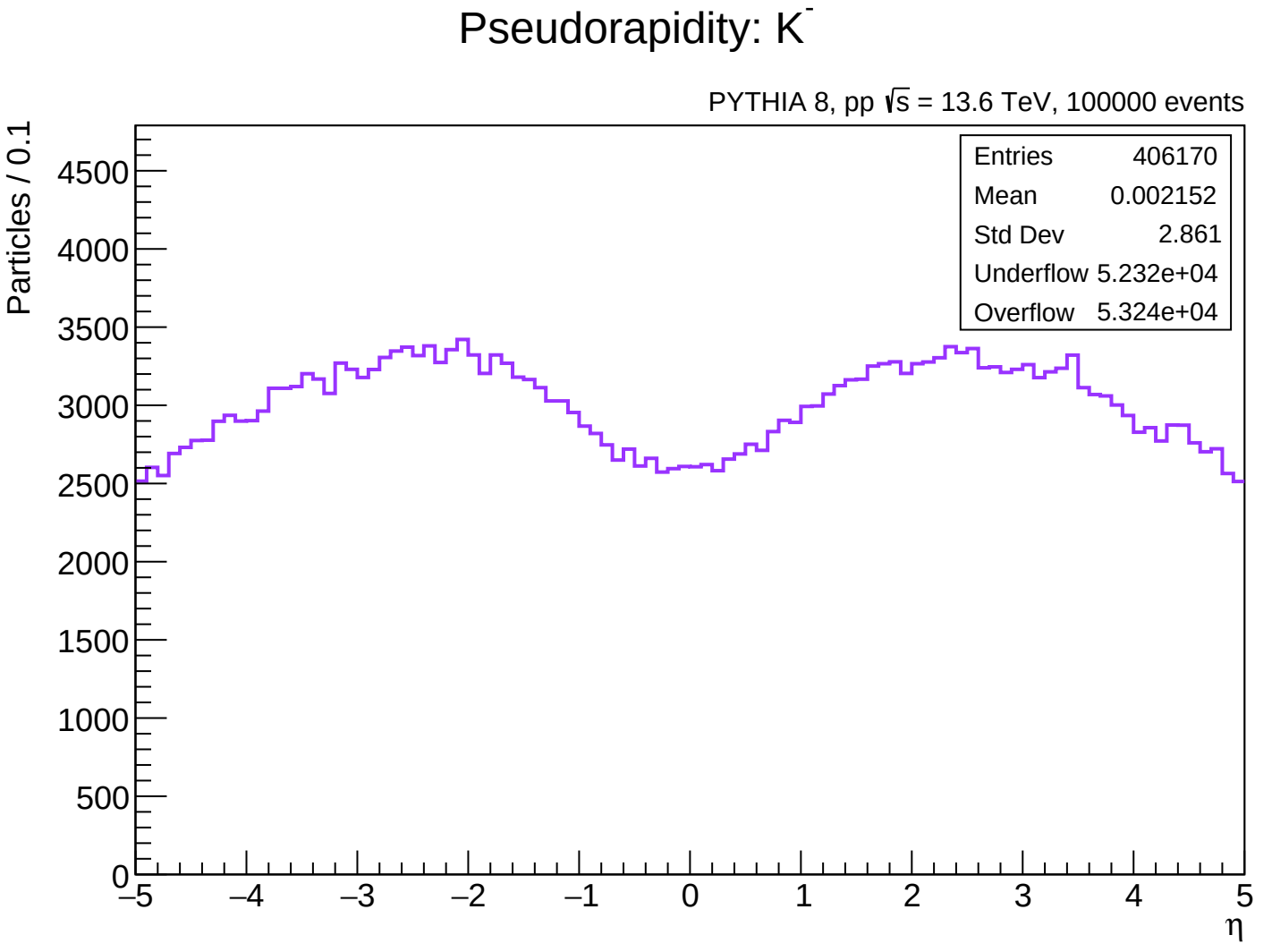
Pseudorapidity: all charged

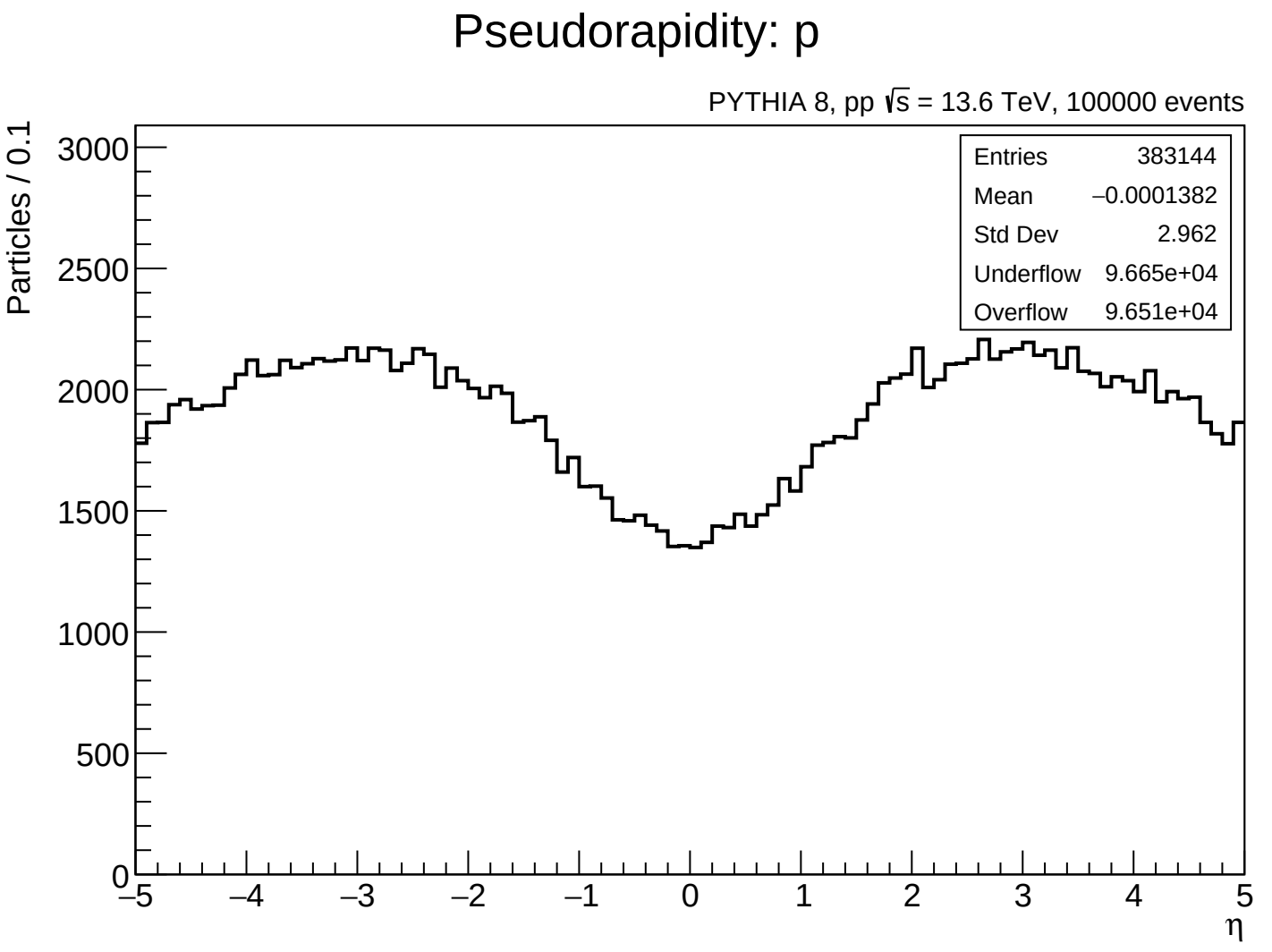


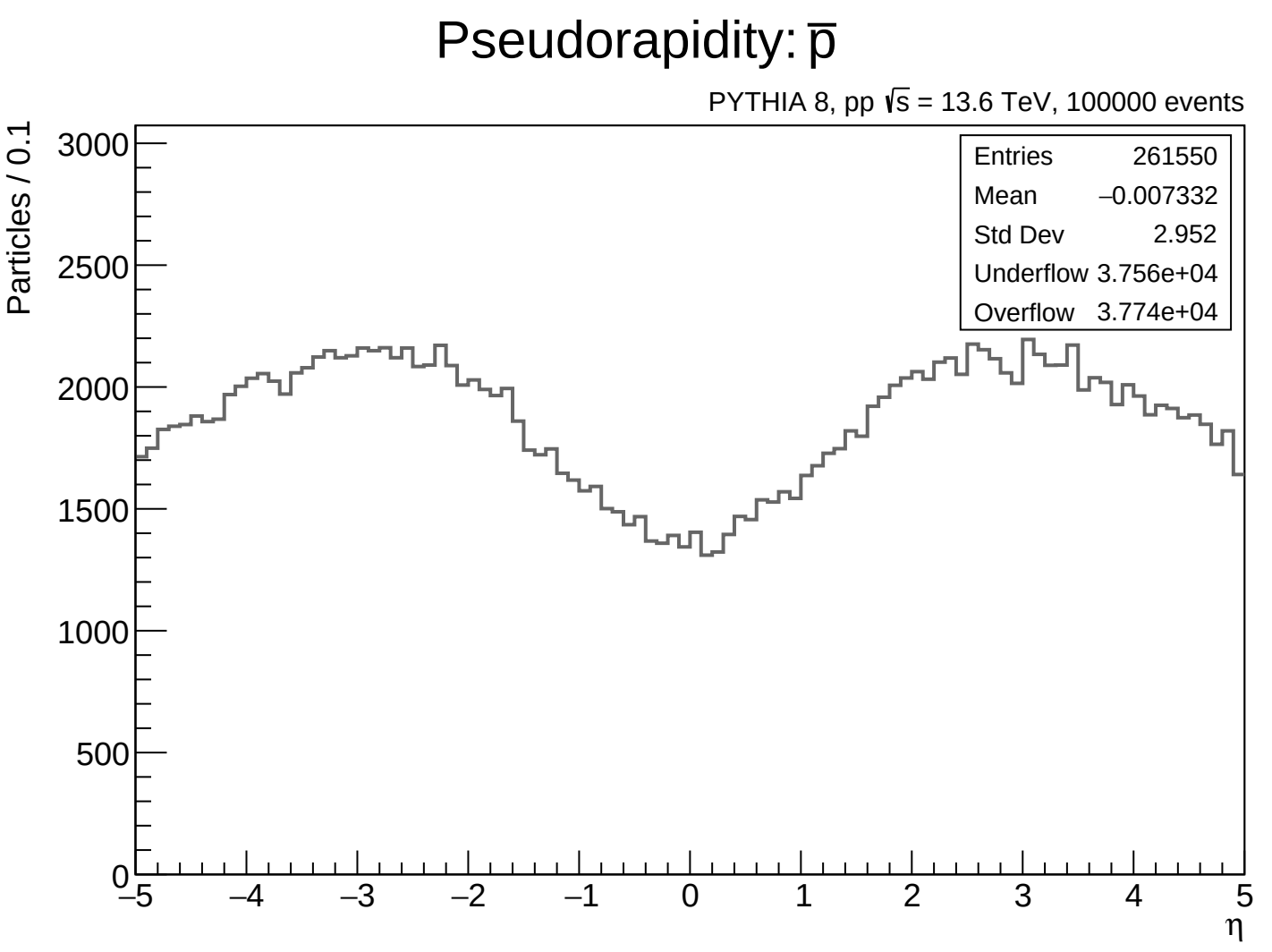

Pseudorapidity: π+
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 3792364
Mean −0.0001818
Std Dev 2.79
Underflow 4.966e+05
Overflow 4.976e+05
Particles / 0.1
40000
35000
30000
25000
20000
15000
10000
5000
0
−5
−4
−3
−2
−1
0
1
2
3
4
5
η


Pseudorapidity: π⁻
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 3726038
Mean 0.001486
Std Dev 2.787
Underflow 4.67e+05
Overflow 4.668e+05
Particles / 0.1
40000
35000
30000
25000
20000
15000
10000
5000
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

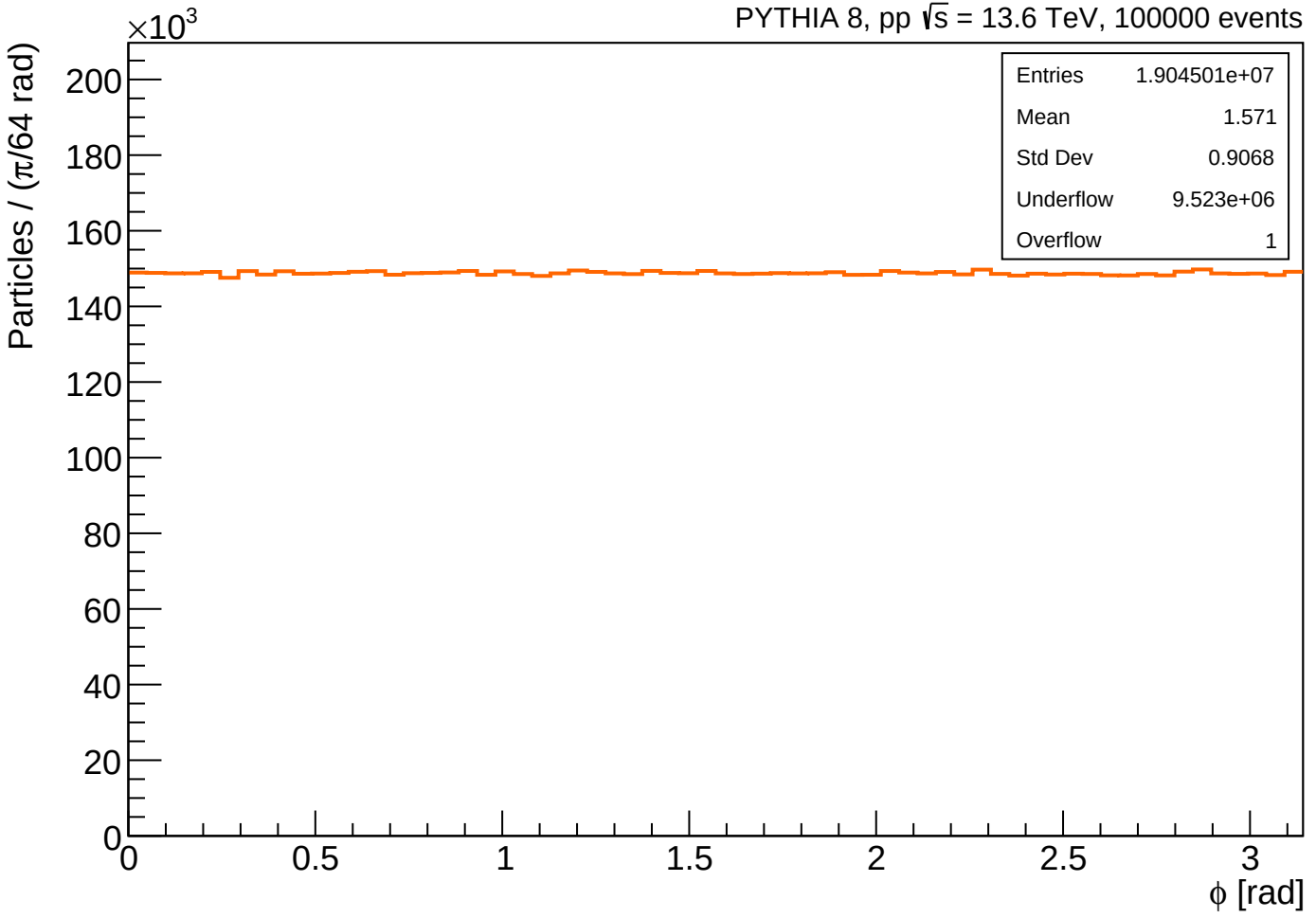

Pseudorapidity: K+
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 418219
Mean 0.0002248
Std Dev 2.868
Underflow 5.808e+04
Overflow 5.797e+04
Particles / 0.1
4500
4000
3500
3000
2500
2000
1500
1000
500
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η


Pseudorapidity: K-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 406170
Mean 0.002152
Std Dev 2.861
Underflow 5.232e+04
Overflow 5.324e+04
Particles / 0.1
4500
4000
3500
3000
2500
2000
1500
1000
500
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

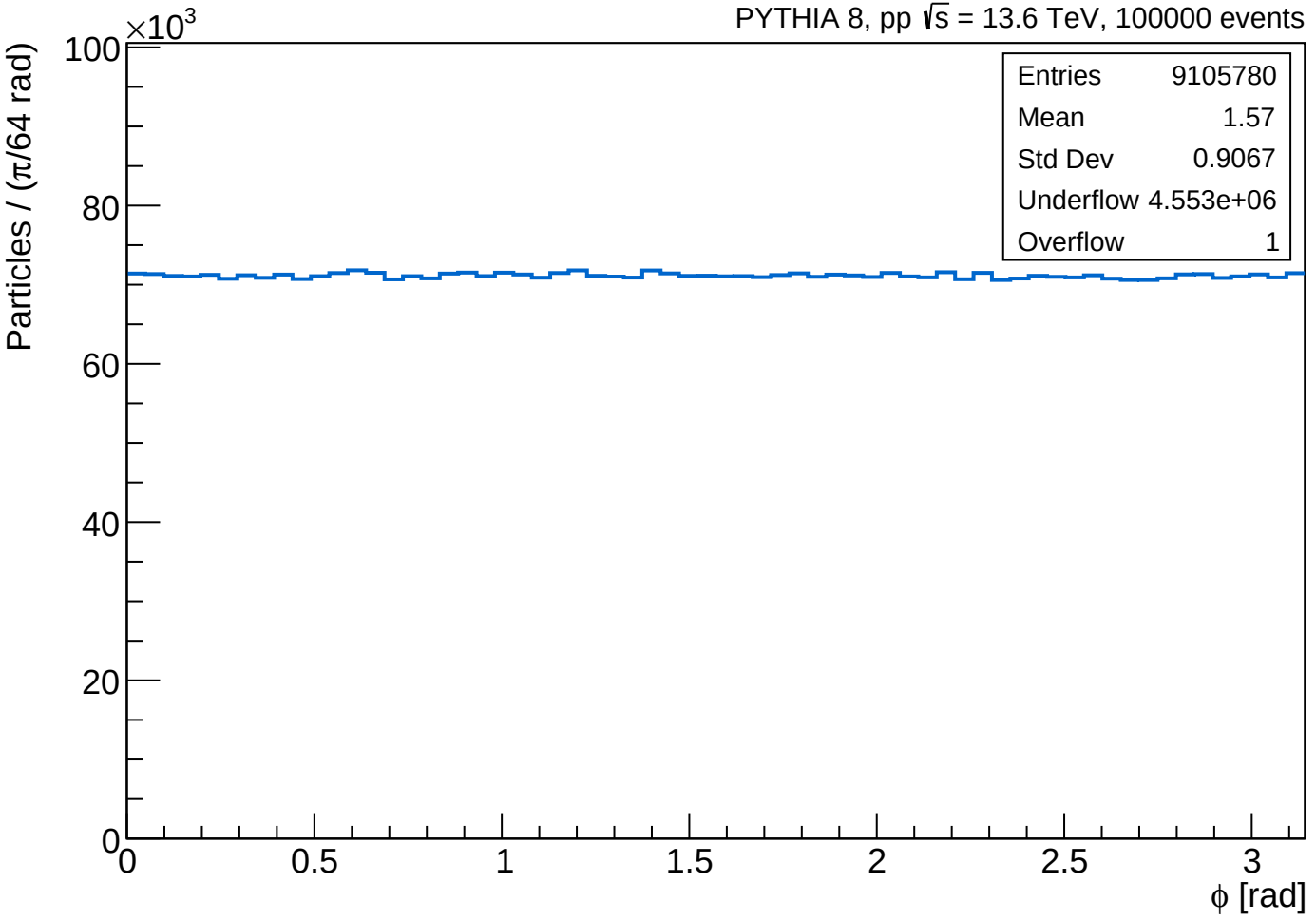

Pseudorapidity: p
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 383144
Mean −0.0001382
Std Dev 2.962
Underflow 9.665e+04
Overflow 9.651e+04
Particles / 0.1
3000
2500
2000
1500
1000
500
0
−5
−4
−3
−2
−1
0
1
2
3
4
5
η

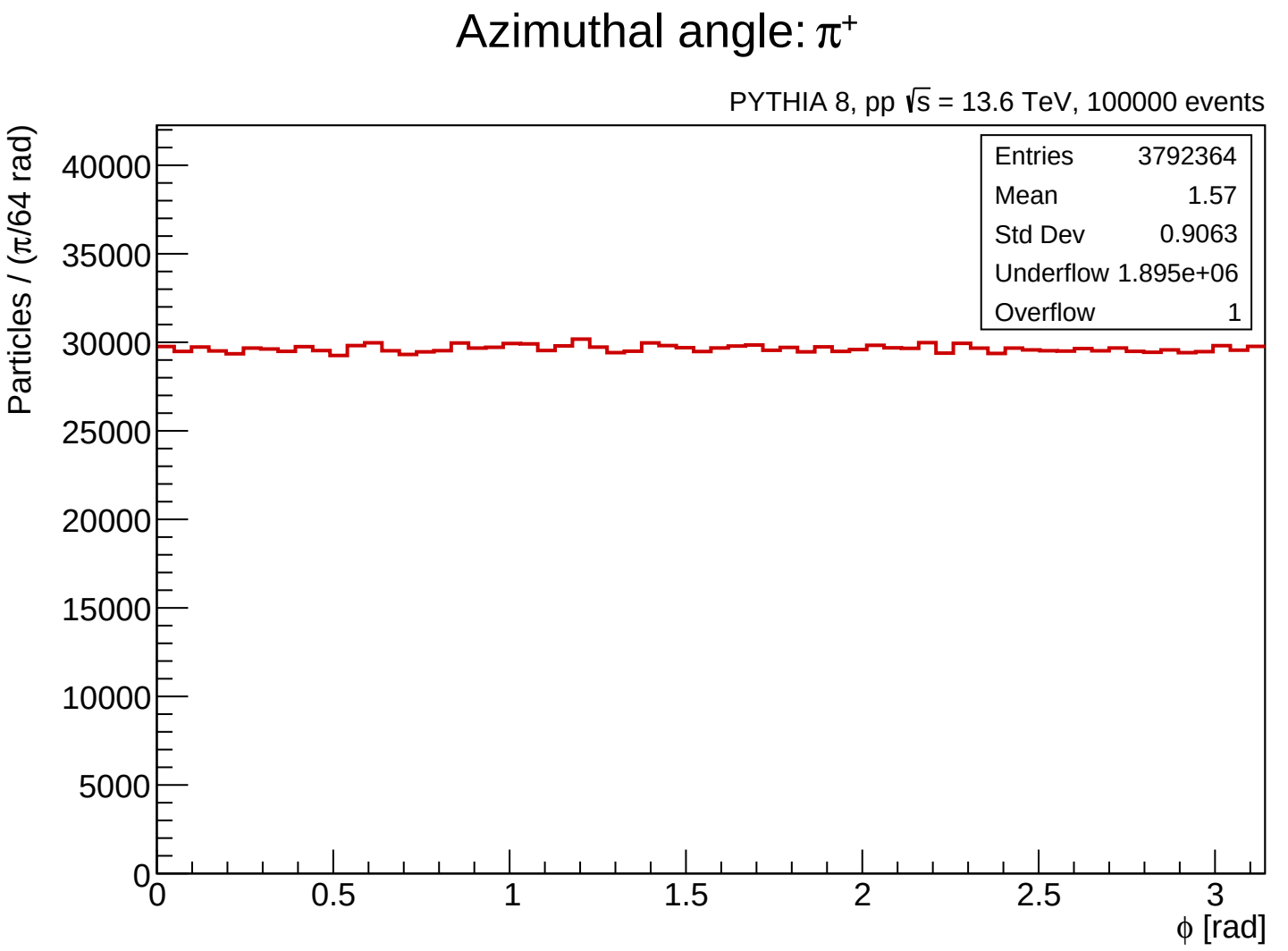

Pseudorapidity: p̄
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 261550
Mean –0.007332
Std Dev 2.952
Underflow 3.756e+04
Overflow 3.774e+04
Particles / 0.1
3000
2500
2000
1500
1000
500
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

Azimuthal angle: all (final particles)

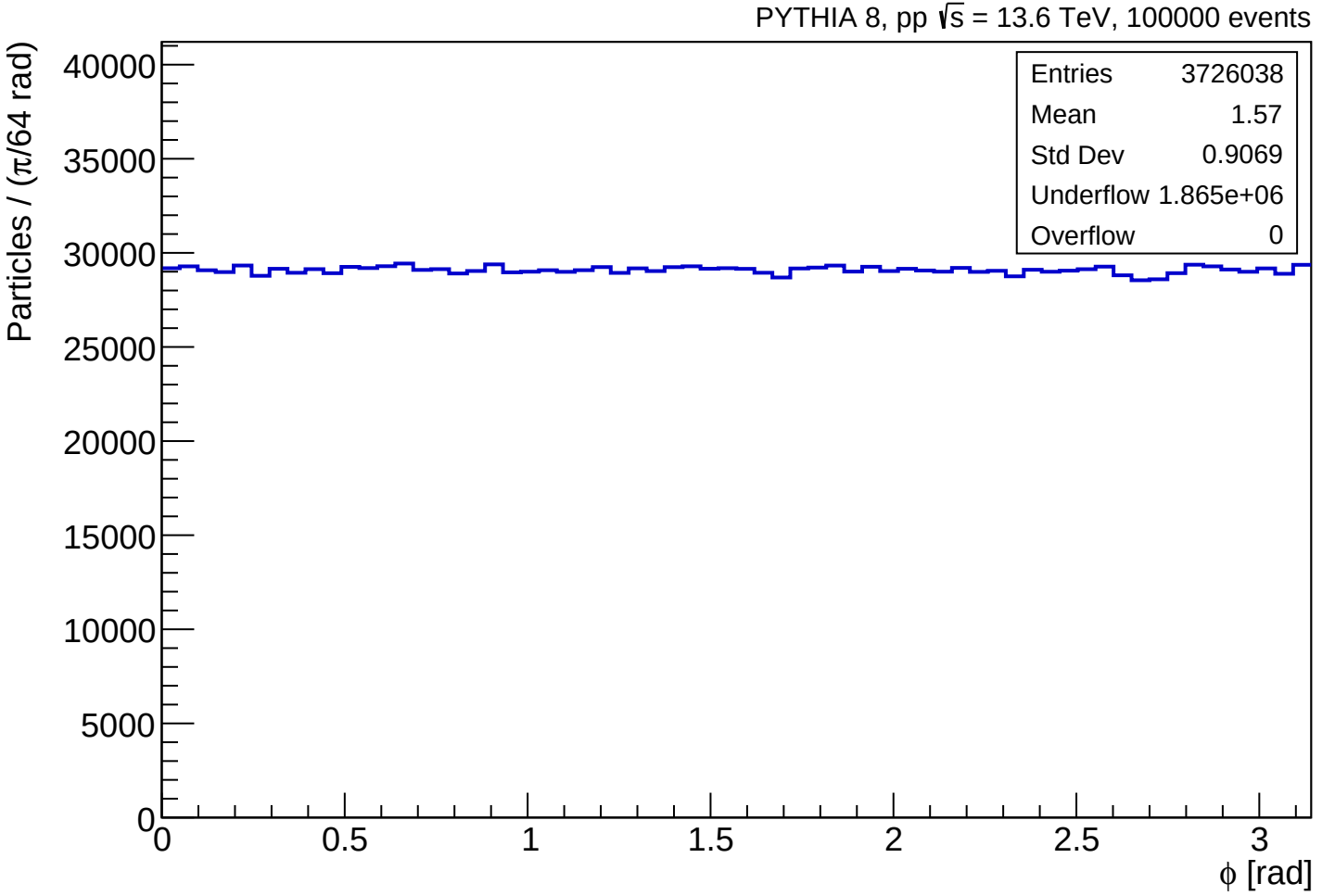


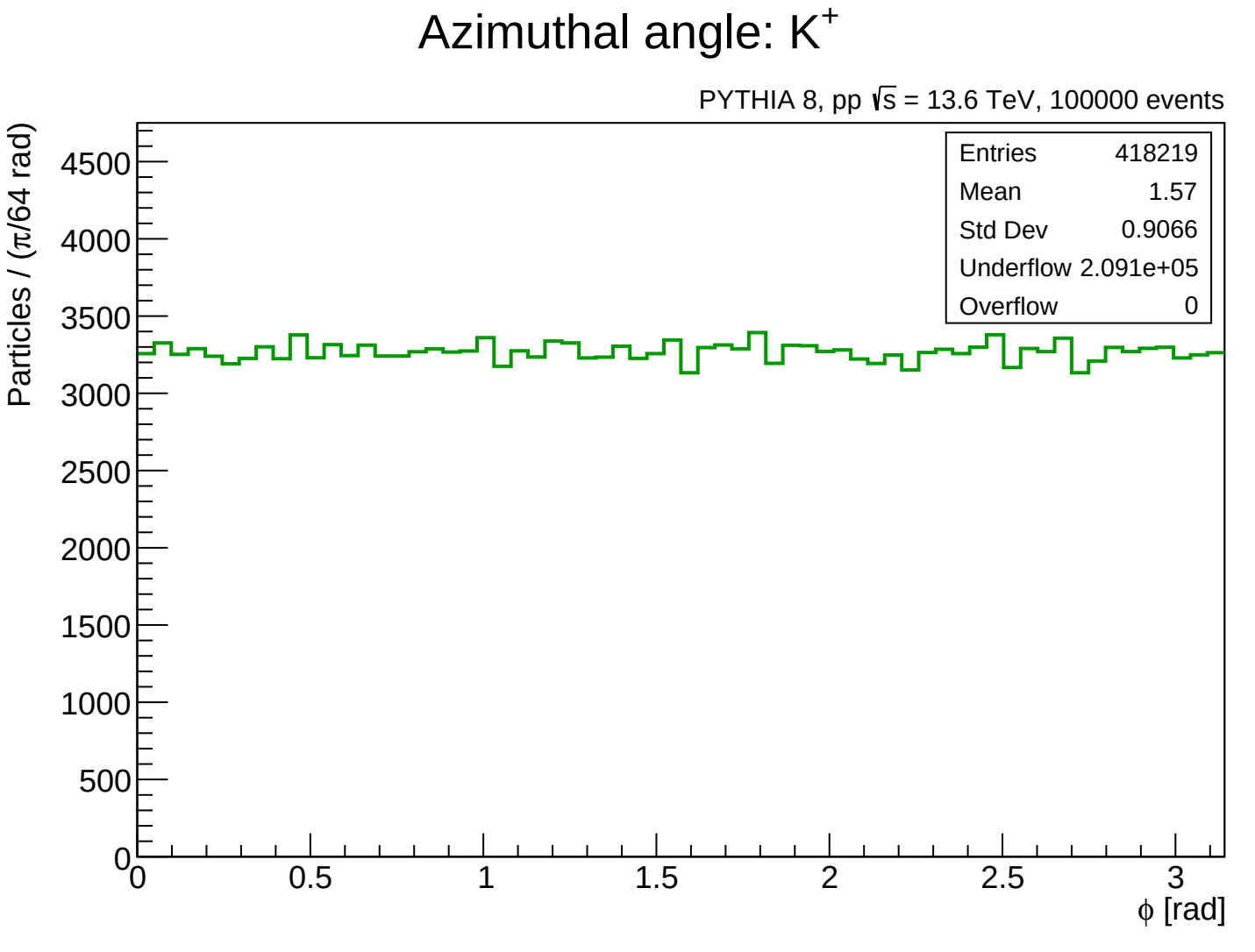
Azimuthal angle: all charged

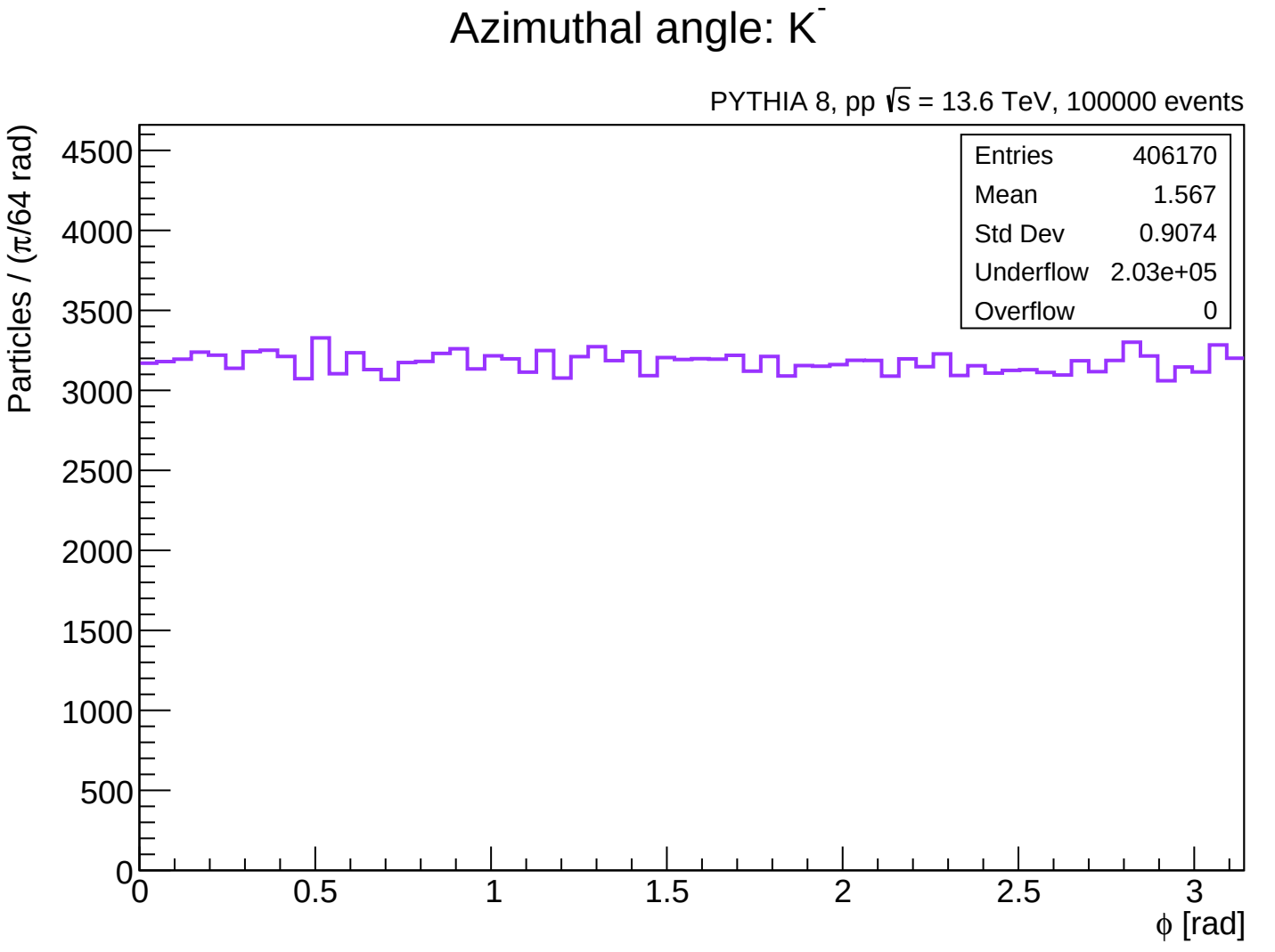



Azimuthal angle: π+
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 3792364
Mean 1.57
Std Dev 0.9063
Underflow 1.895e+06
Overflow 1
Particles / (π/64 rad)
40000
35000
30000
25000
20000
15000
10000
5000
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]

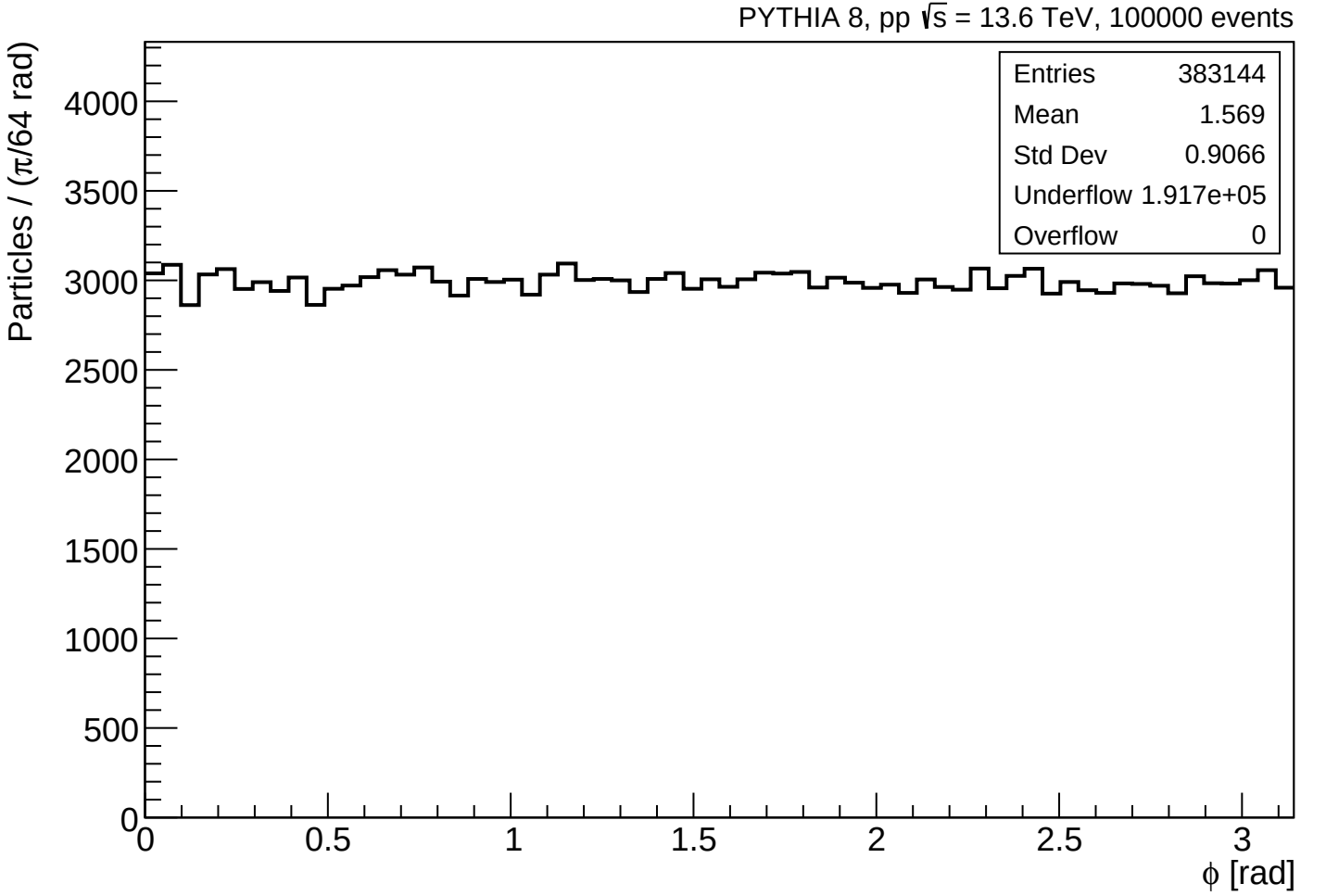
Azimuthal angle: π^-

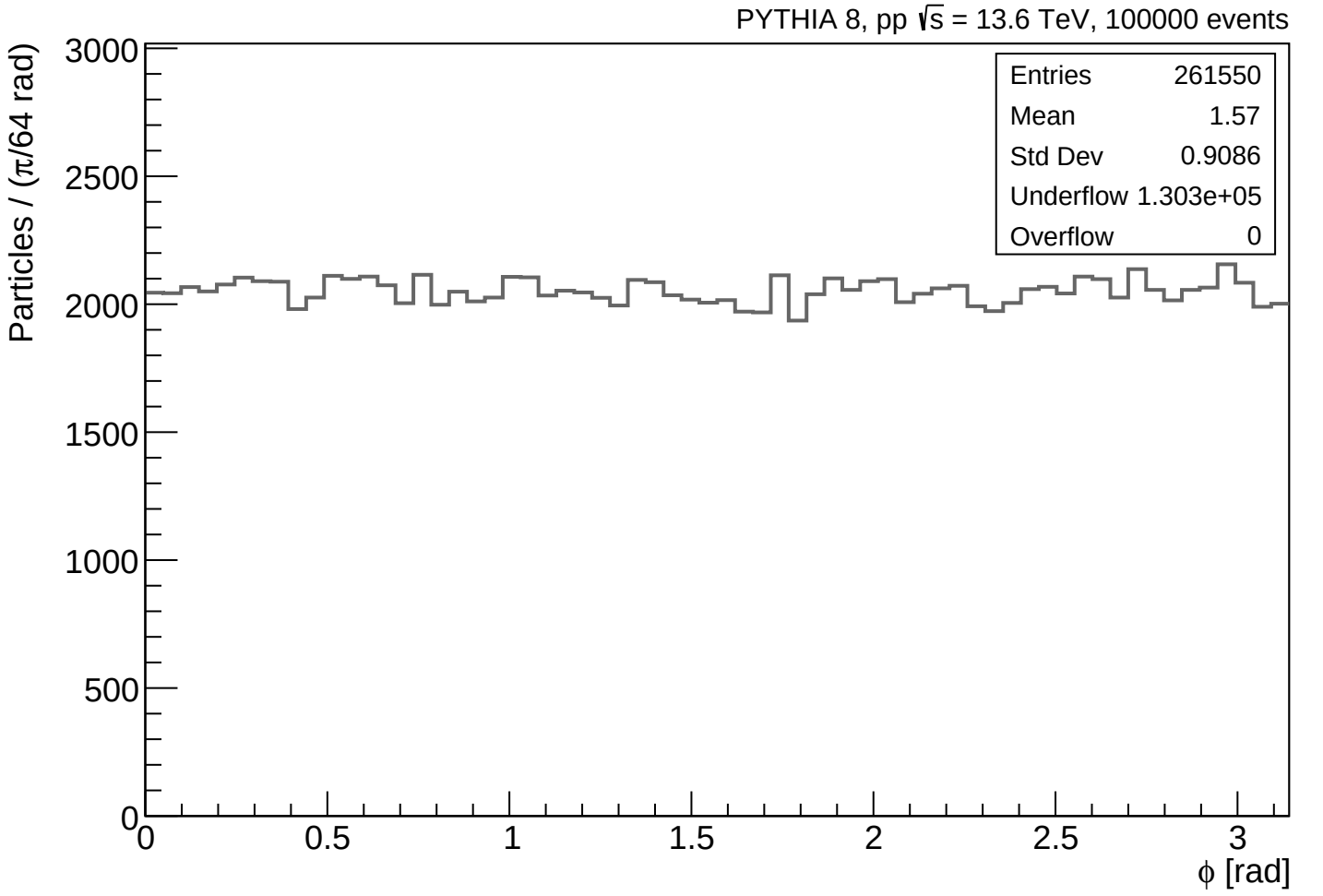


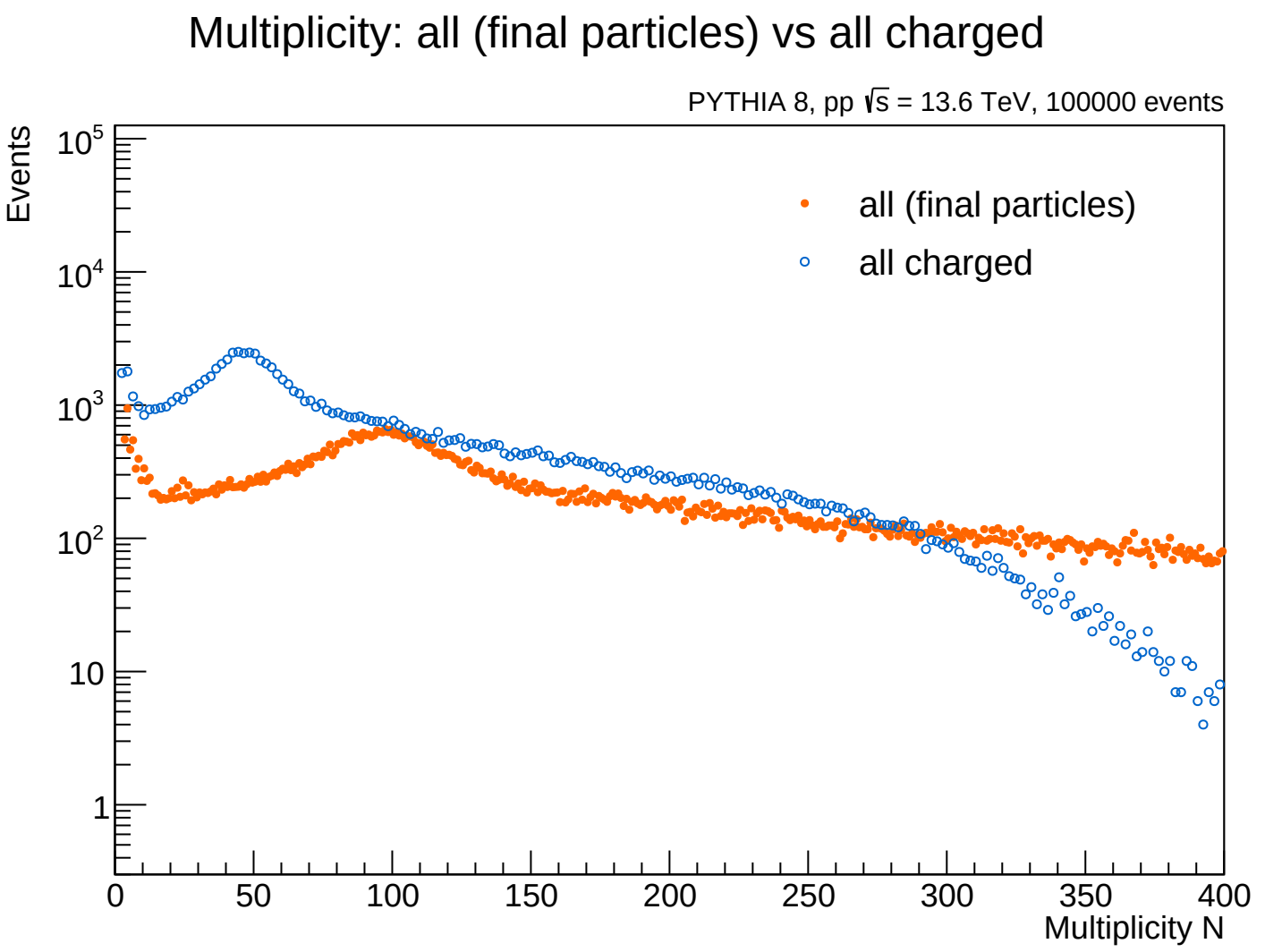

Azimuthal angle: K+
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 418219
Mean 1.57
Std Dev 0.9066
Underflow 2.091e+05
Overflow 0
Particles / (π/64 rad)
4500
4000
3500
3000
2500
2000
1500
1000
500
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]

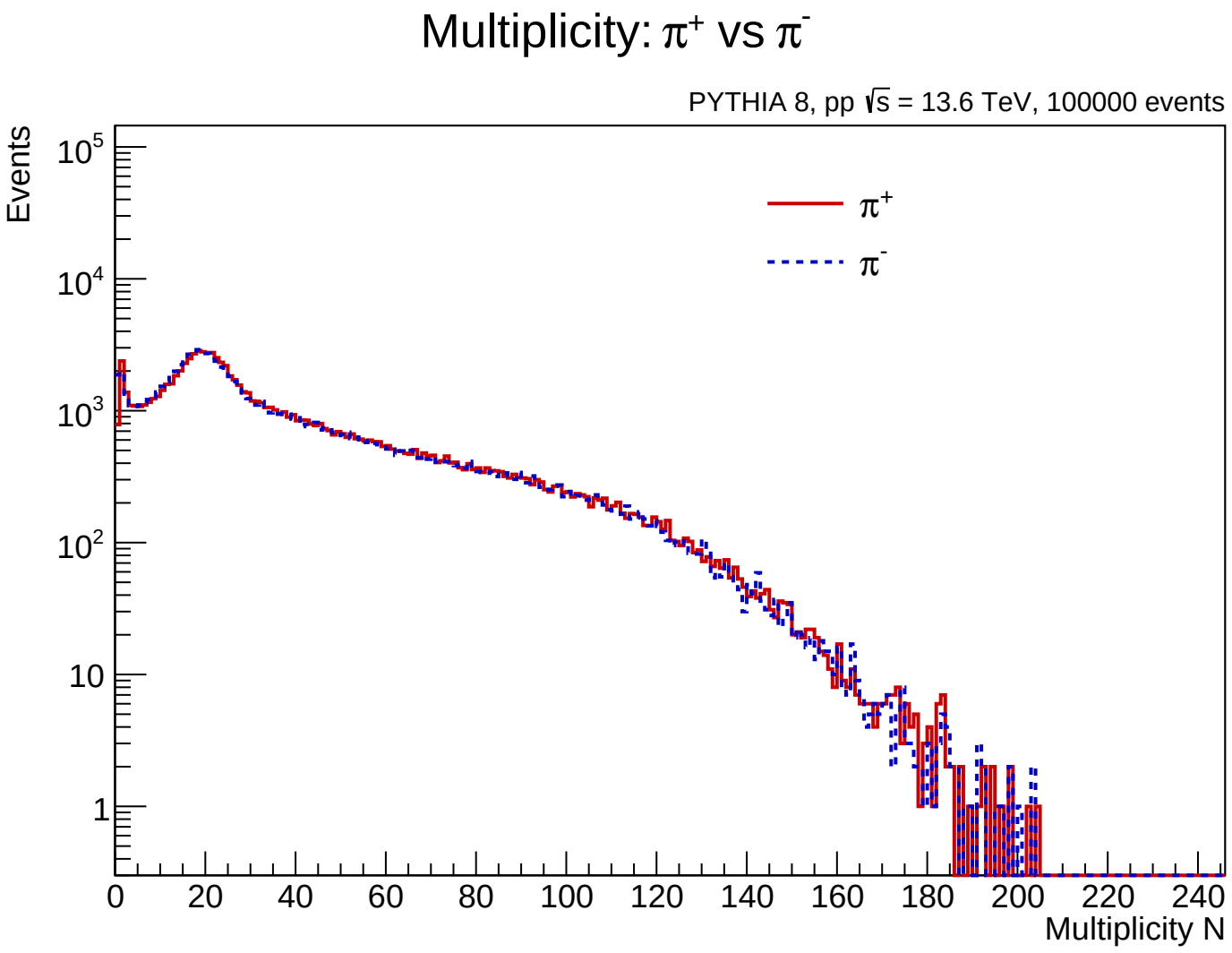

Azimuthal angle: K-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 406170
Mean 1.567
Std Dev 0.9074
Underflow 2.03e+05
Overflow 0
Particles / (π/64 rad)
4500
4000
3500
3000
2500
2000
1500
1000
500
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]

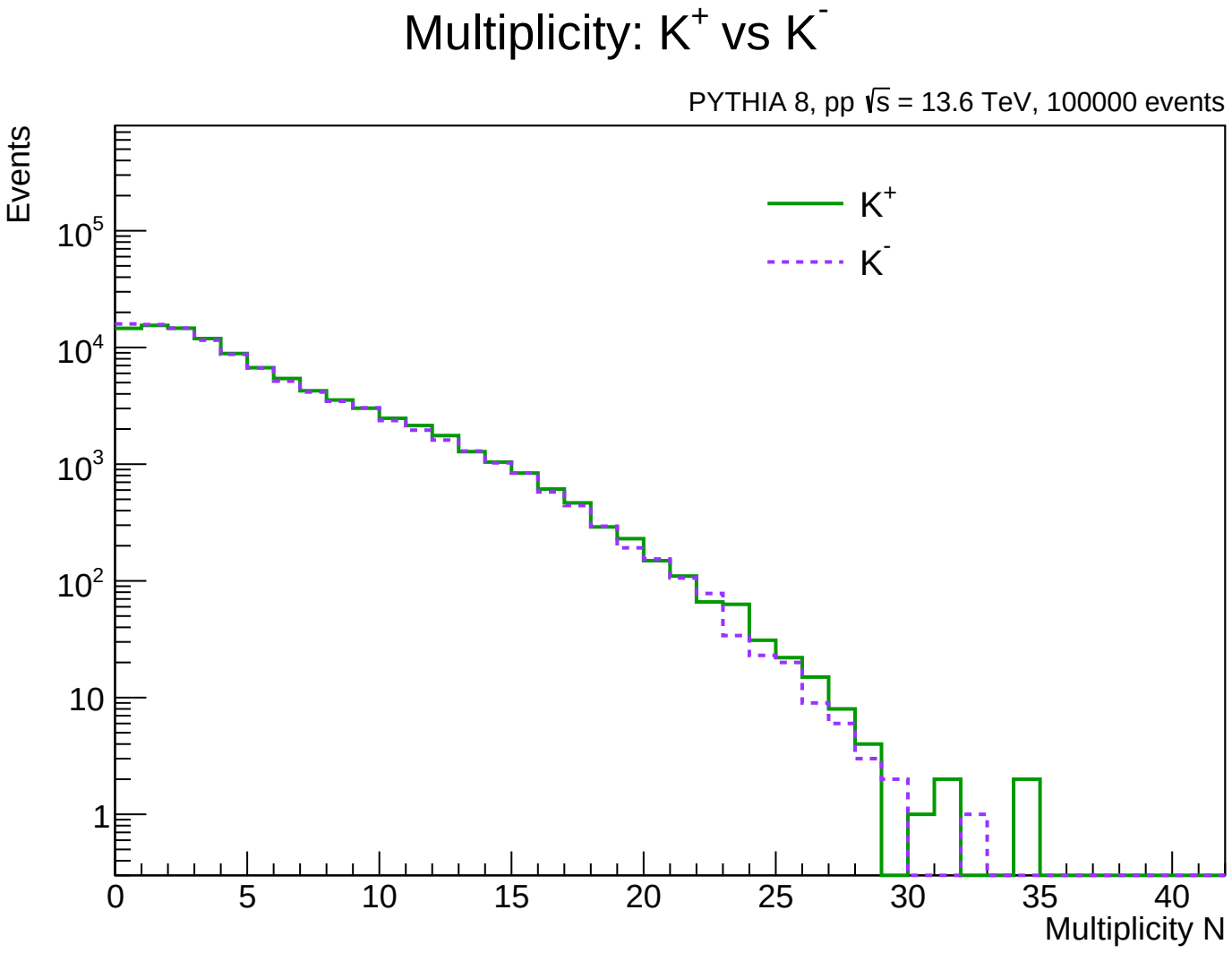
Azimuthal angle: p

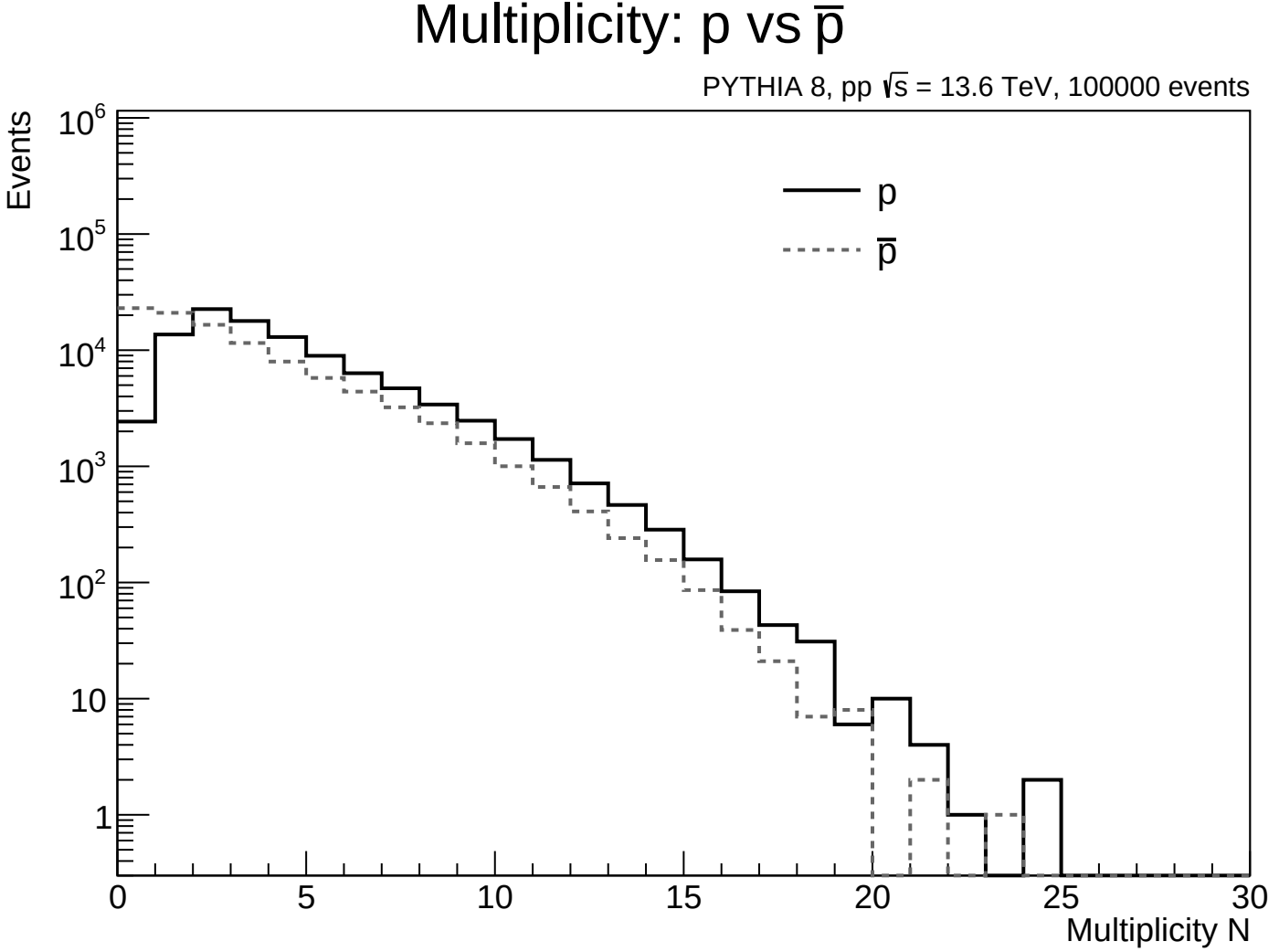


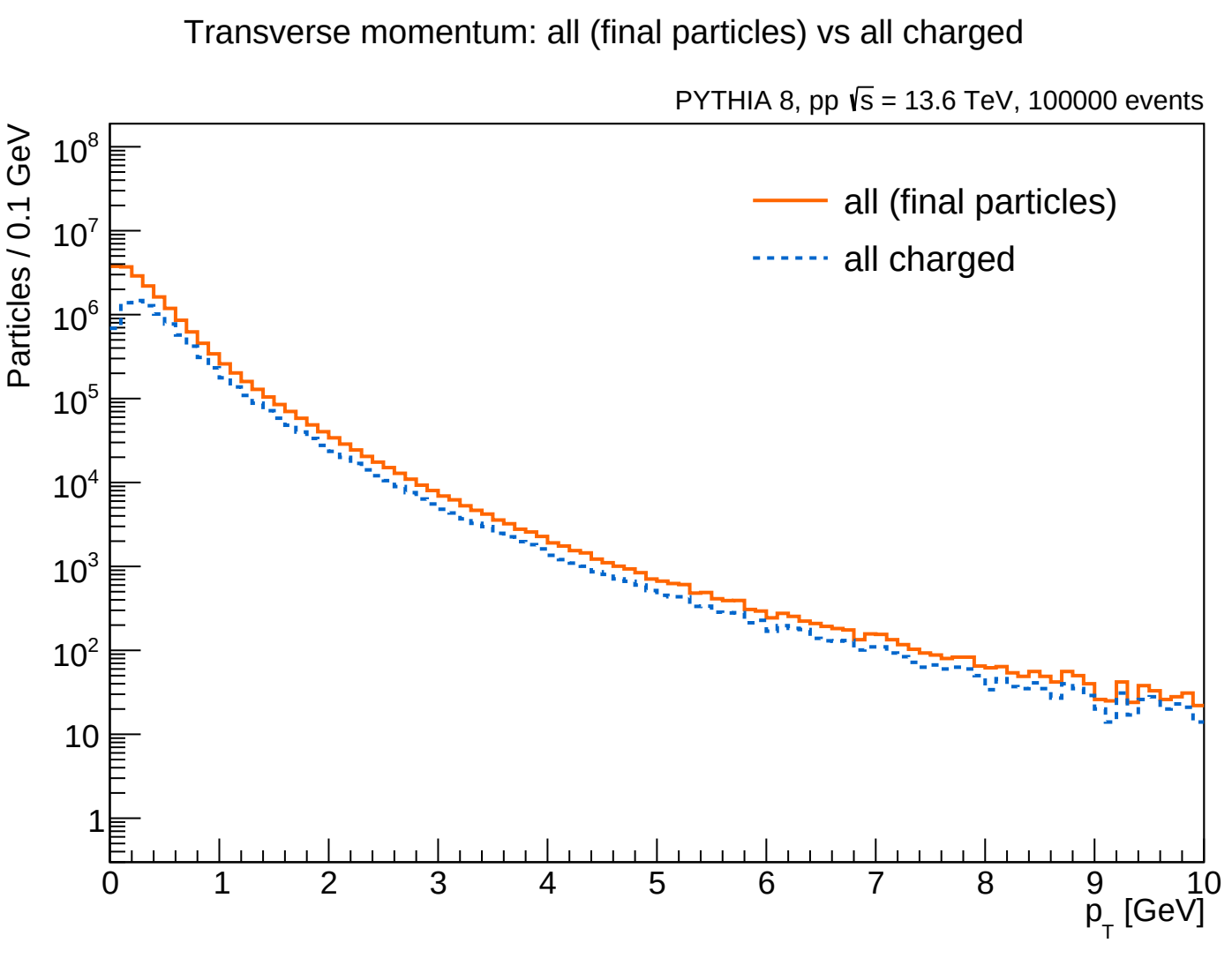

Azimuthal angle: p̄
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Entries 261550
Mean 1.57
Std Dev 0.9086
Underflow 1.303e+05
Overflow 0
Particles / (π/64 rad)
3000
2500
2000
1500
1000
500
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]

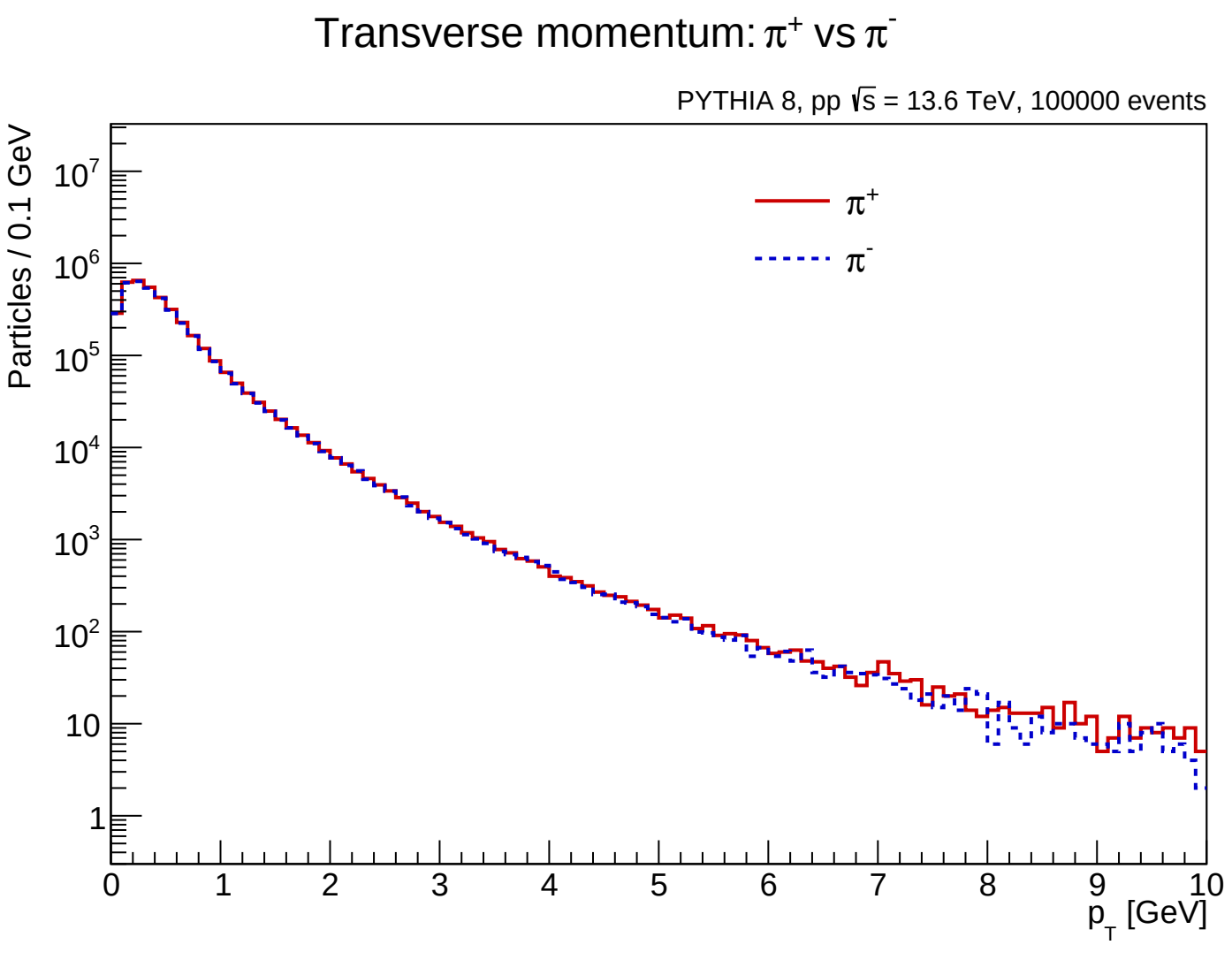

Multiplicity: all (final particles) vs all charged
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Events
Multiplicity N
all (final particles)
all charged
10^5
10^4
10^3
10^2
10
1
0
50
100
150
200
250
300
350
400

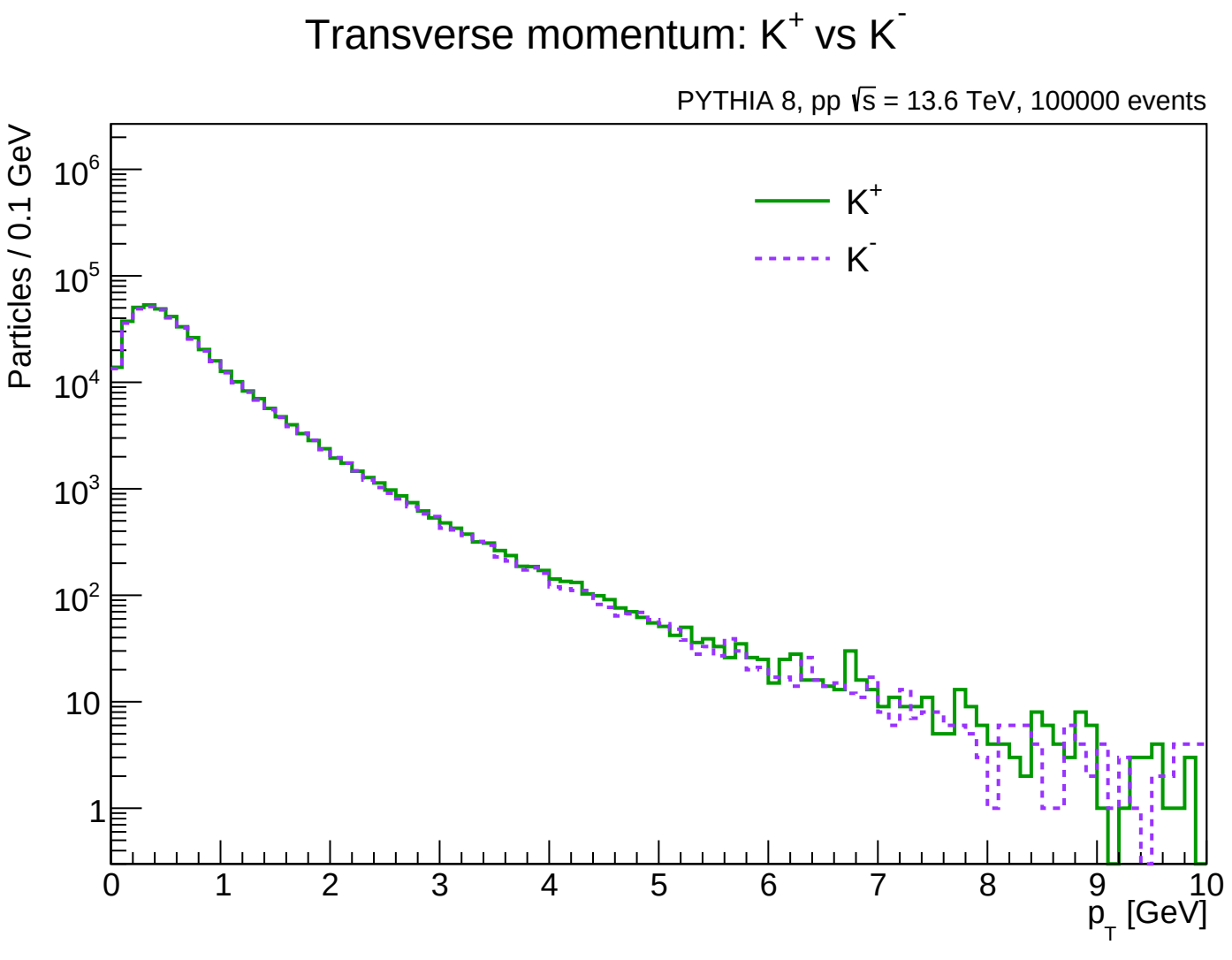

Multiplicity: π+ vs π-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Events
10^5
10^4
10^3
10^2
10
1
π+
π-
0
20
40
60
80
100
120
140
160
180
200
220
240
Multiplicity N

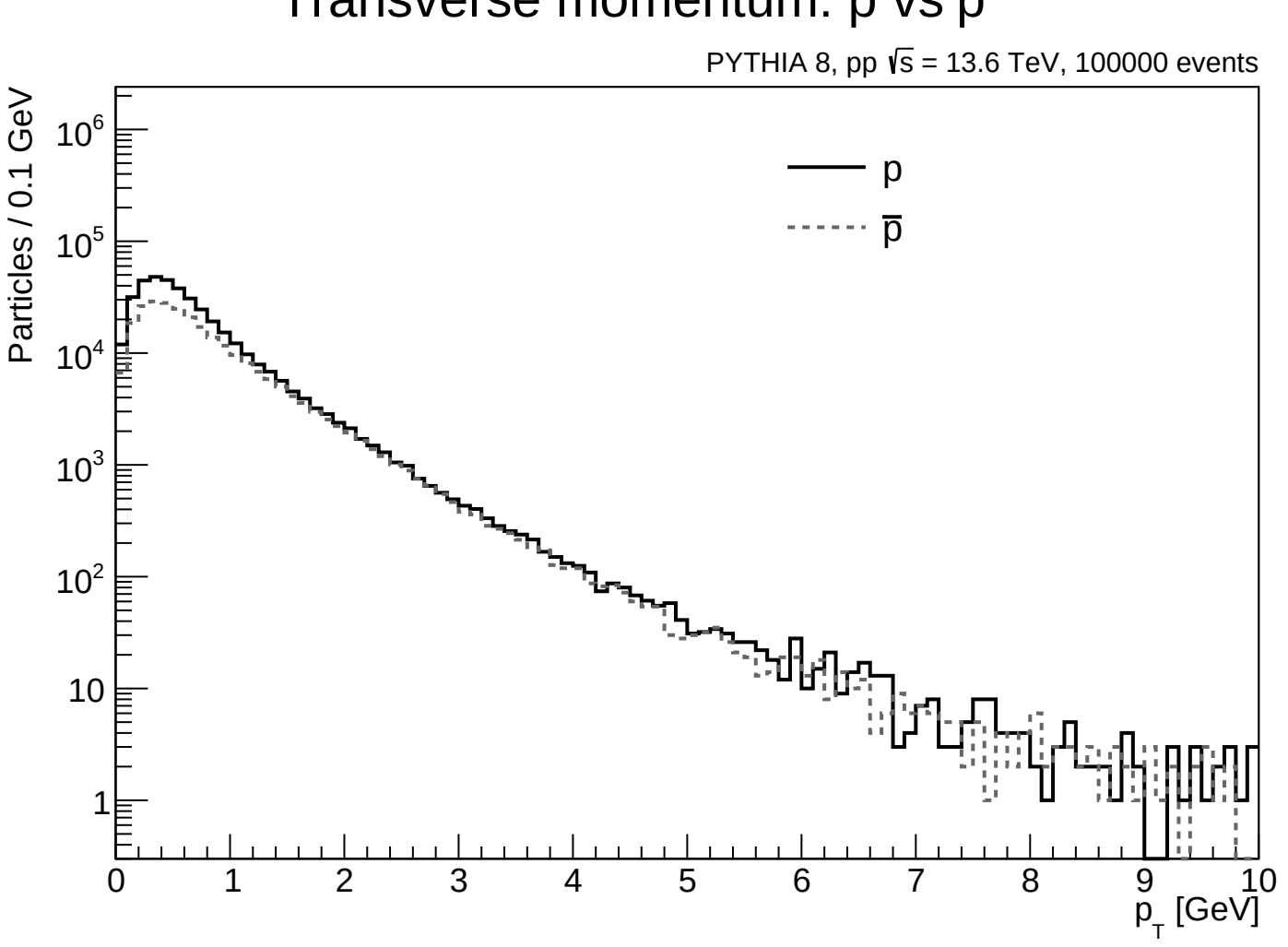

Multiplicity: K+ vs K-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Events
K+
K-
10^5
10^4
10^3
10^2
10
1
0
5
10
15
20
25
30
35
40
Multiplicity N

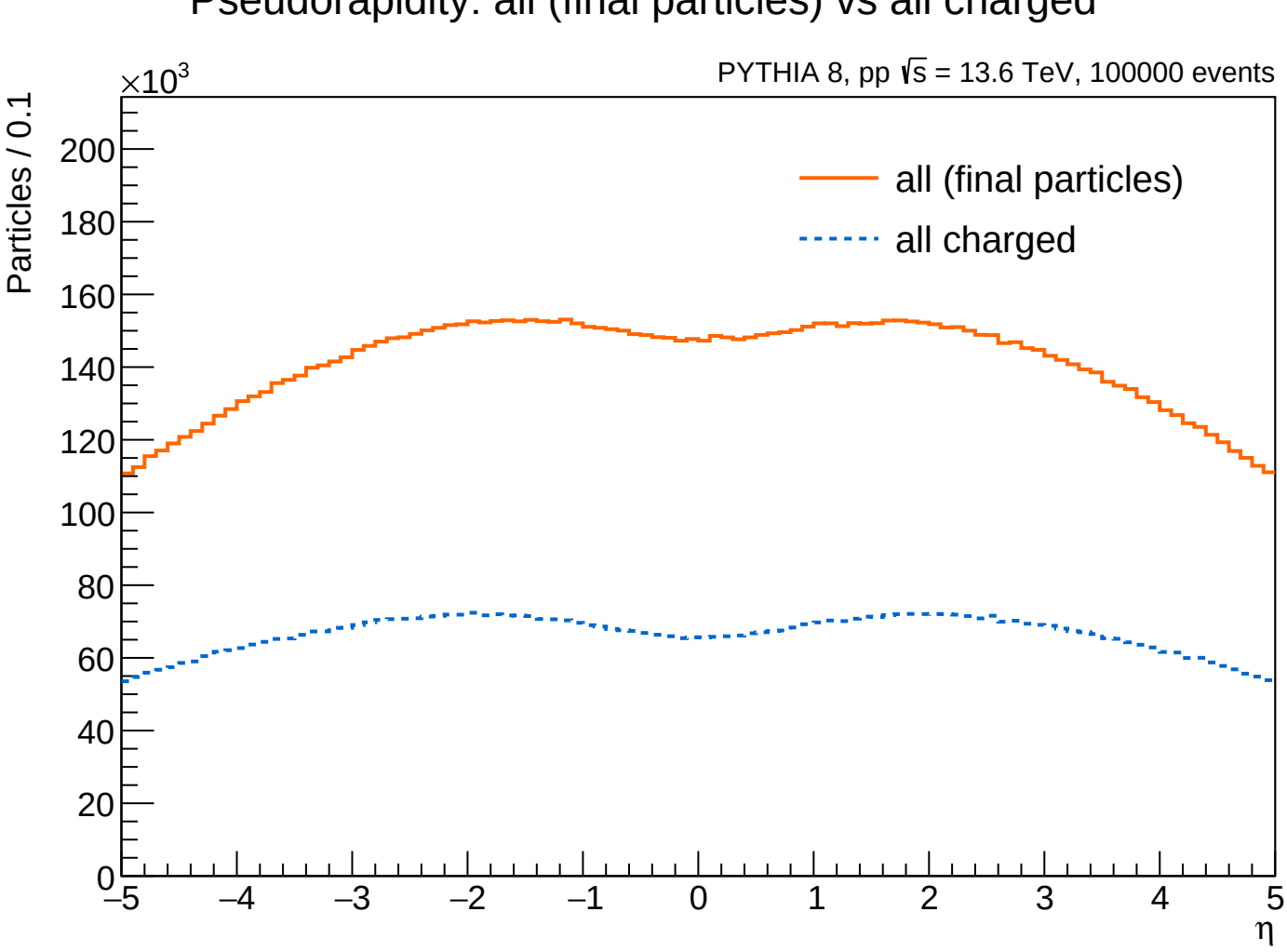

Multiplicity: p vs p̄
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Events
10^6
10^5
10^4
10^3
10^2
10
1
p
p̄
0
5
10
15
20
25
30
Multiplicity N

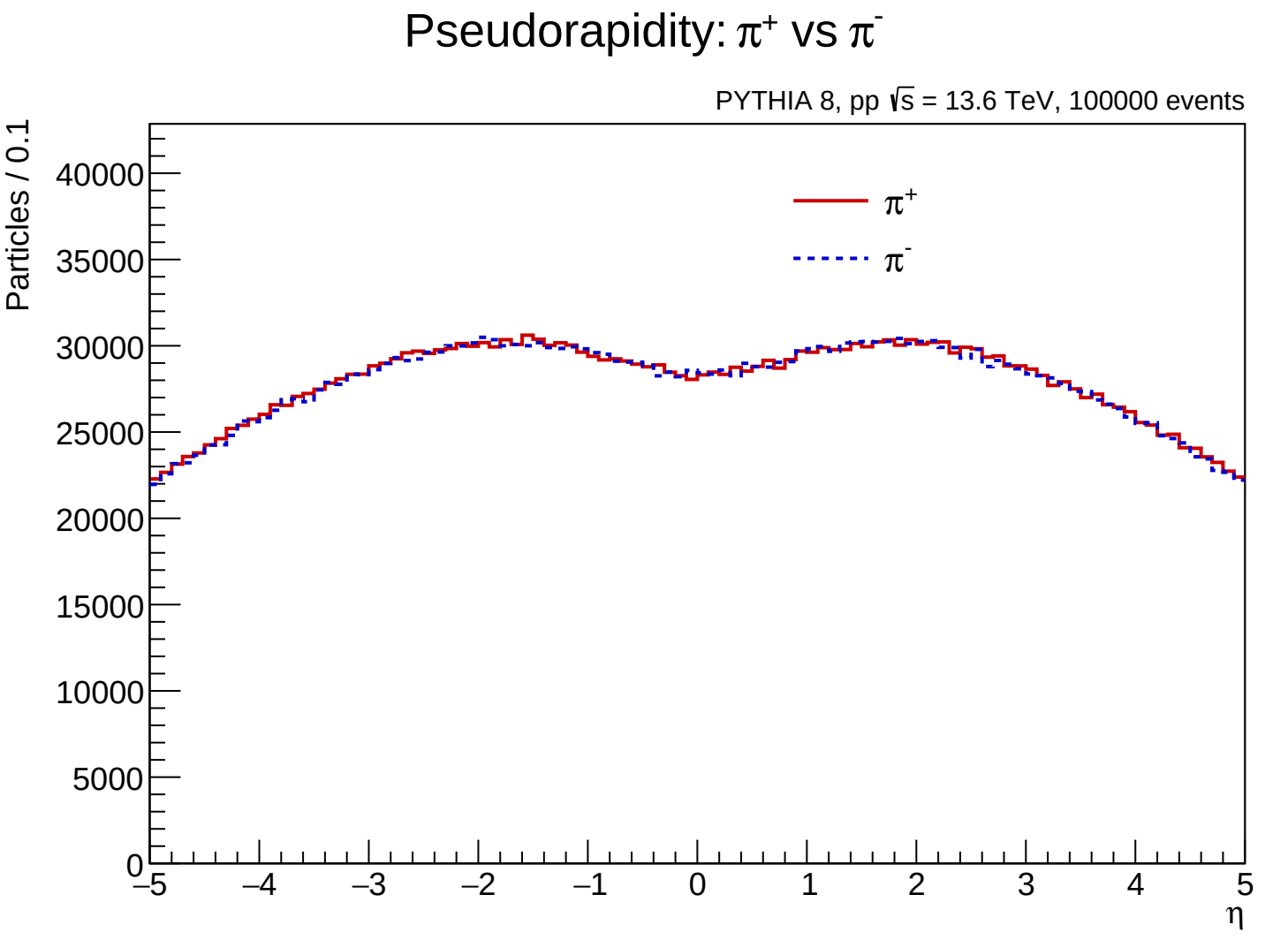

Transverse momentum: all (final particles) vs all charged
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
all (final particles)
all charged
Particles / 0.1 GeV
10^8
10^7
10^6
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]

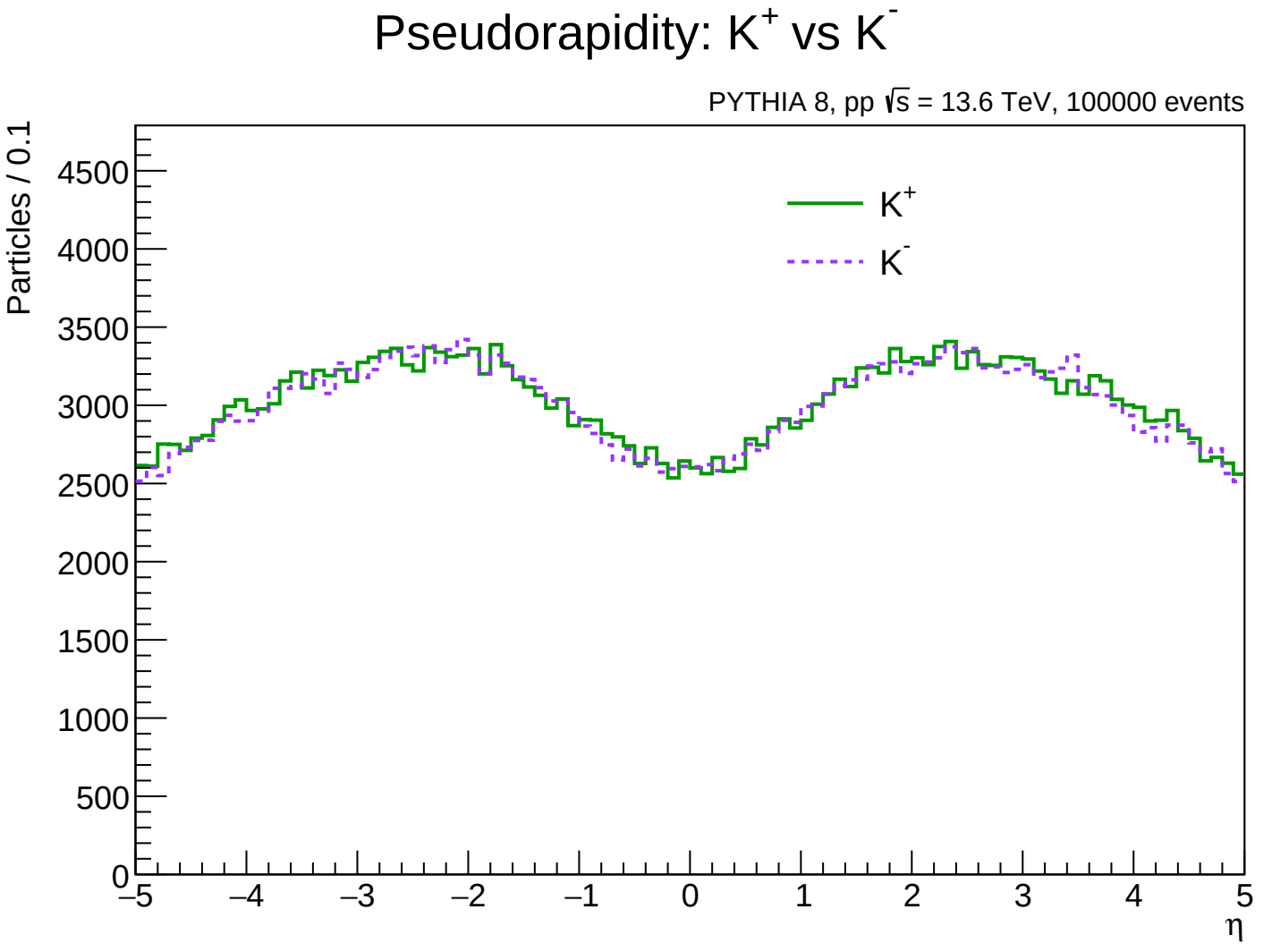

Transverse momentum: π+ vs π-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Particles / 0.1 GeV
10^7
10^6
10^5
10^4
10^3
10^2
10
1
π+
π-
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]

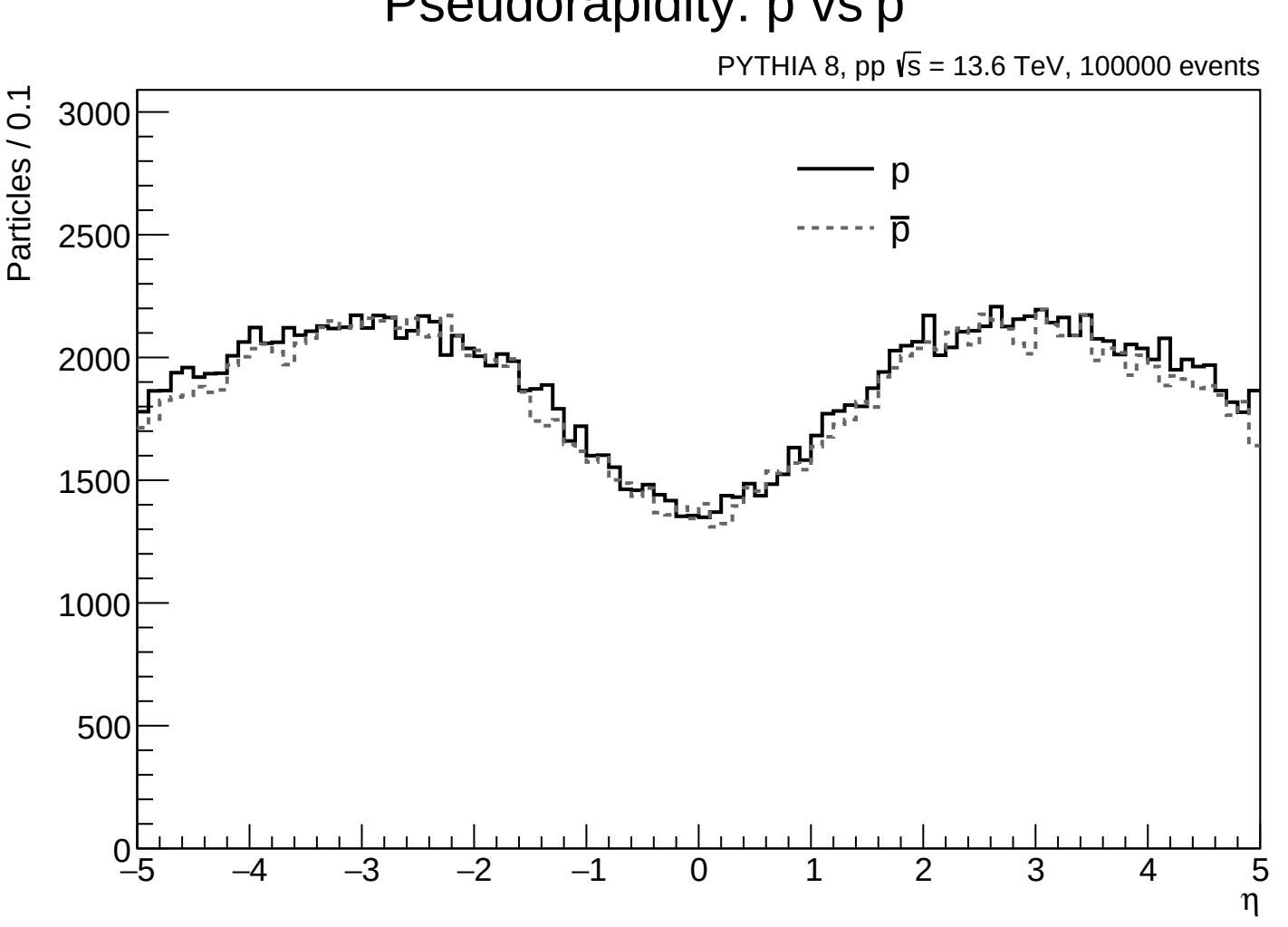

Transverse momentum: K+ vs K-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Particles / 0.1 GeV
K+
K-
pT [GeV]

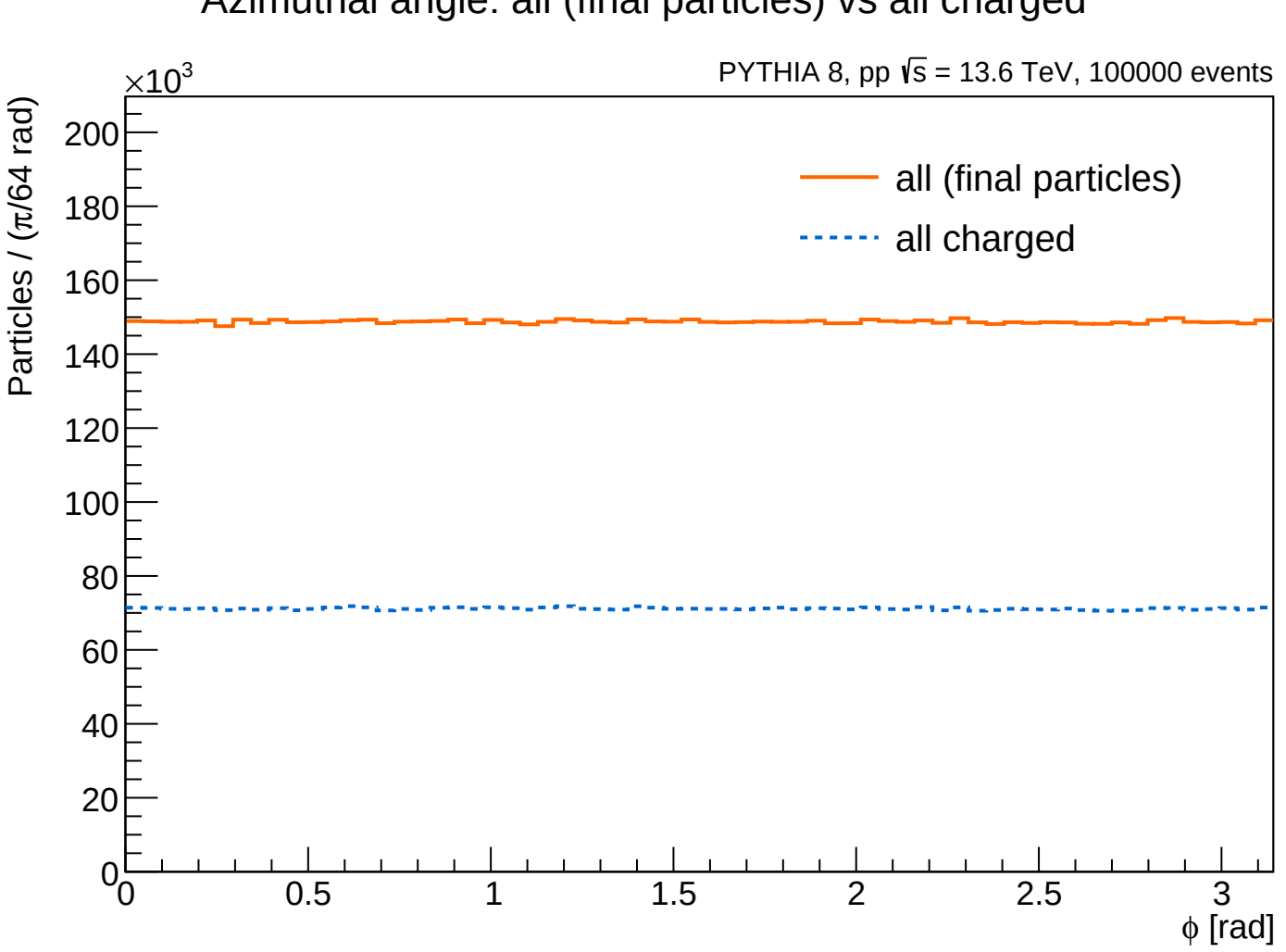

Transverse momentum: p vs p̄
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Particles / 0.1 GeV
10^6
10^5
10^4
10^3
10^2
10
1
p
p̄
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]

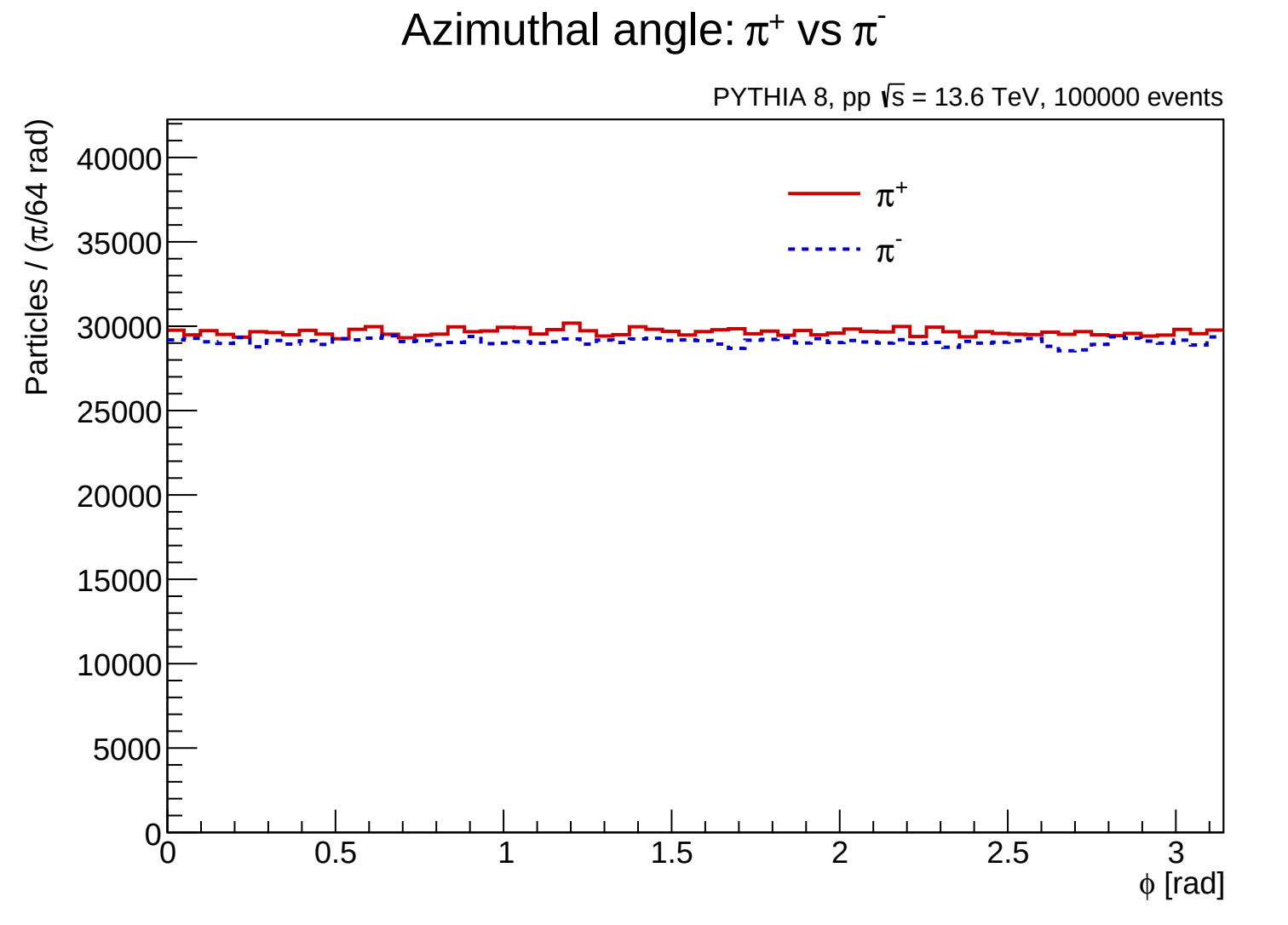

Pseudorapidity: all (final particles) vs all charged
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
×10^3
Particles / 0.1
all (final particles)
all charged
200
180
160
140
120
100
80
60
40
20
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

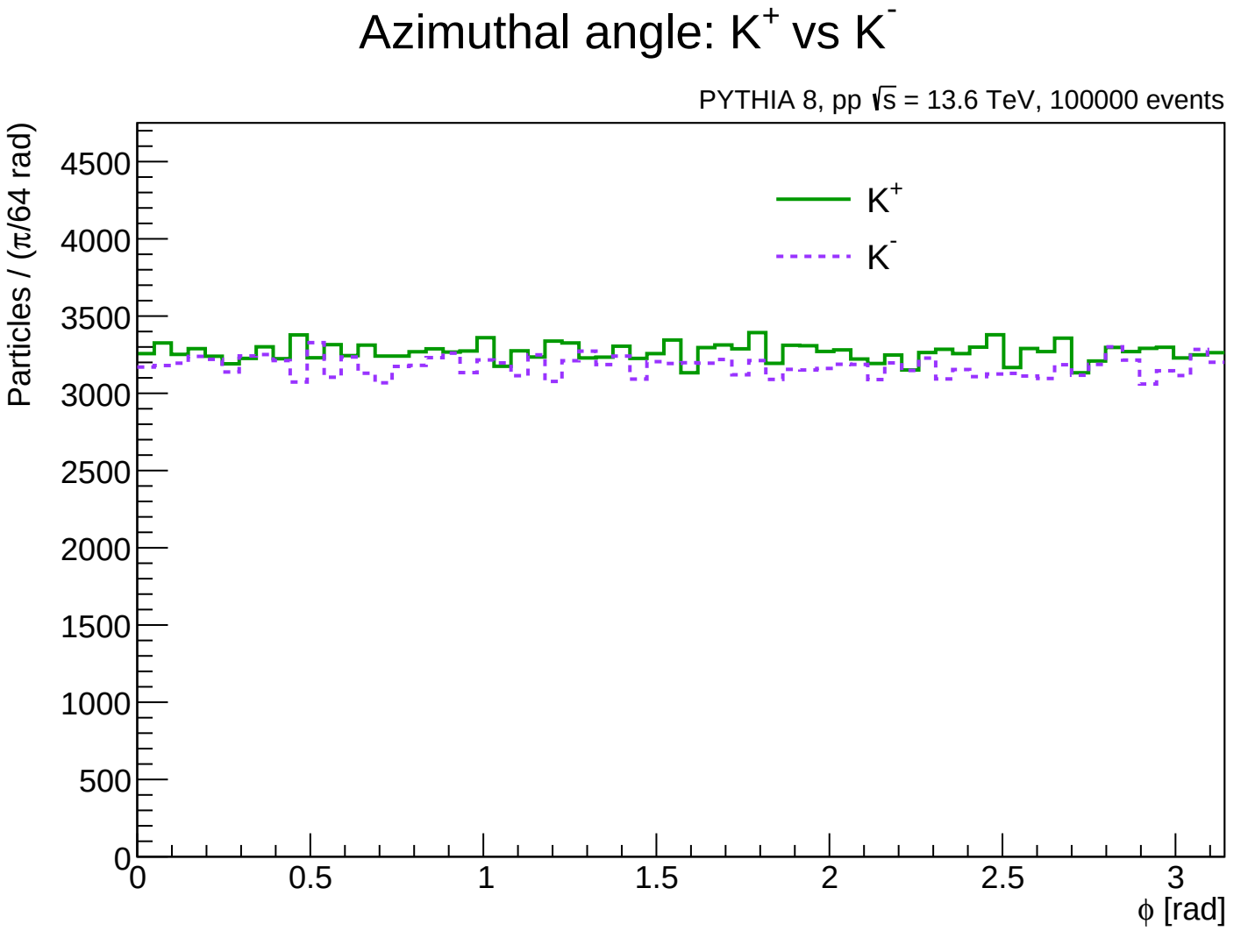

Pseudorapidity: π+ vs π-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Particles / 0.1
40000
35000
30000
25000
20000
15000
10000
5000
0
π+
π-
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

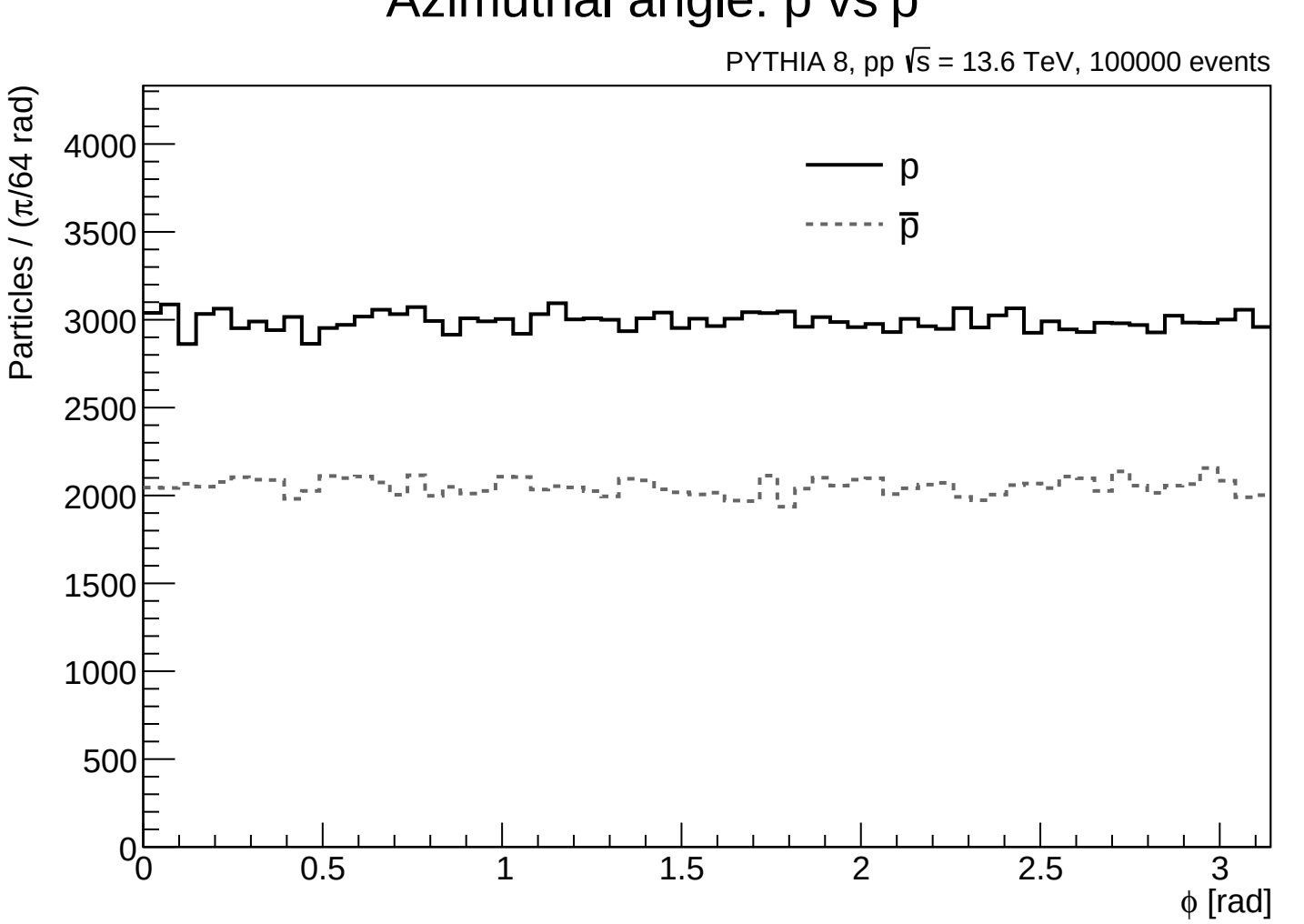

Pseudorapidity: K+ vs K-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Particles / 0.1
K+
K-
4500
4000
3500
3000
2500
2000
1500
1000
500
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η


Pseudorapidity: p vs p̄
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Particles / 0.1
3000
2500
2000
1500
1000
500
0
p
p̄
–5
–4
–3
–2
–1
0
1
2
3
4
5
η


Azimuthal angle: all (final particles) vs all charged
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
×10^3
Particles / (π/64 rad)
200
180
160
140
120
100
80
60
40
20
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]
all (final particles)
all charged


Azimuthal angle: π+ vs π-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Particles / (π/64 rad)
40000
35000
30000
25000
20000
15000
10000
5000
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]
π+
π-


Azimuthal angle: K+ vs K-
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Particles / (π/64 rad)
4500
4000
3500
3000
2500
2000
1500
1000
500
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]
K+
K-


Azimuthal angle: p vs p̄
PYTHIA 8, pp √s = 13.6 TeV, 100000 events
Particles / (π/64 rad)
4000
3500
3000
2500
2000
1500
1000
500
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]
p
p̄